

SECTION 8.0

OBSTETRIC/GYNECOLOGIC SURGERY

CHAPTER 8.1

Gynecologic Oncology

SURGEONS

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STAGING LAPAROTOMY FOR OVARIAN, FALLOPIAN TUBE, AND PRIMARY PERITONEAL CANCER

SURGICAL CONSIDERATIONS

Description

Ovarian carcinoma has the highest mortality rate of all gynecologic malignancies because it is usually discovered in advanced stages with a pelvic mass, omental caking, and ascites. Surgery is used for staging as well as therapy. Studies have demonstrated an inverse relationship between postop residual tumor mass and survival; therefore, the goals of surgery are accurate staging and optimal tumor debulking (<1 cm residual disease). The standard open procedure consists of a meticulous exploration of the abdominopelvic cavity, abdominopelvic cytology, multiple random and targeted biopsies, **total abdominal hysterectomy (TAH)**, **bilateral salpingo-oophorectomy (BSO)**, **pelvic and paraaortic lymph node dissection**, **infracolic omentectomy**, **appendectomy**, and additional cytoreductive procedures. After access to the abdomen is obtained through a midline abdominal incision, cytologic washings of the pelvis, pericolic gutters, lesser sac, and hemidiaphragms are done. The peritoneal cavity is carefully explored. A TAH/BSO is performed by ligating and transecting the round,

infundibulopelvic, broad, cardinal, and uterosacral ligaments on both sides. The specimen is cut away from the vagina and the cuff closed. Pelvic and paraaortic lymph node dissections are then performed. Pelvic lymphadenectomy is performed by opening the pelvic peritoneum, developing the paravesical and pararectal space, identifying the ureter, and removing the lymph node tissue adjacent to the common and external iliac vessels and obturator vein and nerve. Paraaortic lymph node dissection is performed by opening the peritoneum over the great vessels followed by removal of the lymph node bundles from the preaortic, lateral aortic, and retroaortic spaces. All residual tumor is removed, using sharp dissection and/or cavitron ultrasonic surgical aspirator (CUSA) and/or argon beam coagulator (ABC), and then an appendectomy is usually performed. The omentum is clamped, transected, and ligated along its attachment to the transverse colon. A **bowel resection** with possible **colostomy** formation may be necessary to achieve optimal cytoreductive surgery (see **Pelvic Exenteration**). Next, the peritoneal cavity is irrigated copiously. Targeted and random biopsies of bladder, cul-de-sac of Douglas, pericolic gutters, hemidiaphragms, small bowel, large bowel, and anterior abdominal wall are performed. A peritoneal port may be placed subcutaneously for use in future intraperitoneal chemotherapy. Approximately 25% of patients undergoing cytoreductive surgery for advanced stages of ovarian carcinoma require bowel resection with either primary reanastomosis or colostomy. A **splenectomy** is not routinely done unless the spleen is involved with tumor. Some patients with unresectable disease and bowel obstruction will require a **gastrostomy tube** placement at this time. A less extensive surgical procedure may be appropriate if a large volume of unresectable tumor is discovered. Surgery in these cases must be individualized. In addition, **laparoscopic surgical staging** has recently become more common for early-stage lesions (see [Fig. 8.1-1](#)).

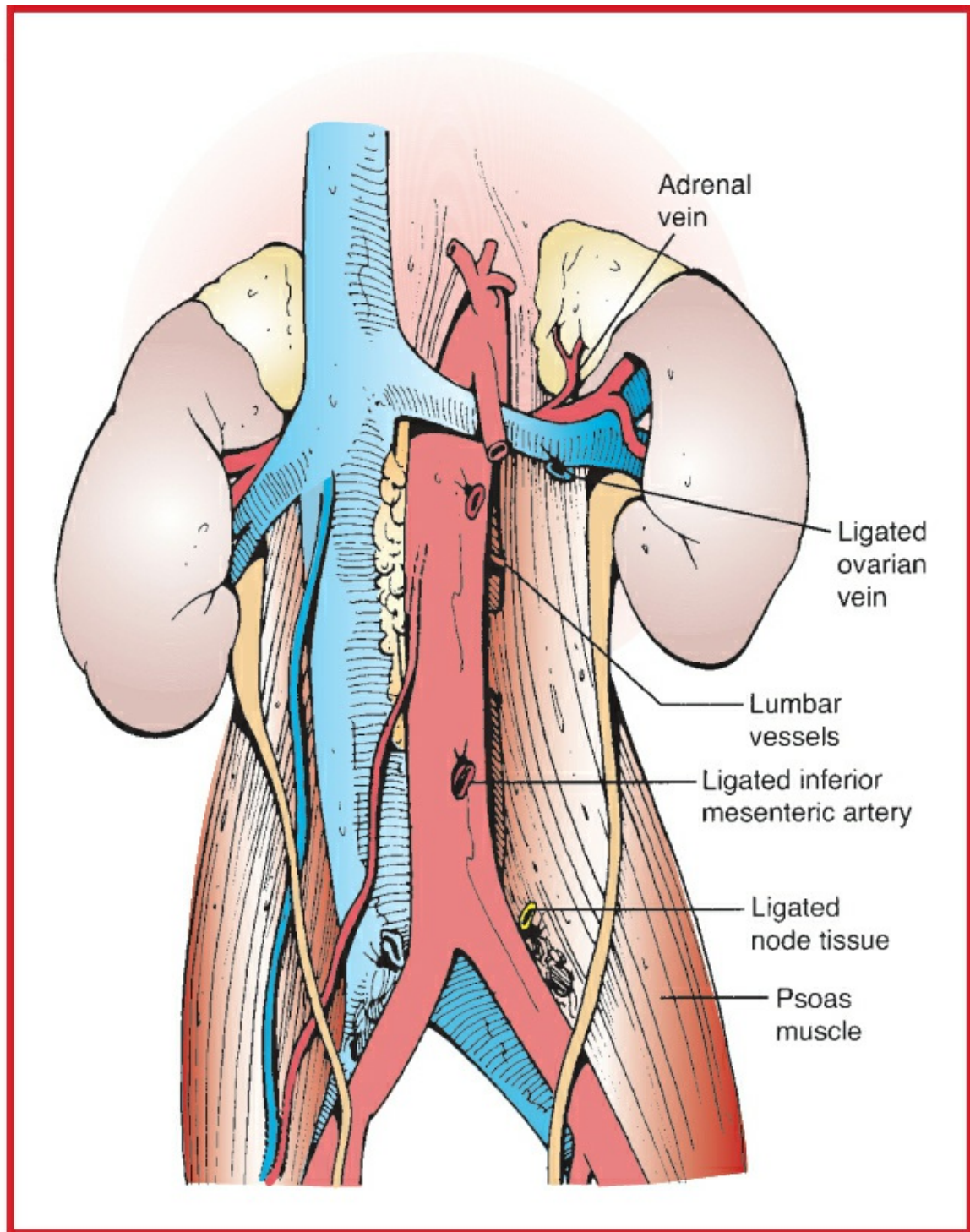


Figure 8.1-1 Aortic node dissection in staging laparotomy for ovarian cancer. In this case, both right and left node dissections have occurred, leaving the kidney, renal hilum, and psoas muscle exposed. Because the dissection is infrarenal and anterior to the lumbar vessels, there is residual fatty and nodal tissue at the posterior limit of the dissection.

In some stage I lesions, a **unilateral salpingo-oophorectomy (USO)** is sufficient therapy. The decision to use this approach depends on cell type, age, reproductive status, and extent of disease. Limited USO staging is usually performed as a fertility-sparing procedure for young women with early-stage tumors. Biopsies, a

retroperitoneal lymph node dissection, omentectomy, and appendectomy are generally performed as well.

Usual preop diagnosis

Ovarian cancer/pelvic mass

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, postop recovery plan, and possibility of urinary catheter postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine, dorsal lithotomy (if anticipating lower bowel resection)
Incision	Midline or paramedian abdominal
Special instrumentation	CUSA, vital view (suction-irrigation device combined with light source) helpful; laparoscopic biopsy forceps (laparoscope, laparoscopic instruments for evaluation of upper abdomen through lower vertical abdominal incision); argon beam coagulator (ABC); TA, GIA, EEA stapling devices
Unique considerations	Removal of large amounts of ascites may → fluid shifts and intravascular volume depletion intraop and postop. Consider preop prophylactic heparin. Some patients will have had bowel prep and may be hypovolemic at the start of surgery. Optimal debulking may require bowel resection, splenectomy, pancreatectomy, or other advanced procedures. During lymphadenectomy, there is a risk of vascular injury that may lead to significant blood loss and need for rapid volume replacement.
Antibiotics	Cefoxitin or cefotetan 2 g iv; redose as indicated by duration and blood loss
Surgical time	1–5 h, including splenectomy and bowel surgery for more advanced stages

Closing considerations	Consider NG tube placement by anesthesiologist (in select cases with extensive bowel involvement or bowel procedures); peritoneal or central venous access for subsequent chemotherapy.
EBL	250–500 mL for stage I lesions; >500–1000 mL for advanced stages
Postop care	Extensive peritoneal raw surfaces → intraperitoneal fluid 3rd-spacing. Patients require aggressive hydration to maintain intravascular volume. Central hemodynamic monitoring and ICU admission are useful in selected patients. Use SCDs and prophylactic heparin for VTE prophylaxis.
Mortality	1–2/1000
Morbidity	Grade 3–4 hemorrhage 7% Postop fever: 14–19% Wound infection: <5% Wound dehiscence: 0.3–3% VTE: 1–2% Ureteral injury: <1% Vaginal vault prolapse: rare
Pain score	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	All age groups; most common, 60–69 yr, mean age 63
Incidence	9.4/100,000 (22,280 new cases/yr); 1.4% lifetime risk of ovarian cancer
Etiology	Unknown
Risk factors	Family Hx of ovarian carcinoma; personal or family Hx of breast cancer (BRCA gene mutations); ↑ age at first pregnancy; nulliparity; advancing age; late menopause; obesity; gonadal dysgenesis; exposure to radiation; environmental factors
Associated	Familial cancer syndromes (e.g., Lynch, BRCA);

conditions	Peutz-Jeghers syndrome, 5% of cases develop gonadal stromal tumor; XY gonadal dysgenesis, gonadoblastomas; multiple nevoid basal cell carcinoma (Gorlin syndrome); ataxia telangiectasia (hereditary, progressive cerebellar lack of muscular coordination associated with recurrent pulmonary infections and ocular and cutaneous telangiectasias)
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ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Ovarian carcinoma may be diagnosed at a late stage, and patients may have developed malignant ascites and pleural effusions, large tumor mass, and omental spread. Surgery is indicated for resection of localized tumor and staging of distant and local metastases. Additional procedures, including bowel resection or lymph node dissection, may be performed at the same time.

Respiratory	Significant ascites and pleural fluid may produce respiratory compromise. The presence of dyspnea, orthopnea, tachypnea, or other chest findings needs to be investigated. Underlying lung diseases, such as asthma, also may be exacerbated by the abdominal distension/ascites. Tests: Consider CXR; $\pm$ ABG; others as indicated from H&P. Consider preop pleural effusion drainage preop if severe respiratory compromise.
Cardiovascular	An ECHO and N-terminal pro-brain natriuretic peptide (NT-proBNP) may be helpful to evaluate cardiac function. Exercise tolerance should be evaluated in every patient and any preexisting cardiac disease detailed in the preop visit. Irreversible, dose-dependent cardiotoxicity may result from doxorubicin chemotherapy. Tests: Consider ECG, NT-proBNP, others as indicated from H&P.
Gastrointestinal	Patient should have adequate preop iv hydration if given a bowel prep overnight. Look for malnutrition ($\downarrow$ wt and albumin).

Neurologic	Not usually significant. Taxol and cisplatin may → peripheral neuropathy.
Hematologic	Bone marrow suppression common following chemotherapy. Carboplatin, commonly used for ovarian cancer, often induces thrombocytopenia. Tests: CBC
Laboratory	CBC, renal function, T&S, consider T&C, and consider hepatic function and PT/PTT. Patients may have had tumor marker studies, including CA 125, CEA, and serum human epididymis protein 4 (HE4).
Premedication	Anxiolysis: consider midazolam 1–2 mg iv. Analgesia: consider acetaminophen 650–1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a TAP or epidural block/analgesia for postop pain management. Consider preop acetaminophen 1000 mg, ± gabapentin 300 mg, ± celecoxib 200 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural analgesia. Typically, a balanced anesthetic with inhalational agents and/or propofol infusion (25–150 mcg/kg/min) and opiates. An epidural catheter may be placed for postop pain management and also may be used intraop to ↓ anesthetic requirements.

Induction	Standard induction, though consider full stomach precautions if ascites present.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Continue muscle relaxation

based on nerve stimulator response. Epidural bupivacaine/opiate combination if catheter is placed preop. Consider NG/OG tube. Patients with combined regional/GA may require increased fluids due to vasodilation and ↓ BP. Consider adding ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion) for pain management.

Emergence

The patient may be extubated at the conclusion of surgery, unless hemodynamically unstable and/or requiring continued vigorous fluid resuscitation. Reverse muscle relaxant with neostigmine (0.07 mg/kg with glycopyrrolate 0.01 mg/kg) or sugammadex (2–4 mg/kg). Consider postop ICU bed for unstable patients or those requiring invasive monitoring for fluid management. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements

Significant blood loss possible
IV: 16–18 ga × 2
BSS goal-directed
5% albumin

Blood loss may be >1 L. Order T&S and consider T&C preop. 5% albumin is useful for rapid volume replacement if Hct is acceptable. If large volumes of ascites are removed, significant ↓ BP may develop. Third-space losses may be significant (≥ 10 – 15 mL/kg/h). Consider use of BSS with fluid warmer for larger volume infusion. Strive to maintain goal-directed euvolemia based on clinical data: BP, PPV, HR, ABG, UO, EBL, and fluid shift estimates, ±CVP. Goal-Directed Fluid Management

		(see Appendix K, p. K-4).
Monitoring	Standard monitors Foley catheter ± Arterial catheter ± CVP catheter	Arterial and CVP catheters may be helpful for extensive surgery and/or patients with significant comorbidities.
Positioning	✓ and pad pressure points, especially important for longer surgeries ✓ eyes SCD	
Complications	Hypothermia Coagulopathy	Consider warming iv fluids. Heating blanket on bed and/or forced-air warming blanket(s) should be used. Preop malnutrition and/or significant blood loss may → coagulopathy

POSTOPERATIVE

Complications	Hemorrhage VTE Ascites/pleural effusion Respiratory insufficiency PONV	Appendix B, p. B-4 and p. B-6
Multimodal pain management	Multimodal approach ± PCA ± Epidural ± PNB/RA	Consider the following multimodal options: po/iv analgesics: consider acetaminophen 1000 mg po/iv q6h, gabapentin 200 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv

		for breakthrough pain. Consider iv ketamine or iv lidocaine for supplemental postop analgesia.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Allen RH, Micks E, Edelman A: Pain relief for obstetric and gynecologic ambulatory procedures. *Obstet Gynecol Clin North Am* 2013; 40(4):625–45.
2. Bauchat JR, Habib AS: Evidence-based anesthesia for major gynecologic surgery. *Anesthesiol Clin* 2015; 33(1):173–207.
3. Kittiwarawut A, Vorasettakarnkij Y, Tanasanvimon S, et al: Serum NT-proBNP in the early detection of doxorubicin-induced cardiac dysfunction. *Asia Pac J Clin Oncol* 2013; 9(2):155–61.
4. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
5. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
6. Nelson G, Altman AD, Nick A, et al: Guidelines for pre- and intra-operative care in gynecologic/oncology surgery: Enhanced recovery after surgery (ERAS®) Society recommendations—Part I. *Gynecol Oncol* 2016; 140(2):313–22.
7. Nelson G, Altman AD, Nick A, et al: Guidelines for postoperative care in gynecologic/oncology surgery: Enhanced recovery after surgery (ERAS®) Society recommendations—Part II. *Gynecol Oncol* 2016; 140(2):323–32.
8. Nelson G, Dowdy SC, Lasala J, et al: Enhanced recovery after surgery (ERAS®) in gynecologic oncology—Practical considerations for program development. *Gynecol Oncol* 2017; 147(3):617–20.
9. Oh TK, Lim MC, Lee Y, et al: Improved postoperative pain control for cytoreductive surgery in women with ovarian cancer using patient-controlled epidural analgesia. *Int J Gynecol Cancer* 2016; 26(3):588–93.

INTERVAL, SECONDARY CYTOREDUCTIVE, AND SECOND-LOOK ASSESSMENT LAPAROTOMY

SURGICAL CONSIDERATIONS

In patients where neoadjuvant chemotherapy was utilized or when optimal debulking could not be performed at initial surgery, an interval (after 3–6 cycles of chemotherapy) or secondary debulking laparotomy is done to achieve optimal cytoreduction. Similar to primary cytoreduction, the surgery involves methodical and meticulous exploration of all of the abdomen and pelvis, multiple cytologies and biopsies, lysis of adhesions, and resection of the residual tumor, as well as the pelvic and periaortic lymph nodes (if not done at time of first surgery). Patients may also be candidates for secondary or tertiary cytoreductive surgical procedures, especially if isolated recurrences are found in the setting of longer disease-free intervals.

Description

Historically, a “**second-look**” **laparotomy** was undertaken to determine whether a patient was surgically and pathologically free of disease, after an appropriate number of treatment cycles with platinum-based (CDDP or carboplatin) chemotherapy. However, 50% of patients with negative second-look surgeries ultimately recur, and the number of second-look surgeries has decreased significantly in favor of noninvasive evaluation including the use of tumor markers (CA 125) and imaging techniques (CT, MRI, PET). Select patients will benefit from second-look surgery to guide further therapy and to perform **secondary debulking** for select patient. Intra-abdominal assessment may be performed in conjunction with placement of an intraperitoneal port.

Usual preop diagnosis

Ovarian carcinoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, postop recovery plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine or dorsal lithotomy
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Incision	Midline or paramedian vertical abdominal
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; redose as indicated by duration and blood loss
Surgical time	2–3 h
Additional considerations	Placement of permanent central venous or intraperitoneal access; Preop consider VTE mechanical and pharmacologic prophylaxis including pneumatic compression stockings and LMW heparin. (e.g. LMW heparin)
EBL	350–700 mL
Postop care	PACU → ward
Mortality	1–2/1000
Morbidity	Postop fever: 14–19% Wound infection: <5% Wound dehiscence: 0.3–3% PE: 1–2% Incisional hernia: 0.5–1% Ureteral injury: 0.5–1%
Pain score	5–6

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients undergoing interval cytoreductive laparotomy have generally been treated with 3–6 cycles of chemotherapy. Depending on the type of adjunctive treatment given, the patient may come to surgery in poor physical condition from malnutrition or toxicity from chemotherapy (see [Table 8.1-1](#)). Peripheral iv access is often preserved due to the use of subcutaneous intravenous access ports (Mediport®, etc.).

Respiratory	Pulmonary function may be impaired by several chemotherapeutic drugs; most commonly bleomycin. Patients often have a Mediport-type catheter, Hickman line, or other central venous access already in place. These ports can be used for induction of anesthesia. Consider preop CXR to assess the
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presence of lung injury if prior chemo given. Consider preop PFTs for patients with dyspnea at rest or with mild exertion. Patients who received bleomycin should not receive $O_2 > 39\%$ intraop, but arterial O_2 saturation ideally should be kept $\geq 93\%$. The pulmonary toxicity of bleomycin is dose-related with a much higher incidence occurring if $>200 \text{ mg/m}^2$ received. Combination chemotherapy with vincristine or cisplatin also increases pulmonary toxicity.

Tests: Consider CXR; others as indicated from H&P

Cardiovascular

Cardiotoxicity is seen with several antineoplastic agents, especially daunorubicin and doxorubicin. The cardiomyopathy produced by these drugs occurs in two forms: (a) acute, ST-T-wave changes and dysrhythmias, which are transient and usually not a serious problem, and (b) chronic, dose-related toxicity manifested by CHF. Total doses of doxorubicin as low as 250 U can cause myocardial damage but is more common at doses $>400 \text{ U}$. Cardiac irradiation, or combination chemotherapy with cyclophosphamide increases the risk of cardiac toxicity. Patients who have received cardiotoxic drugs are usually followed by serial ECHOs, and the results should be reviewed preop. Patients with CHF or ECG changes should have a cardiology consultation preop to optimize their medical condition.

Tests: ECG, consider echo and NT-proBNP; others as indicated from H&P

Neurologic

Peripheral neuropathies are produced by vincristine, cyclophosphamide, Taxol (paclitaxel), 5-fluorouracil, and several other drugs. Vincristine can also $\rightarrow$ SIADH. Other CNS effects include N/V, seizure, and cerebellar dysfunction. A preop neurologic exam is useful for patients with evidence of neurotoxicity. Document presence of neurologic deficits preop for subsequent comparisons.

Endocrine	Corticosteroids (e.g., prednisone) are commonly used with chemotherapeutic agents, as treatment for pulmonary fibrosis and other complications of chemotherapy. The use of >10 mg prednisone (or equivalent) per day for >2 wk suppresses the endogenous secretion of the adrenal cortex, which may take up to 6 mo to recover fully. Hydrocortisone 100 mg iv, preop with an additional 2–3 subsequent doses q8h will provide adequate “stress-dose” coverage perioperatively. The dose is tapered rapidly over 2 or 3 d postop. Diabetes and obesity are common. Tests: FBS and HbA1c (within 3–6 mo) if diabetic; others as indicated from H&P
Renal	Many chemotherapeutic drugs have renal toxicity; therefore, renal function testing is indicated. Patients with impaired renal function should be given appropriate dosages of medications (e.g., antibiotics), which depend on renal excretion. Tests: Renal function
Musculoskeletal	Vincristine produces a neurotoxicity manifested by numbness and tingling in the extremities, weakness, foot drop, loss of reflexes, ataxia, and muscle pains. Muscle weakness in the arms and legs indicates that the drug should be stopped. Muscle weakness may also involve the larynx and extraocular eye muscles. Reduced amounts of NMBs should be used intraop, and a nerve stimulator used to follow twitches.
Gastrointestinal	Consider hydration overnight if given a bowel prep or if there is significant N/V. Look for ascites and malnutrition. Tests: Consider electrolytes, LFTs.
Hematologic	Bone marrow suppression is a common side effect of antineoplastic drugs. The toxicity usually produces a reversible drop in leukocytes, erythrocytes, and platelets, with a nadir 10–14 d posttreatment. Patients with a total neutrophil count of <1000 should be kept in isolation until counts improve. A

	low Plt count (<50,000) is an indication for Plt transfusion preop. Regional anesthesia in patients with thrombocytopenia needs to be considered carefully due to ↑ risk of bleeding complications.
Laboratory	Consider CBC, Cr, albumin, lytes, PT/PTT.
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300 mg po, and/or celecoxib 200 mg po Stress-dose hydrocortisone (100 mg iv) if indicated
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a TAP or epidural block/analgesia for postop pain management. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA or combined GETA/epidural.

Table 8.1-1 Toxicities of Selected Antineoplastic Chemotherapeutic Agents	
Agent	Toxic Effects
Vincristine, vinblastine	Neuropathies, SIADH, myelosuppression
Cyclophosphamide	Prolonged neuromuscular block
Mechlorethamine	Prolonged neuromuscular block
Bleomycin	Pulmonary fibrosis
Doxorubicin, daunorubicin	Cardiotoxicity, GI upset, myelosuppression
Methotrexate	Myelosuppression, GI upset, stomatitis, pulmonary infiltrates

Fluorouracil	Myelosuppression, hepatic and GI alterations, nervous system dysfunction
Mercaptopurine	Myelosuppression
Thioguanine	Myelosuppression
Actinomycin D	Myelosuppression, GI upset, stomatitis
Mitomycin	Myelosuppression, GI upset
Cisplatin	Myelosuppression, GI upset, electrolyte disturbances, nephrotoxicity, peripheral neuropathy
Carboplatin	Myelosuppression, peripheral neuropathy, GI upset, electrolyte disturbances
Paclitaxel	Myelosuppression, peripheral neuropathy, GI upset, arthralgia/myalgias, mucositis, interstitial pneumonitis
Docetaxel	Myelosuppression, peripheral neuropathy, malaise, maculopapular rash, GI upset

General anesthesia

Induction	Standard induction. Consider renal function and surgery duration when deciding on agents.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance: see Anesthetic Considerations for Staging Laparotomy. Consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion) for supplemental analgesia.
Emergence	Extubate when patient is responsive and neuromuscular block is fully reversed. In patients with borderline pulmonary function, extubation may be delayed until patient is in the PACU or ICU and after ABG is checked while the patient breathes spontaneously. Consider PONV prophylaxis. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if

initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include ondansetron.

Regional anesthesia

Epidural	2% lidocaine ± epinephrine 1:200,000 (10–20 mL) or 0.25% bupivacaine (10–20 mL) initially; then at ~3–5 mL/h. Opiates, such as morphine (2–4 mg) or hydromorphone (0.3–0.6 mg), may be given in the epidural for postop pain control.	
Blood and fluid requirements	IV: 16–18 ga × 1–2 BSS @ 7–10 mL/kg/h Keep UO >0.5 mL/kg/h 5% albumin FFP/Plt	Goal-Directed Fluid Management (see Appendix K, p. K-4) Consider PRBC for Hct < 25%. Goal-Directed Fluid Management (see Appendix K, p. K-4) 5% albumin may be used as volume replacement, although no proven advantages over crystalloid solutions. Consider FFP and Plt if evidence of significant coagulopathy (↑ PT, ↑ PTT, ↓ Plt).
Monitoring	Standard monitors Foley catheter ± Arterial line ± CVP catheter	Consider arterial and CVP catheters for patients with compromised cardiac or pulmonary function or patients having extensive surgical procedures.
Positioning	✓ and pad pressure points ✓ eyes	

	Antiembolism stockings and SCD	
Complications	Hypothermia	Warm fluids; keep heating pad on bed; use forced-air warmer.
	Bleeding	✓ PT; PTT, Plts periodically if large blood loss

POSTOPERATIVE

Complications	Bleeding Infection PONV Respiratory insufficiency VTE	
Multimodal pain management	Multimodal approach ± PCA ± epidural ± PNB/RA	Consider the following multimodal options: po/iv analgesics: consider acetaminophen 650–1000 mg po/iv q6h, gabapentin 100–600 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain. Consider iv ketamine or iv lidocaine for supplemental analgesia.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.

Tests	Consider CXR ABG	As indicated by postop clinical findings
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Suggested Reading

1. See suggested reading for Anesthetic Consideration for Staging Laparotomy for Ovarian, Fallopian Tube, and Primary Peritoneal Cancer above ([p. 848](#)).

RADICAL VULVECTOMY

SURGICAL CONSIDERATIONS

Description

Historically, invasive vulvar carcinoma has been treated with en bloc dissection of the inguinal-femoral region and the vulva. The surgery involves bilateral excision of lymphatic and areolar tissue in the inguinal and femoral regions, combined with removal of the entire vulva between the labia-crural folds, from the perineal body to the upper margin of mons pubis ([Fig. 8.1-2](#)). A large surgical wound is created and, if 1° closure without tension is not possible, a skin or myocutaneous graft may be necessary. Deep pelvic nodes are rarely involved with metastases when the superficial and deep groin nodes are free of disease; therefore, a **pelvic lymphadenectomy** is no longer routinely performed. If presence of tumor is documented in the groin nodes, particularly in Cloquet sentinel nodes (the most cephalad, deep inguinal nodes), a **deep pelvic lymphadenectomy** may be performed. **Postop radiation therapy**, however, is widely used instead of a pelvic lymph node dissection to minimize operative morbidity and confer a survival advantage.

A skin incision in the shape of a bull's head ([Fig. 8.1-2](#)) allows access to the inguinal-femoral region. (The incision ideally should extend 2+ cm beyond the tumor margin.) The inguinal ligament and rectus fascia should be cleared bilaterally of all nodal tissues and the fossae ovalis on both sides identified. The lateral aspect of the femoral sheath is incised along the sartorius muscle, with care being taken not to injure the femoral nerve or vessels, and the cribriform fascia is cleaned off the femoral artery. The external pudendal artery, which marks the entrance of the saphenous vein into the fossa ovalis, should be identified and ligated. The proximal and distal segments of the saphenous vein should be ligated and excised as the fibrofatty, lymph-bearing tissue of the femoral sheath is resected. Cloquet nodes at the femoral ring beneath the inguinal ligaments on both sides should be resected and submitted for frozen section pathology evaluation. The deep inguinal lymphatic chain is removed on both sides by opening the inguinal

canal from the external inguinal ring. The vulvar incision is carried down through the labia-crural folds. The internal pudendal vessels at the posterior lateral margin of the vulvar incision are identified as they emerge from Alcock canal, and then they are ligated and incised.

Use of electrocautery in this portion of the procedure usually tends to decrease operative blood loss. The dissection is continued along the periosteum of the symphysis at the level of the fascia of the deep muscles of the urogenital diaphragm. The bulbocavernosus, ischiocavernosus, and superficial transverse perinei muscles are removed. A circumferential vaginal incision, excluding the urethral meatus, is then performed and the vulva is removed. The incisions overlying the groin node dissections should be closed with minimal tension after placement of closed-suction Jackson-Pratt drains. The vulvar surgical wound is closed by slightly undermining the skin of the edges of the incision and suturing them to the vaginal mucosa. A **vulvar reconstruction**, using myocutaneous flaps, also can be performed at this time (see **Pelvic Exenteration**).

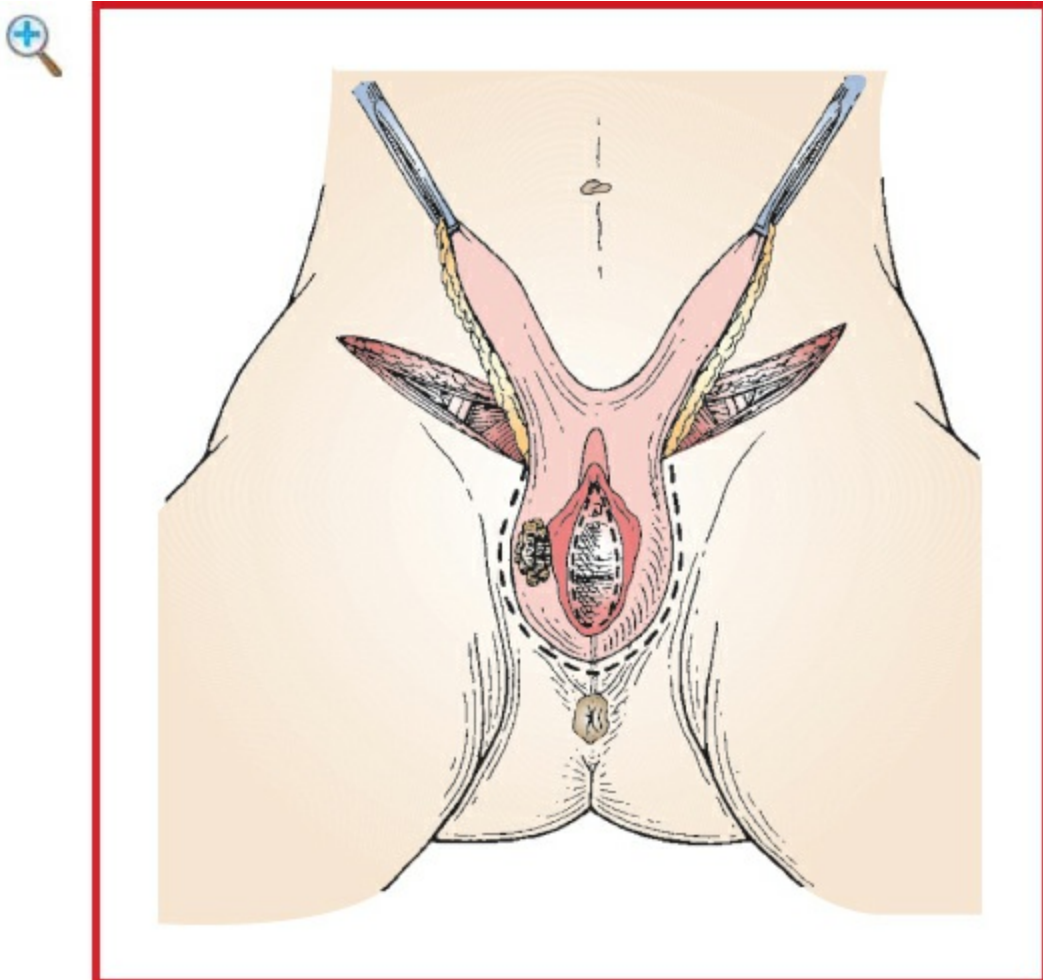


Figure 8.1-2 En bloc radical vulvectomy incisions shown; bilateral inguinal lymphadenectomy is

Variant procedure or approaches

In 1962, Byran and associates popularized a **three-incision technique** first described by Kehrer in 1918. This three-incision technique, with separate vulva and groin incisions, is the most common approach (Fig. 8.1-3). This operative approach has led to a significant decrease in wound infection and breakdown, apparently without increasing tumor recurrence in the inguinal dermal bridge above the symphysis pubis. Another variant is the **hemivulvectomy** (Fig. 8.1-4), in which unilateral radical hemivulvectomy and groin node dissection are performed in selected stage I, non midline, unifocal vulvar cancer patients. This procedure will minimize morbidity, disfigurement, and sexual dysfunction. The observation that almost no contralateral groin metastases occur in the absence of positive ipsilateral groin nodes allows the surgeon to perform only a **unilateral groin node dissection**. Recent trials have shown sentinel lymph node mapping may be an alternative to groin lymphadenectomy in select patients with early-stage disease. For this approach, a radioactive tracer is injected intradermally 20–30 min before groin incision. Blue dye is also injected to improve visualization. This is done after the patient is prepped due to the rapid movement of the dye to lymphoid tissue. Using both radioactivity and direct visualization allows identification of the sentinel lymph node. If a sentinel lymph node(s) is not found, full lymphadenectomy is performed. Benefits of this approach include less dissection of tissue and lower rates of postoperative complications.

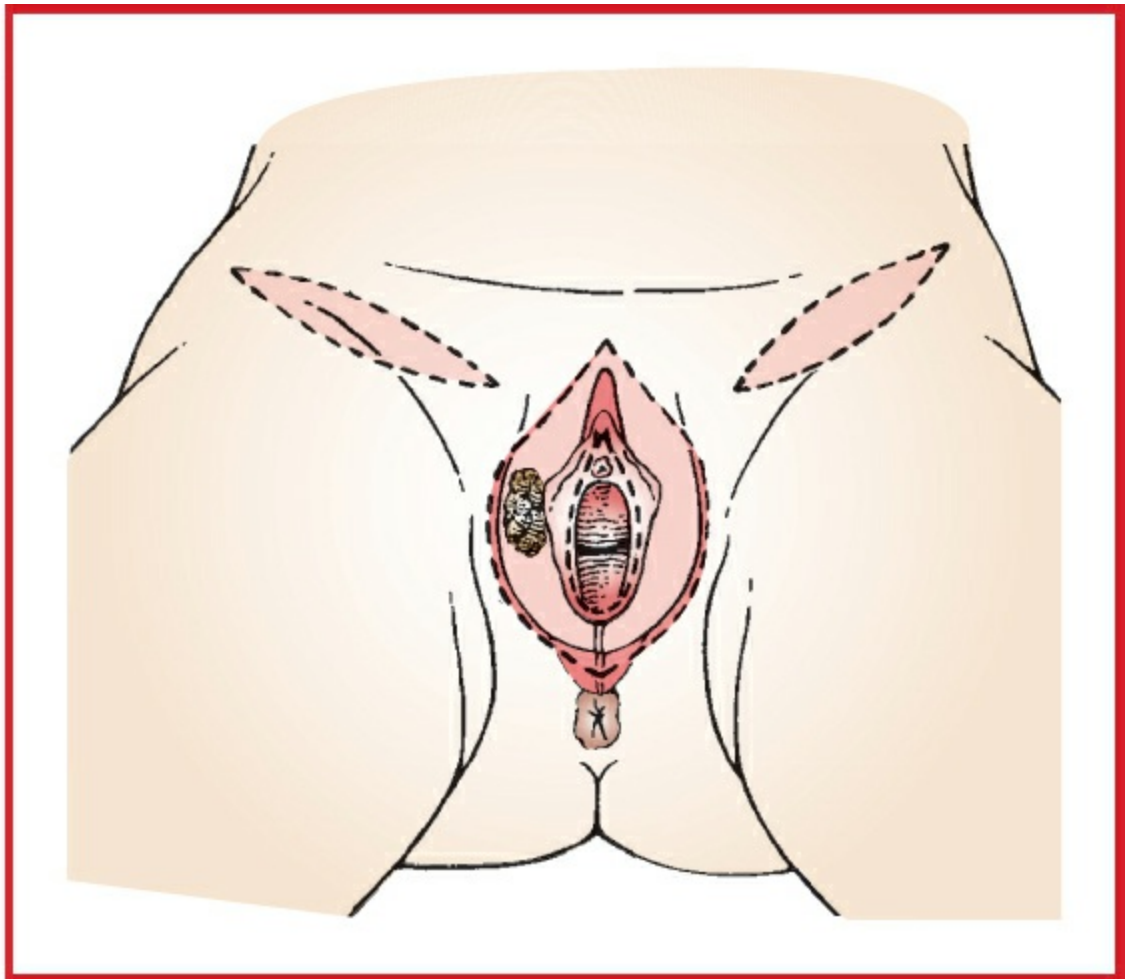


Figure 8.1-3 Three-incision radical vulvectomy and bilateral inguofemoral lymphadenectomy. (Reproduced with permission from Rock JA, Thompson JDTeLinde's Operative Gynecology, 8th edition. Lippincott-Raven, Philadelphia: 1997.)

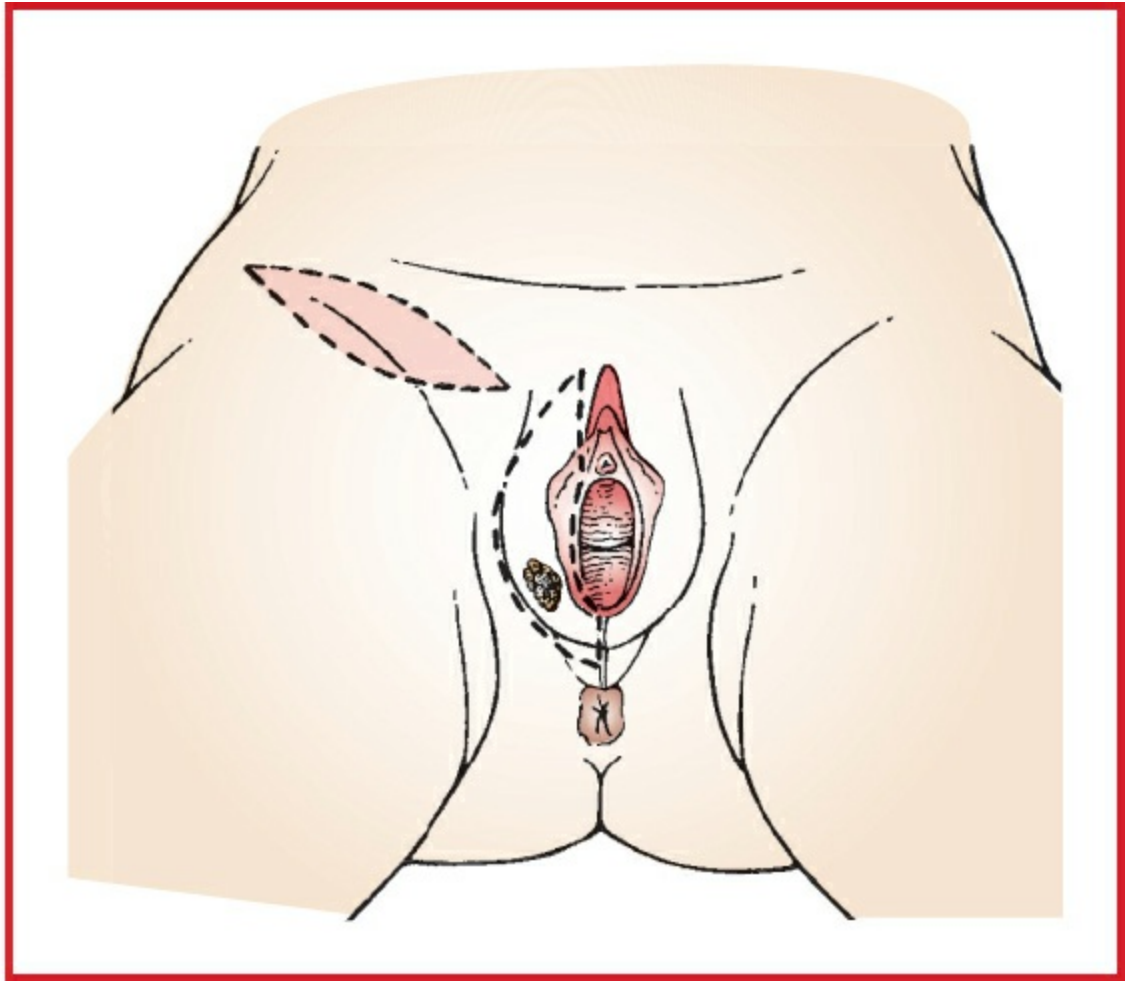


Figure 8.1-4 Unilateral lymphadenectomy for a well-lateralized lesion. (Reproduced with permission from Rock JA, Thompson JD: TeLinde's Operative Gynecology, 8th edition. Lippincott-Raven, Philadelphia: 1997.)

Usual preop diagnosis

Invasive vulvar cancer

ERAS: Preop counseling by the surgeon/anesthesiologist should include: no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, postop recovery plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	En Bloc Dissection	Three-Incision	Hemivulvectomy
Position	Modified	⇐	⇐

	dorsolithotomy in Allen universal stirrups		
Incision	Bull's head, from iliac crest to iliac crest and along labia-crural folds (Fig. 8.1-2)	2 separate groin incisions from iliac crest to pubic tubercle; 1 vulvar incision (Fig. 8.1-3)	1 or 2 separate groin incisions from iliac crest to pubic tubercle; vulvar incision (Fig. 8.1-4)
Special instrumentation	Argon beam coagulator	⇐	⇐
Unique considerations	Two-team approach to minimize surgical time. Consider constipating medications (e.g., Lomotil) to ↓ postop bowel movements. Consider preoperative prophylactic heparin	⇐	⇐
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; redose as indicated by duration and blood loss	⇐	⇐
Surgical time	3–4 h	⇐	2–3 h
Closing considerations	Possible skin graft; vulvar and groin suction drains	⇐	⇐
EBL	500–1000 mL	⇐	250–500 mL

Postop care	PACU or ICU, if necessary; aggressive local wound care. VTE prophylaxis (see Appendix B, p. B-7). NB: trauma to femoral vessels at time of groin lymph node dissection increases risk of VTE and PE	⇐	⇐
Mortality	1–2%	⇐	⇐
Morbidity	Wound infection and breakdown: 40–80%	15%	<15%
	Introital stenosis and dyspareunia: 50%	⇐	⇐
	Lymphedema of lower extremities: 25–30%	⇐	<25% (if deep groin nodes not dissected)
	Lymphocysts: 10%	⇐	⇐
	Genital prolapse: 7%	⇐	1–2%
	Stress incontinence: 5%	⇐	1–2%
	Thrombophlebitis: 3–5%	⇐	1–2%
	Hernia: 1–2%	⇐	⇐
	PE: 1–2%	⇐	⇐
Pain score	5–6	5–6	4–5

■ PATIENT POPULATION CHARACTERISTICS

Age range Median = 70 yr

Incidence	2.2/100,000; 4% of female genital malignancies
Etiology	Two pathways of vulvar cancer are thought to exist, one related to HPV infection and the second related to chronic inflammation or an autoimmune process.
Risk factors	HPV infection, vulvar dystrophies, tobacco use, immunosuppression, HIV
Associated conditions	Bowen disease, condyloma acuminata, diabetes, obesity, HTN, arteriosclerosis, cervical malignancy

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Patients with vulvar carcinoma are often elderly. Consider and evaluate coexisting medical conditions thoroughly. Radical vulvectomy is performed for invasive tumor that has not metastasized to distant sites.

Respiratory	The presence of lung disease and smoking Hx should be discussed with the patient preop. Consider CXR and PFTs or reviewing preexisting studies for patients with significant respiratory disease and symptoms. Tests: Others as indicated from H&P
Cardiovascular	There is an increased incidence of HTN and atherosclerosis in these patients. A cardiology consultation is indicated for angina, recent MI, CHF, or heart murmurs. Tests: ECG; consider NT-proBNP; others as indicated from H&P
Renal	With age, creatinine clearance decreases 2° ↓ renal mass, but serum creatinine remains unchanged because of decreased muscle mass. Tests: Serum creatinine; others as indicated from H&P
Gastrointestinal	Patients should have iv hydration preop if given bowel prep overnight.

Neurologic	Document a neurologic exam if Hx of stroke, seizure, or other neurologic diseases. Hx of peripheral neuropathy or autonomic dysfunction should be assessed in diabetic patients.
Endocrine	Tests: As indicated from H&P Diabetes and obesity are common. Tests: FBS and HbA1c (within 3–6 mo) if diabetic; others as indicated from H&P
Hematologic	Chronic anemia may be present. Tests: Hgb/Hct; Plt count
Laboratory	Cr, if age > 65
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a TAP or epidural block/analgesia for postop pain management. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA or regional anesthesia, alone or in combination.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion) for supplemental analgesia.
Emergence	No special considerations. Discuss with surgeon local

wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

Epidural	2% lidocaine ± epinephrine 1:200,000 (10–20 mL) or 0.25% bupivacaine (10–20 mL) are used; then at ~3–5 mL/h. Opiates such as morphine (2–4 mg) in the epidural for postop pain control	
CSE	Hyperbaric bupivacaine (10–12 mg), ± preservative-free morphine (0.1–0.2 mg) → T10 sensory level	
Blood and fluid requirements	IV: 16–18 ga × 2 BSS @ 6–8 mL/kg/h Warm iv fluids UO > 0.5 mL/kg/h	Goal-Directed Fluid Management (see Appendix K, p. K-4). Occasionally, femoral vessels may be injured, requiring rapid blood replacement. Consider PRBCs for Hct < 21% in healthy patients and <25% in patients with cardiac or pulmonary disease. Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors ± Arterial line ± CVP line Foley catheter	Invasive monitors indicated for patients in poor condition or with cardiovascular or respiratory disease. An arterial catheter is useful for drawing labs in surgery to check Hct, coags, glucose, or ABGs.
Positioning	✓ and pad pressure points ✓ eyes SCD	

POSTOPERATIVE

Complications	Hypothermia Bleeding PONV VTE	
Multimodal pain management	Multimodal approach ± PCA ± epidural ± PNB/RA	Consider the following multimodal options: po/iv analgesics: consider acetaminophen 650–1000 mg po/iv q6h, gabapentin 100–600 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain. Consider iv ketamine or iv lidocaine for supplemental analgesia.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.
Tests	Tests as indicated from postop clinical findings	

Suggested Readings

1. Hoffman MS: Malignancies of the vulva. In: Rock JA, Jones HW, eds. *TeLinde's Operative Gynecology*. Lippincott Williams & Wilkins, Philadelphia: 2015, 1293–350.
2. Huang TJ: Invasive cancer of the vulva. In: DiSaia PJ, Creasman WT, eds. *Clinical Gynecologic Oncology*, 9th edition. C.V. Mosby, St. Louis: 2018, 217–30.
3. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
4. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
5. Moroney JW, Kunos C, Wikinson EJ et al: Vulva. In: Barakat RR, Perelman RC, Markman M, et al, eds. *Principles and Practice of Gynecologic Oncology*, 6th edition. Lippincott Williams, & Wilkins, Philadelphia: 2013, 757–848.

CONIZATION OF THE CERVIX

SURGICAL CONSIDERATIONS

Description

Conization of the cervix can be used for both diagnostic and therapeutic purposes. It is performed in cases of biopsy-proven dysplasia with unsatisfactory colposcopy (inadequate visualization of the endocervical canal) or following endocervical curettage showing dysplasia or atypical glandular epithelial cells ([Fig. 8.1-5](#)). Persistent abnormal cytology associated with normal colposcopy, colposcopic suspicion of invasion, and/or cervical biopsy showing microinvasive cancer is also an indication for this procedure. The surgery consists of the annular removal of a cone-shaped wedge of tissue from the cervix with a scalpel. With the advent of the loop electrosurgical excision procedure (LEEP), most cone biopsies are done under local paracervical/intracervical block in an office setting and do not require the services of an anesthesiologist.

Variant procedure or approaches

In selected patients, a **laser** is used in place of the scalpel. This procedure can be performed under local anesthesia with less blood loss, but operative time is usually longer. The thermal effect of the laser at the cone margins, although usually minimal, may interfere with pathologic interpretation. In pregnant patients, a **shallower cone** is done to minimize complications. Approximately 1% of women with cervical carcinoma are pregnant at the time of diagnosis, and 1/1240 pregnancies is complicated by cervical cancer. Recognition and therapy of preinvasive cervical lesions during pregnancy, therefore, are of paramount importance. Because of the increased vascularity of the pregnant uterus and cervix, conization is usually associated with increased blood loss and morbidity.

Usual preop diagnosis

Cervical dysplasia

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, and postop recovery plan. See [Appendix K, p. K-1](#).

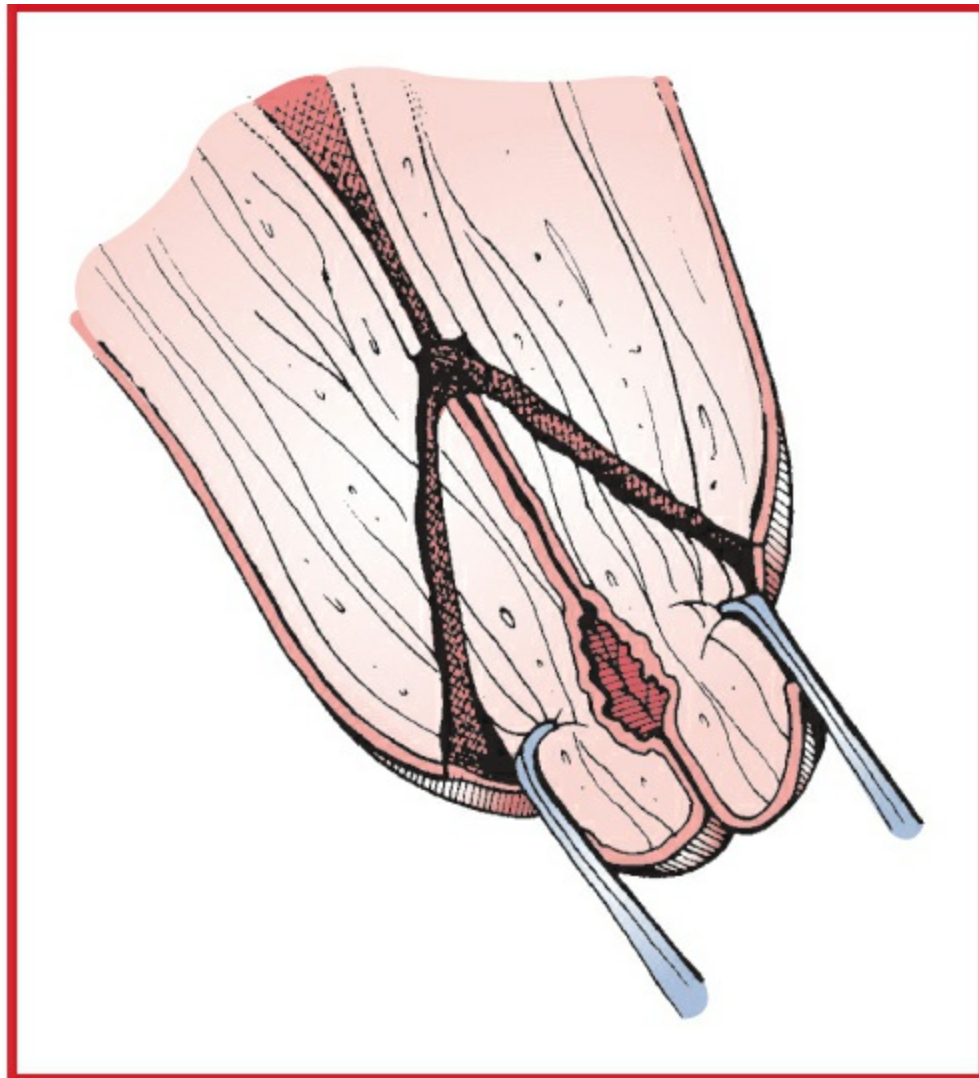


Figure 8.1-5 Cut across endocervix to complete cone excision.

■ SUMMARY OF PROCEDURES

	Conization	Laser Conization	Shallow Cone In Pregnancy
Position	Lithotomy	⇐	Lithotomy; with left lateral tilt in 3rd trimester
Incision	Cervical	⇐	⇐
Special instrumentation	Colposcope	CO ₂ laser; protective eye wear; colposcope	Colposcope

Unique considerations	Infiltration of the cervix with dilute vasopressin or phenylephrine solution. A 1:200,000 epinephrine solution also can be used. Vaginal pack necessary in selected patients	⇐	Phenylephrine, vasopressin, or epinephrine should not be used during pregnancy. Liberal use of hemostatic sutures should be made.
Antibiotics	None	⇐	⇐
Surgical time	30–60 min	30–90 min	⇐
EBL	50–200 mL	50 mL	100–350 mL
Postop care	Monitor for postop bleeding	⇐	⇐
Mortality	<0.01%	⇐	⇐
Morbidity	Hemorrhage: 5–10%	⇐	10–15%
	Preterm delivery: 16%	⇐	⇐
	Cervical stenosis: 2–3%	⇐	⇐
	Dysmenorrhea: rare	⇐	⇐
	Infertility: rare	⇐	⇐
	Injury to rectum and bladder: rare	⇐	⇐
	Pelvic cellulitis: rare	⇐	⇐
	Uterine perforation: rare	⇐	⇐
			Fetal loss: 10–15% (up to 30% in 1st trimester)*

			Premature labor: 5–10% (controversial) Rupture of membrane: 2–5%
Pain score	3	3	3
*The naturally higher incidence of spontaneous miscarriages in the 1st trimester contributes to this figure.			

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive and postreproductive years
Incidence	~5% of Pap smears (+ for dysplasia); 10–17% of patients who undergo colposcopic exams
Etiology	Majority of cervical dysplasia is caused by persistent HPV infections
Risk factors	Smoking; human papillomavirus (HPV); herpes simplex virus (HSV); multiple sexual partners; early age of onset of coitus; multiparity; lower socioeconomic status; HIV; immunocompromised hosts

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Conization is done for diagnosis and treatment of cervical lesions. Occasionally, it is necessary to perform the procedure during pregnancy, which increases the risk of bleeding complications. The effect of anesthetic agents on the fetus (especially 1st trimester) also needs to be considered when choosing anesthetic technique.

Respiratory	Not significant unless underlying lung disease
Cardiovascular	These patients are generally young and, therefore, less likely to have significant heart disease.
Neurologic	Usually not significant unless there is Hx of seizure disorder or other neurologic illness
Hematologic	Consider hemogram.
Laboratory	Consider pregnancy test.

Premedication	Anxiolysis: consider midazolam 1–2 mg iv (though generally held if pregnant). Analgesia: consider acetaminophen 650–1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg, po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Usually a local or MAC anesthetic; occasionally, GETA or spinal. Pregnancy makes it desirable to perform the procedure under local or regional anesthesia if appropriate (1% of cases).

General anesthesia

Induction	Standard induction. If pregnant patient >20 wk of gestation, consider rapid-sequence induction with cricoid pressure.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	No special issues. PONV prophylaxis may include dexamethasone and ondansetron.

Regional anesthesia

Both spinal and epidural techniques are acceptable and may be preferred for pregnant patients who cannot tolerate local anesthesia. Prehydration with 1000 mL BSS before block is recommended. Treat ↓ BP with ephedrine 5–10 mg iv, or phenylephrine 50–100 mcg, titrated to effect.

Blood and fluid	Minimal blood loss	Zero Balance Fluid
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requirements	IV: 18–20 ga × 1 BSS @ 2–4 mL/kg/h	Management (see Appendix K, p. K-2)
Monitoring	Standard monitors Fetal monitoring may be indicated for viable pregnancies.	
Positioning	✓ and pad pressure points ✓ eyes Left uterine displacement Antiembotic stockings + SCD	NB peroneal nerve compression at lateral fibular head → foot drop Left uterine displacement with a wedge under mattress should be used for pregnant patients (after ~20 wk) or earlier gestation if cardiovascular compromise when supine.
Complications	Laser eye damage Fire Premature labor (if pregnant)	If a laser is used, eye protection is required for the patient and all OR personnel; be alert for fire hazards when using a laser.

▼ POSTOPERATIVE

Complications	Peroneal nerve injury (2° lithotomy position) PONV Premature labor Bleeding	Nerve injury manifested as foot drop and loss of sensation over dorsum of foot
Multimodal pain management	po/iv analgesics	Consider the following multimodal options: po/iv analgesics: for example, acetaminophen 1000 mg po/iv q6h, ± celecoxib 200 mg po q12h.
ERAS	PONV protocol (see	Encourage early (within 24 h)

Suggested Readings

1. Copeland LJ, Landon MB: Malignant disease in pregnancy. In: Gabbe SG, Niebyl JI, Simpson JL, eds. *Obstetrics: Normal and Problem Pregnancies*, 7th edition. Churchill Livingstone, New York: 2016, 1255–81.
2. Meyer LA, Lasala J, Iniasta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
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4. Simeoens C, Goffin F, Simon P, et al: Adverse obstetrical outcomes after treatment of precancerous cervical lesions: a Belgian multicentre study. *BJOG* 2012; 119:1247–55.
5. Tewari KS: Cancer in pregnancy. In: DiSaia PJ, Creasman WT, eds. *Clinical Gynecologic Oncology*, 9th edition. C.V. Mosby, St. Louis: 2018, 353–419.

ANESTHETIC CONSIDERATIONS FOR LASER THERAPY TO THE VULVA, VAGINA, AND CERVIX

PREOPERATIVE

Laser therapy is indicated for preinvasive lesions of the vulva, vagina, or cervix. It destroys tissues by the selective application of light energy focused into a beam. Vaporized tissues tend to heal without scarring, and blood loss is minimal due to the cauterizing effect of the laser. Many gynecologic laser procedures are done with local anesthesia in the clinic setting and do not require the services of an anesthesiologist.

Respiratory	Not significant unless there is underlying lung disease
Cardiovascular	In elderly patients, exercise tolerance should be assessed. Tests: ECG if >60 yr
Hematologic	Tests: Hct
Laboratory	Consider pregnancy test in young women.

Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 650–1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of an epidural block/analgesia for postop pain management. Consider preop acetaminophen 1000 mg, ± celecoxib 200 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Usually MAC; GETA/LMA or regional technique may be used. Sedation with propofol, midazolam, and fentanyl in small doses usually is effective.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Muscle relaxation not necessary. A technique with relatively rapid emergence (e.g., propofol and/or sevoflurane/desflurane/N ₂ O combinations) is useful for outpatient surgery.
Emergence	Limit opiates. Discuss with surgeon local wound infiltration or nerve block placement. PONV prophylaxis should include ondansetron.

Regional anesthesia

Spinal or epidural anesthesia or CSE may be used with a sensory level to T10. Provide supplemental O₂ if iv sedation given.

Spinal	A T10 sensory level is desirable; hyperbaric bupivacaine 10–12 mg can be used. Small-diameter spinal needles (e.g., 25-ga Sprotte needles) minimize chance of postdural puncture headache (PDPH).	
Epidural	2% lidocaine, ± epinephrine 1:200,000 (10–15 mL), or 0.25% bupivacaine (10–15 mL) is used; redose as needed with 3–5 mL. Opiates, such as morphine (2–4 mg) or hydromorphone (0.3–0.6 mg), may be given in the epidural for postop pain control.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 2–4 mL/kg/h	Consider prehydrating patient (e.g., 500 mL BSS) if using neuraxial block, especially if NPO period was prolonged. Zero Balance Fluid Management (see Appendix K, p. K-2)
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes Antiembotic stockings + SCDs	
Complications	Eye injury OR fires Aerosolization of viral particles	Goggles should be worn by the patient and all OR personnel during laser use to prevent injury to eyes. Whenever laser is in use, be prepared for fires: Keep $\text{FiO}_2 \leq 0.3$, know where fire extinguisher is located, and watch for improper handling of lasers.

POSTOPERATIVE

Complications	PONV	PDPH may require epidural
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	PDPH	blood patch for treatment.
Multimodal pain management	Multimodal approach ± epidural	
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition and mobilization.

Suggested Readings

1. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
2. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.

SUCTION CURETTAGE FOR GESTATIONAL TROPHOBLASTIC DISEASE

SURGICAL CONSIDERATIONS

Description

Suction curettage is the most efficient method of evacuating a gestational trophoblastic neoplasm (mole). The procedure involves dilation of the cervix by instruments or by laminaria tents, followed by insertion of suction cannula of appropriate diameter into the uterine cavity. Standard negative pressures used are in the range of 30–70 mm Hg. Intravenous oxytocin—to maintain uterine contraction and minimize blood loss—is started after a moderate amount of tissue has been removed. Suction curettage is followed by gentle, sharp curettage of the uterus to ensure adequate evacuation. Curettage may be performed under ultrasound guidance to decrease the risk of uterine perforation and to ensure adequate evacuation. Paracervical injection of dilute vasopressin solution or 1% lidocaine with 1:200,000 epinephrine may decrease operative blood loss (in cases not complicated by thyrotoxicosis or HTN) ([Fig. 8.1-6](#)).

Variant procedure or approaches

Evacuation of a mole >16 wk gestation size is associated with a significant risk of trophoblastic embolization and cardiorespiratory compromise (2° pulmonary

HTN/edema, cyanosis, ↓ CO, ↓ BP, right heart failure). Consider hemodynamic monitoring with TEE or a CVP catheter to manage cardiovascular changes associated with trophoblastic embolization and to prevent inadvertent fluid overload. In general, earlier diagnosis of molar pregnancies has decreased the incidence of complications such as HTN and thyrotoxicosis.

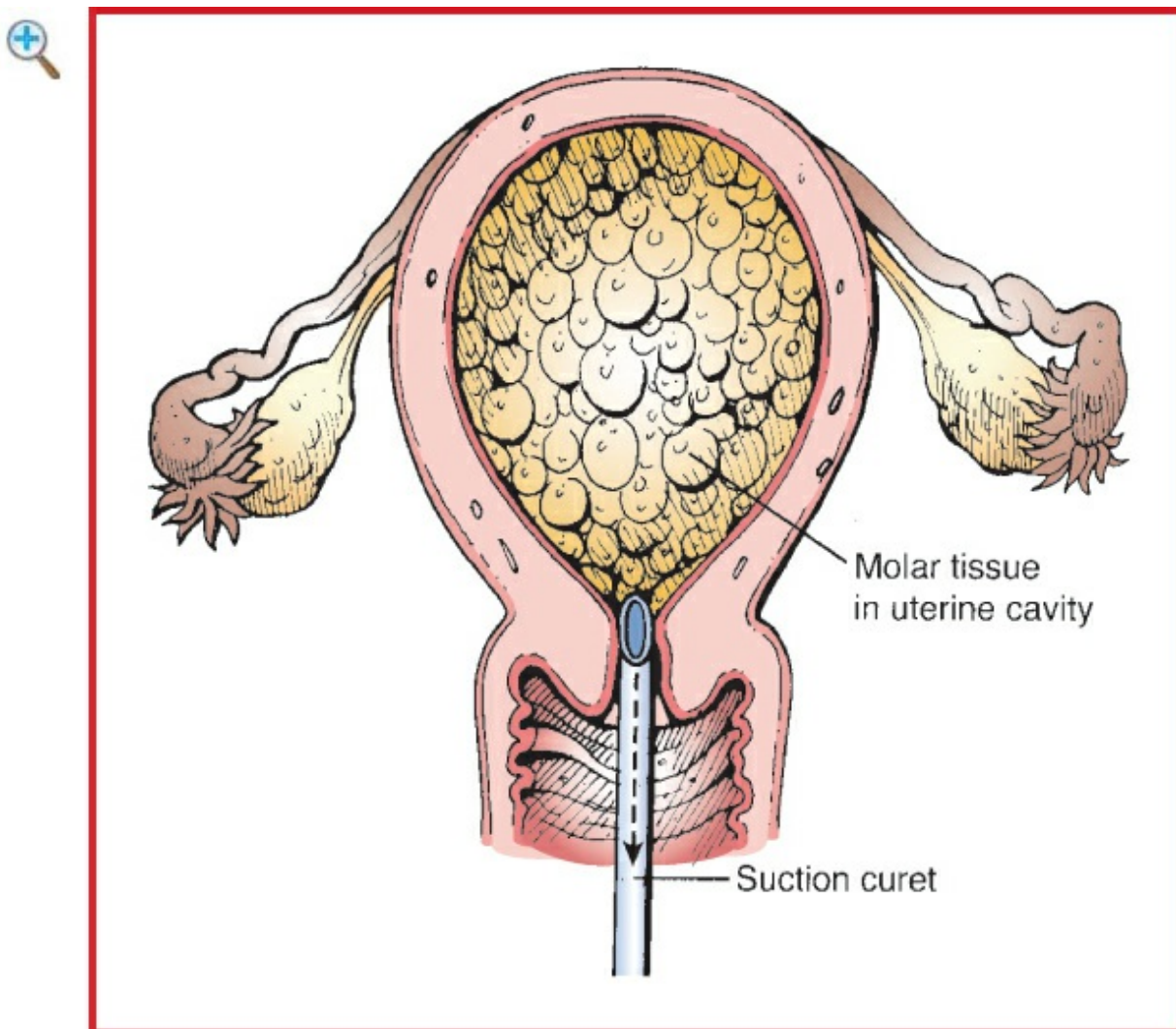


Figure 8.1-6 Suction curettage of a molar pregnancy.

Usual preop diagnosis

Gestational trophoblastic disease (GTD)

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, and postop recovery plan. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Small Mole <16 Wk Size	Large Mole >16 Wk Size
Position	Lithotomy	⇐
Incision	None	⇐
Special instrumentation	Suction evacuation kit	⇐
Unique considerations	If mole >12 wk size, laparotomy setup should be readily available. Oxytocin drip, anesthesia, or surgery may precipitate thyroid storm. If hyperthyroidism is suspected before induction of anesthesia, beta-blockers should be used. In PACU, monitor for signs of trophoblastic embolization or amniotic fluid embolus. If patient is Rh negative, give RhoGAM.	⇐ ± central hemodynamic monitoring with TEE or CVP catheter; avoid overzealous use of fluids.
Antibiotics	Cefotetan 2 g iv or doxycycline 200 mg po before procedure and 100 mg po after procedure	⇐
Surgical time	30–60 min	⇐
EBL	200–400 mL	⇐
Postop care	Outpatient (usually)	Consider inpatient observation or ICU admission.
Mortality	Rare	⇐
Morbidity	Trophoblastic embolization: rare	11–27% 10%

	Excessive bleeding: 2%	⇐
	Infection: <2%	1–2%
	Uterine perforation: <1%	Acute pulmonary edema: 2–
Pain score	3	11% 3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age group
Incidence	1/1000 deliveries in the United States
Etiology	Genetics, androgenous (all chromosomes in true moles are paternal in origin); nutritional deficiency, protein, folic acid, carotene (vitamin A)
Risk factors	Extremes of reproductive age, prior GTD (incidence 0.6–2.0%), Asian, Hispanic populations
Associated conditions	Hyperemesis gravidarum (2° ↑ levels of HCG); preeclampsia in 1st trimester; thyrotoxicosis (because of its analogy to TSH molecule, ↑ levels of HCG can bind TSH receptors and cause thyrotoxicosis)

■ ANESTHETIC CONSIDERATIONS

■ PREOPERATIVE

Some 80% of cases are diagnosed at 12–18 wk of development. Most patients have an unusually large uterus for the length of the pregnancy, and vaginal bleeding is common. Trophoblastic disease is classified by the degree of invasiveness; **retained mole** is the most common type and the least invasive; **invasive mole** involves the wall of the uterus, and **metastatic mole** involves more distant sites. Chemotherapy with methotrexate and actinomycin D usually is given postop for invasive or metastatic disease. (See [Table 8.1-1](#) for toxicities of chemotherapeutic agents.)

Respiratory	Pulmonary edema may complicate preeclampsia, which occurs in 25% of patients with GTD. If respiratory distress is present, it also may be 2° embolization of tumor to lungs. Avoid overhydration → pulmonary complications in patients with large moles.
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	Tests: Consider preop ABG if pulmonary function impaired and in patients at high risk of developing trophoblastic embolization. Others as indicated by H&P
Cardiovascular	Dehydration and/or ↓ blood volume may exist due to hyperemesis and vaginal bleeding. Adequate hydration should be given preop if patient shows Sx of hypovolemia (e.g., tachycardia, orthostatic ↓ BP, low UO). Preeclampsia also may complicate this disease and can be diagnosed by HTN, proteinuria, ± edema. If the patient has severe preeclampsia, invasive monitoring of BP may be advisable. Also, if patient is receiving Mg⁺⁺ therapy, a serum level should be ✓'d preop. Mg⁺⁺ therapy may inhibit myocardial contractility in high doses. Ca⁺⁺ is the preferred antidote for myocardial depression. MgSO₄ → uterine atony → ↑ blood loss.
Neurologic	Seizure prophylaxis with Mg⁺⁺ is indicated for women with preeclampsia. Magnesium 4–6 mg loading dose iv over 20 min and then 1–2 g/h maintenance infusion. A 2nd bolus of 2 g over 5 min can be administered if the initial bolus dose is complete. If seizure occurs, a small dose of propofol (20–40) mg or midazolam (1–2 mg) should be given iv and respiration assisted with supplemental O₂ by mask. The trachea should be intubated for airway protection in patients with full stomachs and in those who are difficult to ventilate by mask.
Musculoskeletal	✓ reflexes if patient has received Mg⁺⁺. Reduce amount of muscle relaxant to compensate for the effects of Mg⁺⁺ on muscle strength.
Hematologic	Anemia may be masked by hypovolemia. Rh– patients with Rh+ partners should receive 300 mcg of Rh immune globulin (RhoGAM) within 72 h postop to ↓ possibility of Rh isoimmunization in future pregnancies. Tests: CBC; ✓ Plt in preeclamptics. If patient has

	received chemotherapy recently, ✓ HCT; complete blood count (WBC, Plt).
Endocrine	Hyperthyroidism occurs in 5% of women with hydatidiform moles and is 2° the thyroid-stimulating effects of HCG. Tests: Thyroid function tests should be ✓'d preop in women with Sx of hyperthyroidism and managed before surgery. FBS and HbA1c (within 3–6 mo) if diabetic; others as indicated from H&P
Laboratory	Serum HCG level; consider thyroid function tests; LFTs; PT; PTT; Plt count; Mg ⁺⁺ level; UA—as indicated from H&P, T&S or T&C
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 650–1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Usually GETA, although may be carried out under spinal or epidural anesthesia.

General anesthesia

Induction	A rapid-sequence induction with cricoid pressure should be used. For RSI, consider supplementation with labetalol, hydralazine, esmolol, or remifentanyl to obtund exaggerated laryngoscopic hypertensive response in preeclampsia.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and

	normothermia.
Maintenance	Standard maintenance. Control BP, if preeclamptic, with labetalol, hydralazine, or SNP. Try to keep DBP at 90–100 mm Hg. NDMR potentiated by magnesium
Emergence	Extubate when fully awake and protective airway reflexes have returned. Consider PONV prophylaxis. Discuss with surgeon local wound infiltration for analgesia. PONV prophylaxis should include ondansetron.

Regional anesthesia

Spinal	A T10 sensory level is desirable; hyperbaric bupivacaine (10–12 mg) may be used.	
Epidural	Use 2% lidocaine ± epinephrine 1:200,000 (10–20 mL) or 0.25% bupivacaine (10–15 mL).	
Blood and fluid requirements	Possible large blood loss IV: 16–18 ga × 1 BSS @ 2–4 mL/kg/h	Goal-Directed Fluid Management. One large-volume iv line should be placed and blood readily available. The usual causes of bleeding are uterine perforation, cervical laceration, or uterine atony. Cautious use of fluids if on magnesium infusion due to risk of pulmonary edema. Zero Balance Fluid Management (see Appendix K, p. K-2)
Control of bleeding	Oxytocin (30–40 U/L) infusion ↑ % volatile anesthetic may ↓ uterine tone Methylergonovine maleate 0.2 mg im	Oxytocin is begun about halfway through procedure at 250 mL/h (consult obstetrician). Large oxytocin boluses may → ↓ BP. Try to keep anesthetic < 1 MAC to prevent uterine relaxation. Methylergonovine may be given for severe bleeding. Because this drug can cause HTN, it is contraindicated in any hypertensive etiology (e.g.,

		preeclampsia).
Monitoring	Standard monitors ± Arterial catheter ± CVP/PA catheter Foley catheter	An arterial catheter ± CVP are indicated in cases of thyrotoxicosis, preeclampsia, or significant hemorrhage. The use of vasodilators, such as SNP, is also an indication for invasive monitoring.
Positioning	✓ and pad pressure points ✓ eyes SCDs	* NB: peroneal nerve compression at lateral fibular head → foot drop
Complications	Embolization of trophoblastic material	Embolization may occur, especially if >16 wk gestation. Significant respiratory and cardiac compromise may occur, requiring postop ventilation and PEEP, hemodynamic support, and ICU admission.

POSTOPERATIVE

Complications	Bleeding PONV HTN ↓ BP Peroneal nerve injury (2° to lithotomy position)	Continue oxytocin infusion. Significant hemorrhage should be evaluated by surgeons for possible perforation, laceration, or atony. Consider PONV prophylaxis with ondansetron 4 mg. BP should be monitored closely in preeclampsics. Consider ICU admission if unstable. Consider trophoblastic embolization and manage aggressively. Nerve injury manifested as foot drop and loss of
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		sensation over dorsum of foot
Multimodal pain management	Multimodal approach	Consider the following multimodal options: po/iv analgesics: for example, acetaminophen 650–1000 mg po/iv q6h, ibuprofen 200–800 mg po/iv q6h, or celecoxib 200 mg po q12h.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition.

Suggested Readings

1. Brown J, Naumann RW, Seckl MJ, et al: 15 years of progress in gestational trophoblastic disease: scoring, standardization, and salvage. *Gynecol Oncol* 2017; 144(1):200–07.
2. Heller DS: Update on the pathology of gestational trophoblastic disease. *APMIS* 2018; 126(7):647–54.
3. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
4. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.

PELVIC EXENTERATION

SURGICAL CONSIDERATIONS

Description

Pelvic exenteration was introduced by Brunschwig as an ultraradical surgical approach for advanced and radioresistant cervical cancer. Although advanced vaginal and vulvar carcinomas occasionally have been treated with this procedure, its most important role is in the management of centrally recurrent, surgically resectable, radioresistant cervical carcinoma. **Total pelvic exenteration** involves **en bloc resection** of all pelvic tissues, including the uterus, cervix, vagina, bladder, and rectum. Involvement of the distal vagina may require resection of the vulva and groin nodes. The goal of this procedure is curative with removal of all cancer tissue and reconstruction of

appropriate diversions for the urine and stool, if the colon cannot be reanastomosed to the rectum. It is rare for cervical and vaginal cancer to involve the lower 5 cm of the rectum and anus. It is, therefore, often possible to mobilize the descending colon and anastomose it primarily to the distal rectum. Otherwise, an **end colostomy** is created. A **continent or incontinent urinary diversion, omental pelvic carpet or sling**, and **gracilis myocutaneous flaps** for vaginal and perineal reconstruction are performed. A **rectus abdominis muscle flap** also can be used for vaginal reconstruction. This type of flap yields excellent aesthetic and functional results. In cases where an omental sling (shown in [Fig. 8.1-7](#)) cannot be developed, an absorbable synthetic mesh is sutured to the pelvic peritoneum to create a pelvic lid (like a hammock) to keep the small bowel off the denuded pelvic peritoneum, thus decreasing the possibility of small bowel obstruction. In general, an additional 2–3 h of surgical time and an additional 300 mL of EBL are expected when a vaginal reconstruction is undertaken. An **exploratory laparotomy** is needed before initiation of the exenterative procedure to exclude spread of disease outside the pelvis and/or extension to pelvic sidewalls and pelvic lymph nodes, all of which are absolute contraindications to this procedure.

Resectability is evaluated by examination of pelvic and paraaortic lymph nodes, liver, hemidiaphragms, and peritoneal surfaces of the upper abdomen and pelvic wall. Washings are obtained for cytology. Evaluation of lymph nodes is then performed, and all suspicious nodes are submitted for frozen section pathologic examination. The pararectal, paravesical, and presacral spaces are then developed. If the patient is found to be inoperable, CUSA can be used to decrease the tumor burden. Consideration also should be given to **intraoperative radiation therapy**, which may provide an adjunctive treatment to radical surgery in the setting of (a) recurrent disease close to the pelvic sidewalls and (b) microscopically positive surgical margins or “close” resection margins. If the patient is deemed operable, the space of Retzius is developed. The round ligaments are then transected close to the pelvic sidewall bilaterally, and the infundibulopelvic ligaments also are ligated and transected. The ureters are divided as close to the bladder as possible. The superior hemorrhoidal vessel is ligated and transected, and the colon is transected at the appropriate level. The anterior divisions of the internal iliac artery on both sides are ligated and divided, and the web of tissue is clamped close to the pelvic sidewall, transected, and suture ligated. Following this, sharp dissection can be done to free up the specimen from the low attachments to the levator muscles. In larger tumors, the levator muscle is partially removed with the specimen to provide adequate margins. From a combined perineal and abdominal approach, the distal vagina, urethra, and perineum ($\pm$ rectum) can be resected. The specimen is handed off the field, hemostasis is achieved, and the bladder is reconstructed with a continent urinary diversion using an ileocolonic segment, ileal

loop, or transverse colon conduit. An ileal loop or other urinary conduit is performed, a colostomy or low rectal anastomosis is done, vaginal reconstruction is undertaken, and the pelvic floor is covered. (Figs. 8.1-7 and 8.1-8 show the completed exenteration.)

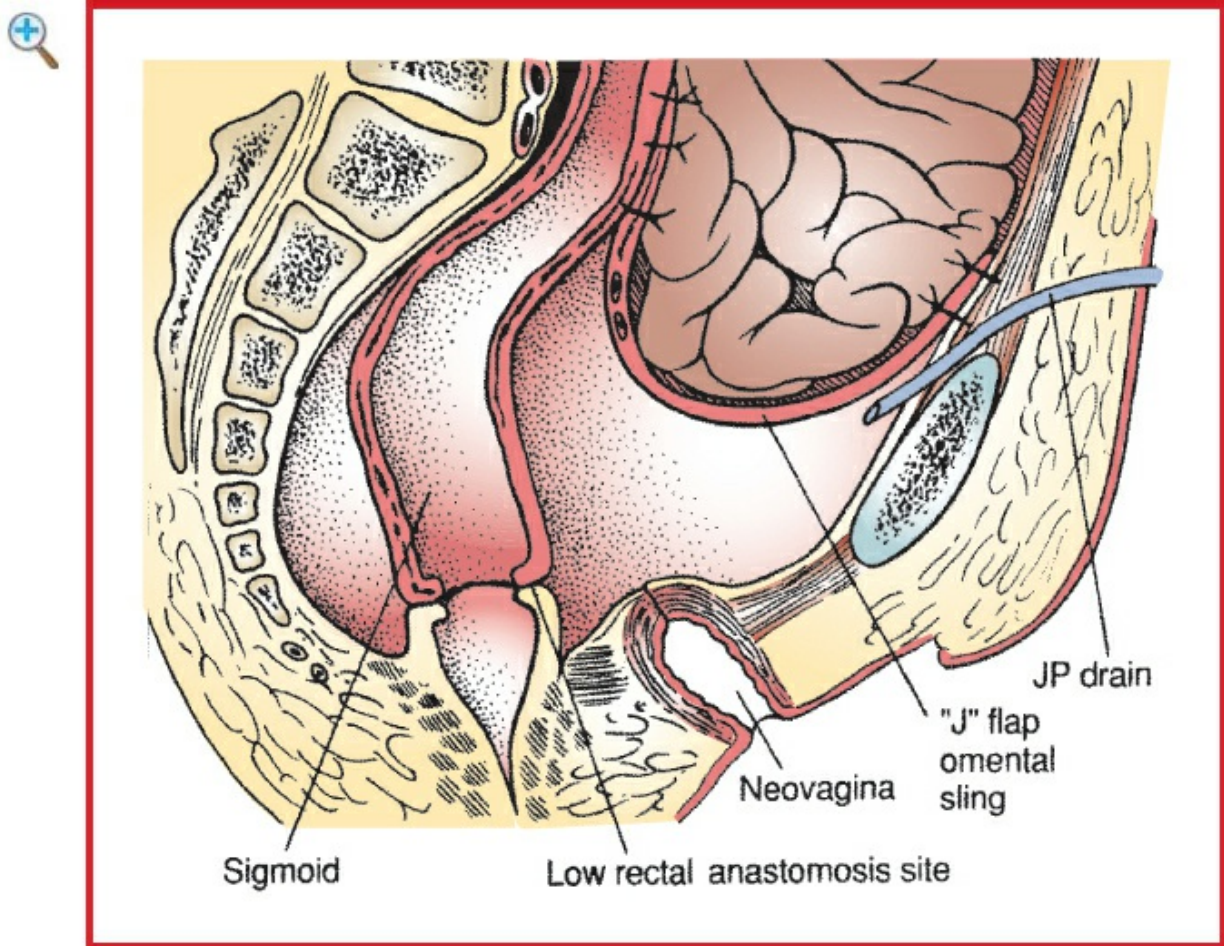


Figure 8.1-7 Sagittal section of the pelvis after a total pelvic exenteration. Note that all the reproductive organs, along with their supporting structures, the rectum and the bladder, have been resected. Drains can be placed through separate stab incisions in the abdomen. The small bowel is kept away from the operative site by a pelvic lid. The urinary and fecal diversions are not diagrammed. (Reproduced with permission from Wheeless CR: Atlas of Pelvic Surgery, 2nd edition. Lea & Febiger, Philadelphia: 1988.)

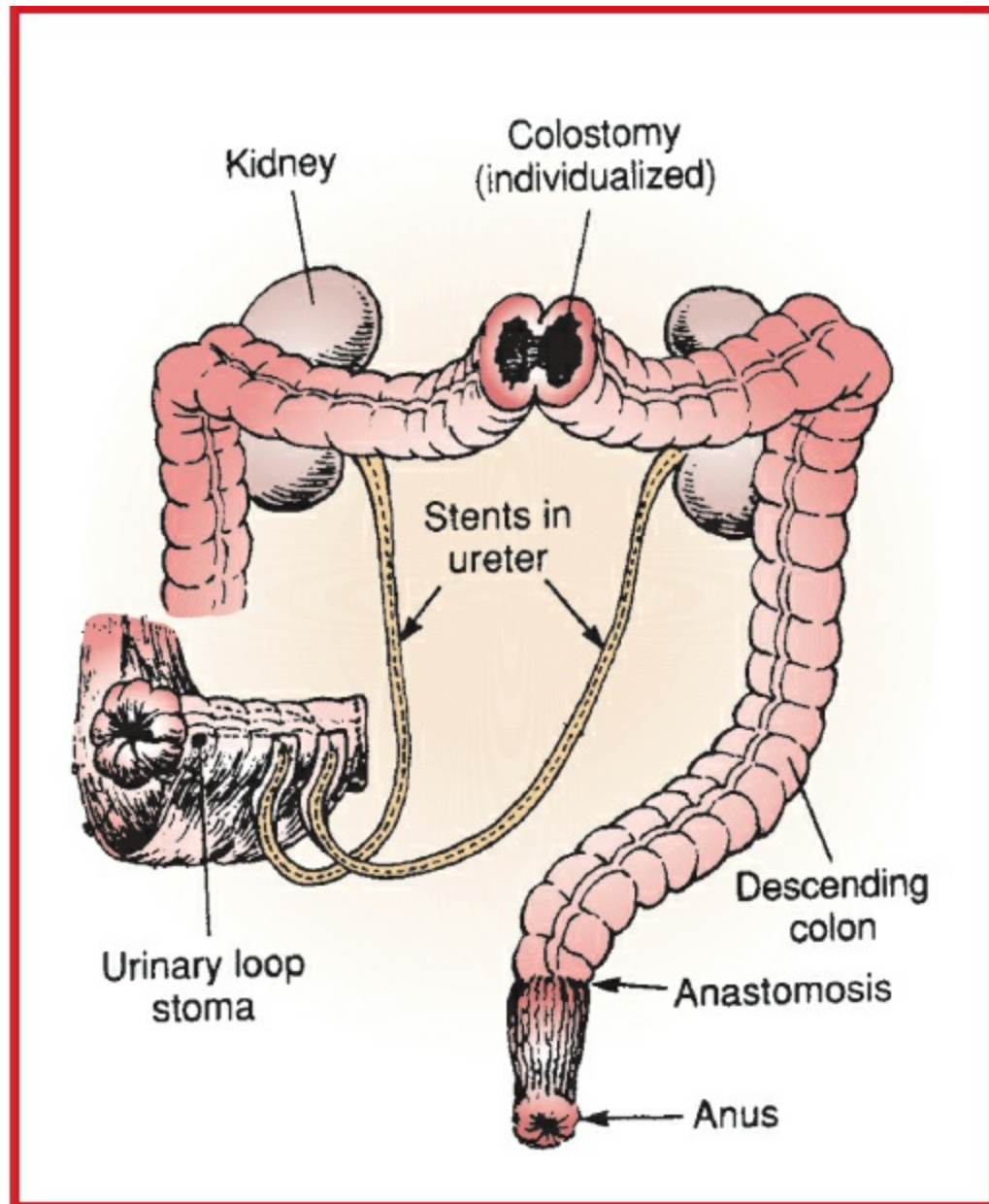


Figure 8.1-8 Conceptual drawing shows the urinary and fecal diversions after a total pelvic exenteration. (Reproduced with permission from Wheeler CR: Atlas of Pelvic Surgery, 3rd edition. Williams & Wilkins, Baltimore: 1997.)

Variant procedure or approaches

Anterior pelvic exenteration is technically similar to a total pelvic exenteration, except the rectum is left intact. **Posterior pelvic exenteration** involves preservation of the bladder. Posterior exenteration has proved more useful in cancer of the vulva or vagina than in cervical cancer. Anterior and posterior exenterations are used in selective cases because of the increased risk of an incomplete tumor resection and multiple complications and malfunctioning of the preserved organ.

Usual preop diagnosis

Recurrent cervical carcinoma following radiotherapy

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, postop recovery plan, possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Modified dorsolithotomy with Allen stirrups
Incision	Midline longitudinal, perineal
Special instrumentation	Vital view, ABC may be helpful; EEA, GIA, TA staplers; robo retractor or similar devices
Unique considerations	NG tube placement intraop. VTE prevention with preop SCDs and prophylactic heparin. Consider preop Greenfield IVC filter placement to avoid PE for high-risk patients. Consider intraop radiation of tumor bed and/or of resection margins. Abort case for positive lymph nodes, extrapelvic metastases and/or tumor extension to pelvic sidewalls noted.
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; redose as indicated by duration and blood loss. Alternatively, ertapenem 1 mg iv or a combination of ampicillin (1 g), gentamicin (80 mg), and metronidazole (500 mg) may be given.
Surgical time	8–12 h (2-team approach); 5–10 h (for anterior and posterior exenteration)
Closing considerations	Abdominal drains; colostomy; ureteral stents; urostomy; intraop radiation therapy. Triple-lumen central line placement. Copious irrigation of the operative sites.
EBL	1200–4000 mL
Postop care	ICU: 2–3 d. Correction of electrolyte imbalance. Extensive peritoneal raw surfaces → intraperitoneal fluid 3rd-spacing. Patients require good hydration to

maintain intravascular volume. SCDs and heparin VTE prophylaxis. Early and aggressive use of TPN is important. Remove ureteral stents 1–2 wk postop, when the edema at the ureterointestinal site has subsided.

Mortality

<5%

Morbidity

Intraop hemorrhage requiring a median of 5 U PRBC

Infectious:

Wound breakdown: 20–30%

Nonspecific: 25%

Flap necrosis: 1–15%

Pelvic abscess: 9–17%

Pyelonephritis: 14–27%

Wound infection: 35%

Sepsis: 5–27%

Psychiatric:

Confusion: 24%

Intestinal:

Ileus: 2–20%

GI fistula: 10–32%

Stoma breakdown: 3%

Small bowel obstruction: 4–9%

Renal:

Ureteral stricture: 8%

Ureteral fistulae: 4–8%

Failure: 1–5%

Cardiovascular:

Venous thrombosis: 7–11%

DIC: 3%

Dysrhythmia: 3%

MI: 3%

Pulmonary:

Pneumonia: 9%

PE: 2%

Neurologic:

CVA: 2%

Pain score

5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	All age groups
Incidence	1.5–17% of cervical cancers treated with radiation therapy (depending on stage and cell type)
Associated conditions	Advanced or recurrent gynecologic malignancies; radiation injury

■ ANESTHETIC CONSIDERATIONS

▲ PREOPERATIVE

This procedure is performed for recurrent rectal, cervical, or other gynecologic cancers and involves removal of all pelvic organs. Occasionally, the bladder or rectum is preserved, if not involved with tumor. Most patients have undergone preop radiation or chemotherapy.

Respiratory	Usually not significant unless there is Hx of smoking or lung disease. Consider prior chemotherapy. (See Anesthetic Considerations for Second-Look/Reassessment Laparotomy for Ovarian Cancer.) Tests: Consider CXR; others as indicated from H&P
Cardiovascular	Exercise tolerance should be assessed. Underlying CAD or CHF should be medically optimized preop. Any exposure to cardiotoxic chemotherapy should be investigated; consider ECHO. Tests: Consider ECG and NT-proBNP; others as indicated from H&P
Neurologic	Any Hx of stroke, seizure, carotid artery disease, or other neurologic disease should be evaluated.
Endocrine	Any endocrine disease, such as diabetes, should be optimized in consultation with the patient's primary care physician or

	endocrinologist. Ask about recent corticosteroid use. Tests: FBS and HbA1c (within 3–6 mo) if diabetic; others as indicated from H&P
Gastrointestinal	Consider ascites and malnutrition.
Hematologic	Look for anemia from chronic disease, chemotherapy, or malnutrition. Consider T&S/T&C. Tests: CBC, PT, PTT
Laboratory	Additional testing as indicated from preop H&P
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300 mg, and/or celecoxib 400 mg po Detailed explanation about the procedure and the potential for postop events including intubation and mechanical ventilation is worthwhile
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a TAP or epidural block/analgesia for postop pain management. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± supplemental epidural anesthesia.

General anesthesia

Induction	Standard induction, unless otherwise indicated by patient condition. Keep in mind that possible surgery could be
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	significantly shortened if patient deemed inoperable during initial exploration.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Balanced anesthetic technique based on comorbidities, use of concomitant epidural anesthesia, and likelihood of needing ICU admission postop. Epidural anesthetic and opiates (morphine or hydromorphone) can be given to reduce anesthetic requirements. Consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion).	
Emergence	If patient is hemodynamically stable, with favorable oxygenation and pulmonary function, normothermic, and responsive at the end of surgery, extubation may be appropriate. Any patient who is unstable and hypothermic, has high-oxygen requirements and significant edema of the face or airway, and is acidotic or who has ongoing fluid requirements should be admitted to the ICU and remain intubated. Due to the large fluid shifts and significant complication rates, postop monitoring in the ICU is appropriate. Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	
Epidural	Consider epidural for intraop and/or postop use: 2% lidocaine (10–15 mL), or 0.25% bupivacaine (10–20 mL), with or without hydromorphone (0.3–0.8 mg), may be given in the epidural for postop pain control. Intraop use may ↑ hemodynamic instability in setting of ↑ bleeding and fluid shifts.	
Ventilation	5–7 cm H ₂ O PEEP may help prevent atelectasis	✓ ABGs during surgery. Adjust ventilation to keep normocarbic. Correct the underlying cause of any metabolic acidosis.
Blood and fluid	Potential large blood loss	Consider access to rapid

requirements	IV: 14–16 ga × 2 BSS @ 10–15 mL/kg/h Consider PRBCs when Hct < 25–27%. Maintain UO of 0.5–1 mL/kg/h. Ionized Ca⁺⁺ FFP/Plt	infuser. Strive to maintain euvolemia based on BP, PPV, HR, UO, ABG, invasive monitoring (CVP, CO, art line tracing), and estimates of ongoing EBL and fluid shifts. Goal-Directed Fluid Management (see Appendix K, p. K-4 p. K-4) Measure ionized Ca⁺⁺ and K⁺ after rapid administration of blood products; replace Ca⁺⁺ as necessary. Plt or FFP for coagulation abnormalities. Monitor values at regular intervals based on PT stability and events.
Monitoring	Standard monitors ± Arterial line ± CVP Foley catheter ± TEE	Invasive hemodynamic monitoring with arterial and CVP catheters usually helpful in these cases, which are of very long, and associated with major bleeding, fluid shifts, ICU admission postop, and potential need for vasoactive infusions. TEE allows intraop assessment of myocardial function and may be appropriate in selected patients.
Positioning	✓ and pad pressure points ✓ eyes	NB: peroneal nerve compression at lateral fibular head → foot drop

	SCD	
Complications	Hypothermia	Use forced-air and fluid warmers and monitor temperature.
	VTE Coagulopathy Trauma to kidney ± Ureters Bleeding Metabolic acidosis Peripheral nerve injury	See Appendix B, p. B-7 . Watch for hematuria or ↓ UO. Monitor hemoglobin and coagulation status periodically during long surgery. Keep adequate reserve blood products available. Prolonged surgery: carefully pad, secure, and monitor all extremities

▼ POSTOPERATIVE

Complications	Bleeding	✓ Hct and coags periodically. Be prepared for continued increased fluid requirements for 24 h postop.
	Fluid overload Hypothermia PONV VTE Peripheral nerve injury	Maintain euvolemia, if significant fluid gain occurs; consider ICU postop ventilation. If extubated For prolonged surgery, carefully pad, secure, and monitor all extremities.
Multimodal pain management	Use multimodal analgesia ± PCA	po/iv analgesics: for example, acetaminophen 1000 mg

	± epidural ± TAP block	po/iv q6h, gabapentin 200 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis.
Tests	Follow CBC, chemistry panel, coags, and ABG as indicated by postop course	Others as indicated from postoperative course. Patient will likely need ongoing critical care/ICU management.

Suggested Readings

1. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
2. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
3. Nelson G, Altman AD, Nick A, et al: Guidelines for pre- and intra-operative care in gynecologic/oncology surgery: Enhanced recovery after surgery (ERAS®) Society recommendations—Part I. *Gynecol Oncol* 2016; 140(2):313–22.
4. Nelson G, Altman AD, Nick A, et al: Guidelines for postoperative care in gynecologic/oncology surgery: Enhanced recovery after surgery (ERAS®) Society recommendations—Part II. *Gynecol Oncol* 2016; 140(2):323–32.
5. Nelson G, Dowdy SC, Lasala J, et al: Enhanced recovery after surgery (ERAS®) in gynecologic oncology—Practical considerations for program development. *Gynecol Oncol* 2017; 147(3):617–20.
6. Oh TK, Lim MC, Lee Y, et al: Improved postoperative pain control for cytoreductive surgery in women with ovarian cancer using patient-controlled epidural analgesia. *Int J Gynecol Cancer* 2016; 26(3):588–93.
7. Types of Urostomies and Pouching Systems. American Cancer Society [Internet]. [cited 2018 Aug 13]. Available from: <https://www.cancer.org/treatment/treatments->

EXPLORATORY LAPAROTOMY OR LAPAROSCOPY AND HYSTERECTOMY/BSO FOR UTERINE CANCER

SURGICAL CONSIDERATIONS

Description

Currently, endometrial cancer is the most common gynecologic malignancy in the United States. The first step in the management of this cancer is surgical staging via laparotomy or laparoscopy. The objective, aside from 1° therapy, is to obtain as much surgical and pathologic staging data as feasible for determination of adjuvant postop therapy. Traditionally, surgery for uterine cancer was performed via laparotomy; however, laparoscopy has become a preferred approach for early-stage uterine cancers after recent trials have shown decreased operative morbidity and similar survival outcomes compared to laparotomy. In laparoscopy, or an open approach, the first step is careful exploration, which is carried out for evidence of omental, liver, peritoneal, and adnexal metastases. Aortic and pelvic areas are evaluated for metastases, and suspicious nodes are removed. In patients with low-grade histology and no or minimal suspected invasion, complete lymph node dissection remains an area of controversy, and treatment is generally individualized based on clinicopathologic factors. In patients with more extensive disease, a pelvic and periaortic lymph nodes dissection is performed. The lymph node sampling may be omitted in the treatment of some uterine sarcomas. A **total hysterectomy** with **BSO** is then performed in the usual manner (see discussion of TAH/BSO in Staging Laparotomy for Ovarian Cancer).

As a result of recent clinical trials, minimally invasive surgery via laparoscopy or robotic-assisted laparoscopy has become the first approach for uterine cancer staging. The GOG conducted a randomized prospective study evaluating exploratory laparotomy with open staging versus laparoscopic staging (GOG LAP2) to compare outcomes and safety. In 2009, the GOG published the data regarding perioperative morbidity and showed that the laparoscopic approach was associated with a shorter hospital stay and fewer moderate-to-severe postoperative complications. In 2012, the GOG published the survival data, which did not show statistical equivalence between two approaches relative to survival; however, a recent Cochrane meta-analysis has shown equivalent recurrence-free and overall survival with laparoscopy compared to laparotomy. When proceeding with a laparoscopic approach, it is important to note that adhesions of

variable severity may be present from prior surgery or radiation. Care should be taken to avoid possible bowel injury at time of trocar insertion. Patient needs to be in steep Trendelenburg position for duration of the procedure, and both arms should be tucked in at the patient's sides. Because argon is a heavy gas, prolonged use of the endoscopic ABC in a patient in steep Trendelenburg can → significant facial and neck subcutaneous emphysema. As with the open technique, the lymph nodes being removed are in the immediate proximity of the great pelvic vessels. The surgeon and anesthesiologist should be mindful of the potential for severe hemorrhage if these vessels are injured.

Usual preop diagnosis

Endometrial carcinoma

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, postop recovery plan, possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES		
	Open Technique	Laparoscopic and Robotic Technique
Position	Supine	Modified dorsolithotomy in Allen stirrups
Incision	Midline longitudinal abdominal/transverse	Vertical infraumbilical and multiple small transverse incisions. For robotic surgery, midline incision is supraumbilical.
Special instrumentation	None	Uterine manipulator, videolaparoscopy equipment. Endoscopic GIA staplers, endoscopic ABC, endoscopic vascular clips, and CO ₂ laser. For robotic surgery, bean bag positioner, additional padding for shoulders, neck, and arms
Unique	None	Mechanical bowel prep; steep

considerations		
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; redose as indicated by duration and blood loss	Trendelenburg ⇐
Surgical time	2–4 h	2–3 h
Closing considerations	± NG tube placement	Release the pneumoperitoneum completely. Closure of fascia at trocar sites ≥10 mm diameter
EBL	400–750 mL	100–500 mL
Postop care	Consider SCDs and heparin for VTE prophylaxis.	Begin early ambulation and feeding. Patients generally can be discharged on POD 1 or 2.
Mortality	0.1%	⇐
Morbidity	Hemorrhage requiring transfusion: 15%	4–7%
	Thrombophlebitis: 7%	⇐
	UTI: 7%	⇐
	Paralytic ileus: 2–5%	1–2%
	Wound infection: 3–7%	Rare
	VTE: 1–3%	⇐
	Pelvic infection: 1–2%	⇐
	Wound dehiscence: 1%	N/A
	Bowel injuries: <1%	1%
	Urinary tract injuries: <1%	1–5%
		Trocar site herniation: 0.5–2%
		Trocar site tumor: <1%
Pain score	7	2–3

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive and postreproductive ages (average = 61 yr)
Incidence	24/100,000; 47,130 new cases in 2012, lifetime risk 2.6%
Etiology	Exposure to unopposed endogenous or exogenous estrogen; ↑ extraglandular conversion of androstenedione to estrone; sequential oral contraceptive pills; exposure to radiation (sarcomas)
Risk factors	Nulliparity, late menopause, early menarche, family hx, chronic anovulation, PCOS, tamoxifen use
Associated conditions	Obesity; diabetes; HTN; nulliparity; late menopause; early menarche; family Hx; Stein-Leventhal syndrome; chronic anovulation; ovarian and colon cancer; granulosa cell ovarian tumors; arthritis; hypothyroidism; HNPCC (Lynch) syndrome

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Endometrial cancer usually is diagnosed in postmenopausal women who present with vaginal bleeding. Exploratory laparotomy or laparoscopy and TAH/BSO are commonly performed for removal of the primary tumor as well as staging of metastatic disease. Occasionally, preop radiation therapy may be in progress, with consequent systemic effects.

Respiratory	✓ or Hx of lung disease or smoking Tests: Others as indicated from H&P
Cardiovascular	Hx of CAD, HTN, or CHF Sx (e.g., angina, dyspnea, or peripheral edema) should be investigated. Assess patient's exercise tolerance and current medications. Tests: Consider ECG within 6–12 mo for age >65. Consider NT-proBNP, if significant CHF or cardiomyopathy present.
Neurologic	Screen during preop eval. Evaluation as indicated by

	H&P
Endocrine	✓ for diabetes and hypothyroidism, which have been associated with this tumor. If the patient has received corticosteroids within the previous 6 m → consider supplemental dose of hydrocortisone (100 mg iv q12h × 2 d). Tests: FBS and HbA1c (within 3–6 mo) if diabetic; others as indicated from H&P
Hematologic	If vaginal bleeding has been profuse or of long duration, significant anemia may occur. Consider preop iron supplements or iv iron therapy. Tests: CBC or hemogram; consider T&S and T&C.
Laboratory	Additional testing based on preop H&P
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300 mg po, and/or celecoxib 200 mg po
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or epidural block/analgesia for postop pain management. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural or spinal analgesia/anesthesia.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Consider ketamine (0.5

	mg/kg bolus + 5 mcg/kg/min infusion).
Emergence	Reverse muscle relaxant with neostigmine (0.07 mg/kg with glycopyrrolate 0.01 mg/kg) or sugammadex (2–4 mg/kg). Provide PONV prophylaxis. (See Appendix B, p. B-4 and p. B-6) Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

Epidural (open procedures)	2% lidocaine (10–20 mL), ± epinephrine 1:200,000, or 0.25% bupivacaine (15–20 mL) is used; then at ~3–5 mL/h. Opiates, such as morphine (2–4 mg) or hydromorphone (0.3–0.6 mg), may be given in the epidural for postop pain control.	
Blood and fluid requirements	IV: 16–18 ga × 1 BSS @ 4–6 mL/kg/h Consider PRBC for Hct < 25%.	Goal-Directed Fluid Management (see Appendix K, p. K-4). Crystalloid is used for volume replacement. If anemia is present preop, it may be necessary to give PRBCs to keep Hct > 25%. Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors Foley catheter	Direct monitoring of arterial pressure is indicated in patients with CAD, severe HTN, or lung disease.
	± Arterial line, CVP OGT/NGT for laparoscopic/robotic procedures	Consider art line and/or CVP if significant comorbidities exist.

Positioning	✓ and pad pressure points ✓ eyes	
Complications	SCD Hypothermia Trauma or obstruction of ureter	Warm iv fluids; use forced-air warmer. Heating pad on OR table. Watch for hematuria or ↓ UO.

▼ POSTOPERATIVE

Complications	PONV VTE Hypothermia Bleeding	See Appendix B, p. B-4 and p. B-6 . See Appendix B, p. B-7 . Use forced-air warming in recovery.
Multimodal pain management	Multimodal approach ± PCA ± epidural ± PNB/RA	Consider the following multimodal options: po/iv analgesics: consider acetaminophen 650–1000 mg po/iv q6h, gabapentin 200 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Coronado PJ, Herrai MA, Magrina JF, et al: Comparison of perioperative outcome and cost of robotic-assisted laparoscopy, laparoscopy, and laparotomy for endometrial cancer. *Eur J Obstet Gynecol Repro Biol* 2012; 165(2):289–94. doi: 10.1016/j.ejogrb.2012.07.00.
2. Galaal K, Bryant A, Fisher AD, et al: Laparoscopy versus laparotomy for the management of early stage endometrial cancer. *Cochrane Database Syst Rev* 2012; (9):CD006655.
3. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
4. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
5. Walker JL, Piedmonte MR, Spirtos NM, et al: Recurrence and survival after random assignment to laparoscopy versus laparotomy for comprehensive surgical staging of uterine cancer: Gynecologic Oncology Group LAP2 Study. *J Clin Oncol* 2012; 30:695–700.
5. Zakhari A, Czuzoj-Shulman N, et al: Laparoscopic and robot-assisted hysterectomy for uterine cancer: a comparison of costs and complications. *Am J Obstet Gynecol* 2015; 213(5):665.e1–7.

RADICAL HYSTERECTOMY

SURGICAL CONSIDERATIONS

Description

Radical hysterectomy is the preferred mode of therapy for young women with Stage IA, IB, or non bulky IIA cervical carcinoma, who want to preserve ovarian function. It is also appropriate with Stage II endometrial and Stage I vaginal carcinoma. The operation involves the removal of the uterus, along with the upper vagina and all the parametrial tissues to the pelvic sidewall. A pelvic and paraaortic **lymph node dissection** usually is performed at the beginning of the procedure. Suspicious nodes are submitted for pathologic frozen section evaluation. The paravesical and pararectal spaces are then developed, with the “web” of tissue between these two spaces being palpated carefully ([Fig. 8.1-9](#)). In cases of cervical or vaginal cancer, if parametrial tumor extension is noted and/or the lymph nodes are positive on frozen section, the hysterectomy may be aborted. The radical hysterectomy is performed after the

pararectal, and paravesical spaces have been developed. The uterine arteries are divided at their origin from the anterior division of the internal iliac artery. The ureters are dissected free of the parametrial tissues, which are then transected close to the pelvic sidewall. Next, the rectovaginal space is developed (Fig. 8.1-10). The uterosacral ligaments are transected between their uterine and sacral attachments. The upper third of the vagina is cross-clamped and divided in such a manner as to provide a 3-cm margin, and the specimen is delivered en bloc. Note that during this procedure the ureters and bladder are dissected free and left intact. With the increasing use of minimally invasive surgery in gynecologic oncology, radical hysterectomy is frequently performed laparoscopically or robotically. Similar to laparoscopic surgery for endometrial cancer, steep Trendelenburg is required during the procedure, and special care is taken to support the head, shoulders, and arms.

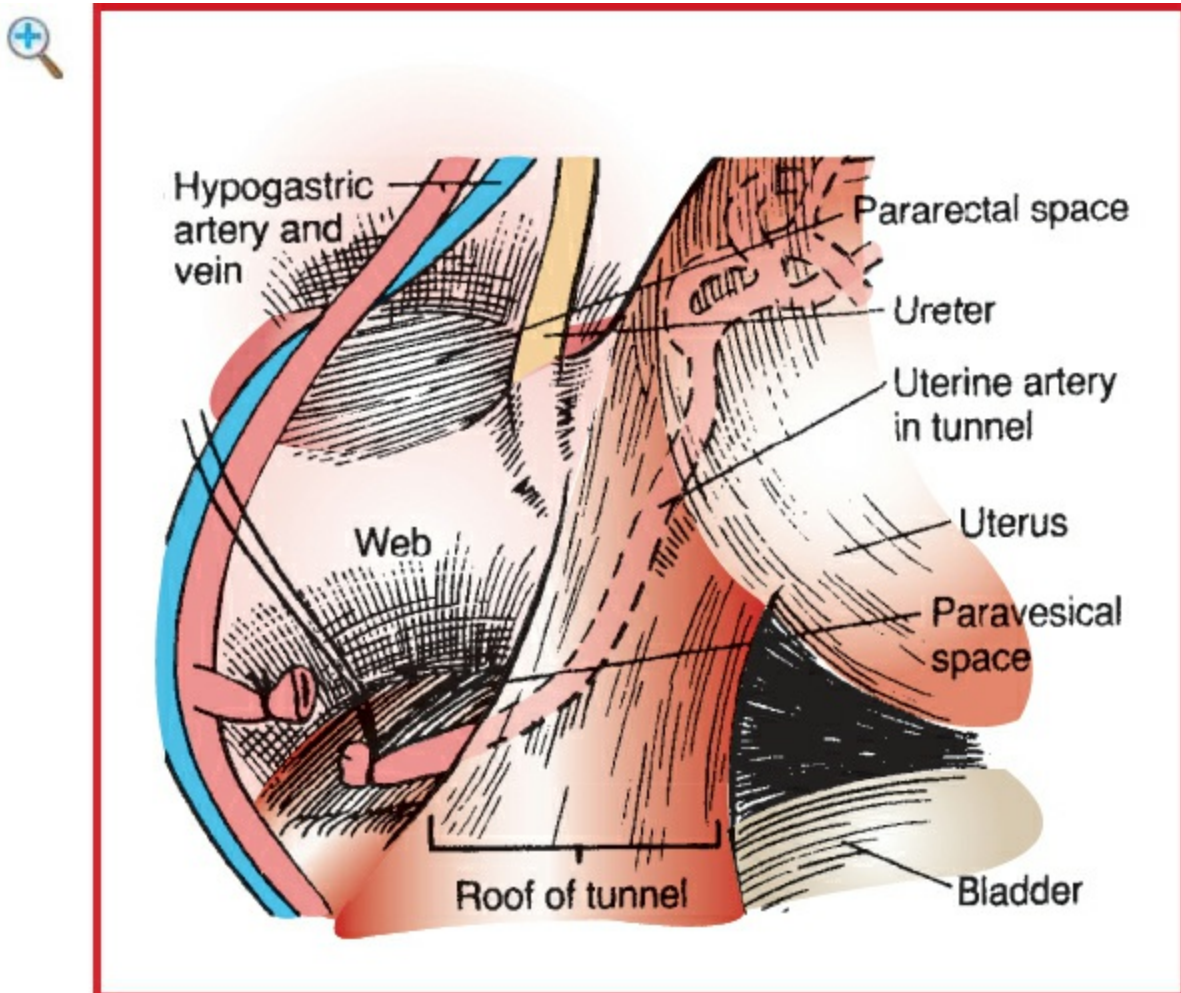


Figure 8.1-9 View of parametrial area, showing positions of ureter and uterine artery to web, hypogastric vessels, paravesical, and pararectal spaces. (Reproduced with permission from Wheeler CR: Atlas of Pelvic Surgery, 3rd edition. Williams & Wilkins, Baltimore: 1997.)

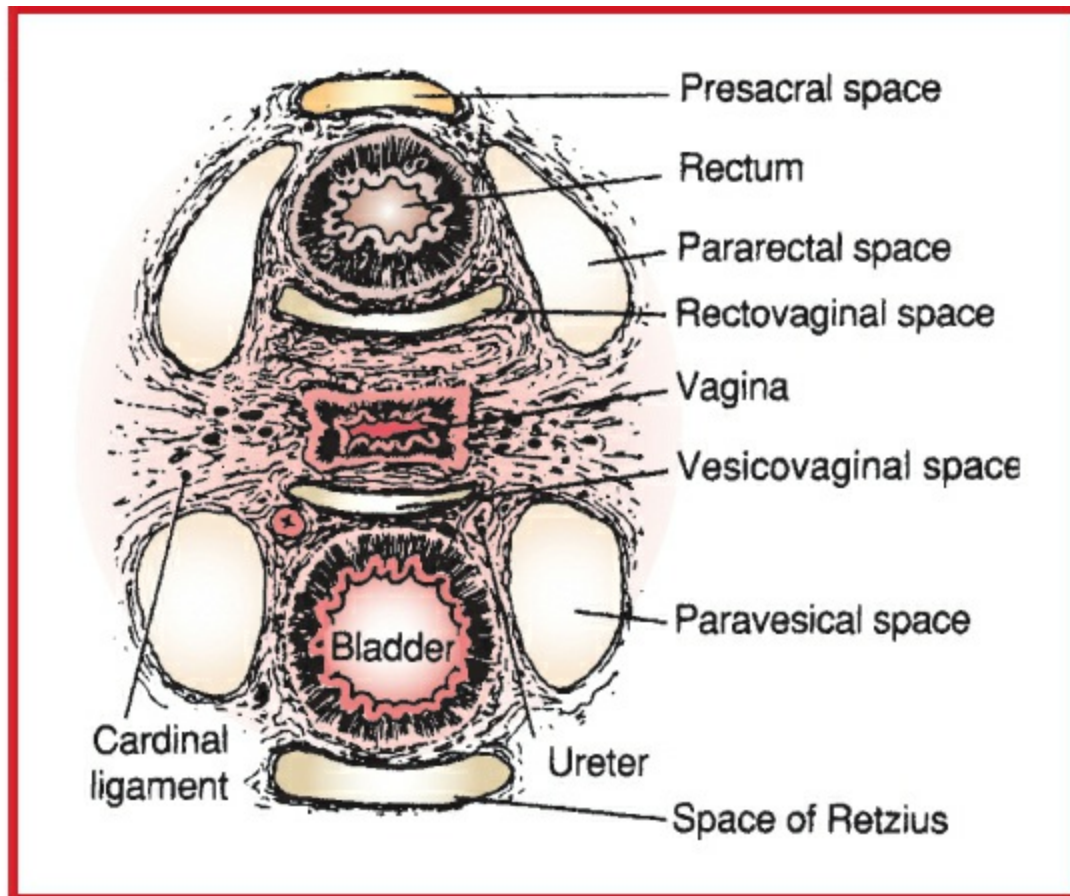


Figure 8.1-10 Schematic diagram of a cut in the anterior/posterior plane with positions of all spaces in relation to pelvic organs. (Reproduced with permission from Wheeler CR: Atlas of Pelvic Surgery, 3rd edition. Williams & Wilkins, Baltimore: 1997.)

In reproductive age women, who may require postop radiotherapy, ovarian function is preserved by performing an **oophoropexy**. This is accomplished by severing the utero-ovarian ligament and mobilizing the ovarian vessels as they course through the infundibulopelvic ligament. The ovaries are then sutured outside the radiation therapy field and marked with metal clips for future identification.

Variant procedure or approaches

Stallworthy, Dolstad, Novak, Rutledge, Wertheim, and other surgeons have proposed several modifications in an effort to reduce the incidence of ureteral and bladder fistulae. These approaches include preservation of blood supply to the terminal 2 cm of the pelvic ureter by widely displacing ureters (not dissecting them from their fascial beds) and limiting parametrial dissection to the proximal 1/3 or 1/2.

Usual preop diagnosis

Stage IA, IB, or nonbulky IIA cervical carcinoma; Stage II endometrial or Stage vaginal carcinoma (less common)

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, recovery plan, possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Laparotomy	Laparoscopy or Robotic-Assisted Laparoscopy
Position	Supine/modified lithotomy	Modified dorsal lithotomy
Incision	Midline longitudinal or low transverse abdominal (Maylard/Cherney)	Supra- or infraumbilical midline incision and multiple small transverse incisions
Special instrumentation	ABC, vital view (suction, irrigation, and light source combined in 1 instrument) helpful	
Unique considerations	Abort case if positive paraaortic lymph nodes found on frozen section or if obvious parametrial involvement noted. Consider intraop radiation therapy (investigational), if patient is inoperable.	⇐ ± Mechanical bowel prep, steep Trendelenburg
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; redose as indicated by duration and blood loss	⇐
Surgical time	3–6 h	2–4 h
Closing considerations	Vaginal and abdominal drains; ± NG tube placement; suprapubic bladder catheter placement; oophoropexy; copious irrigation	Release the pneumoperitoneum completely. Closure of fascia at trocar sites >10 mm diameter

EBL	500–1500 mL	100–500 mL
Postop care	Transient ileus very common. Advance diet slowly. Transient bladder dysfunction very common; therefore, continued bladder drainage for 1+ wk may be necessary. Consider using SCDs and heparin for VTE prophylaxis.	Early ambulation and feeding. Patients can be discharged POD#1 or 2. Transient bladder dysfunction common; urinary catheter kept in place 5–7 d. Consider preoperative prophylactic heparin.
Mortality	0.3–2.0%	0.1–0.2%
Morbidity	Paralytic ileus: 3–11% Pelvic lymphocyst: 2–3% Intraop hemorrhage: 3–5% Thrombophlebitis: 7% Pneumonia: 1.5–4% Wound infection: 3.5% Vesical injury: 2–3% Pelvic infection: 1–2% PE: 1–2% Vesical fistulae: 1–2% Small bowel obstruction: 1–2% Ureteral fistulae: 1–2% Uretral injury: 1–2% Wound dehiscence: 1% Postop lymphedema: <1%	Ileus: 1–2% Lymphocyst: 2–6% Intraop hemorrhage: 1–3% Thrombophlebitis: 7% Pneumonia: 1.5–4% Wound infection: rare Vesical injury: 2–3% Ureteral injury: 1–3% Pelvic infection: 1–2% PE: 1–3% Small bowel obstruction: rare Wound dehiscence: rare Postop lymphedema: rare Conversion to laparotomy: 1–2% Cuff dehiscence: 5–8%

	Rectal injury: <1%	Trocar site herniation: 0.5–2%
	Pelvic urinoma: rare	Bladder dysfunction: 10–20%
Pain score	8	5–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive and postreproductive years
Incidence	10/100,000
Etiology	Human papillomavirus (HPV), subtypes 16, 18 (most common), and others
Associated conditions	Smoking; venereal warts; genital herpes; multiple sexual partners; early-age onset of coitus; multiparity; lower socioeconomic status; HIV infection

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

This surgery usually is performed for cervical carcinoma in young women who wish to preserve ovarian function and whose tumor has not spread beyond local invasion. Lymph nodes are removed to confer a therapeutic advantage and to plan postop adjuvant therapy, if any.

Respiratory	These tumors have been associated with cigarette smoking. ✓ for preexisting lung disease. Tests: As indicated from H&P
Cardiovascular	Tests: Consider ECG within 6–12 mo for age >65. Consider NT-proBNP if significant CHF or cardiomyopathy present. Others as indicated by H&P
Hematologic	✓ for preop anemia Tests: Hemogram or CBC; consider T&S and T&C.
Laboratory	Renal panel if age ≥ 65 yr, HTN, renal disease, DM
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300 mg po, and/or celecoxib 200 mg po

ERAS

Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of TAP or epidural block/analgesia for postop pain management. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA ± epidural or spinal analgesia for postop pain control.

	Laparotomy	Laparoscopic or Robotic Assist
Induction	Standard induction	⇐
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	⇐
Maintenance	Standard maintenance. Consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion).	Standard maintenance
Emergence	Discuss with surgeon local wound infiltration or nerve block (e.g., TAP block) placement or reinforcement if initially placed >4 h ago. Consider iv acetaminophen if po dose given >4–6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.	Reverse muscle relaxant with neostigmine (0.07 mg/kg with glycopyrrolate 0.01 mg/kg) or sugammadex (2–4 mg/kg). Provide PONV prophylaxis. (See Appendix B, p. B-4 and p. B-6)

Blood and fluid requirements	500–1500 mL (laparotomy) 100–500 mL (laparoscopic) IV: 16–18 ga × 1–2 BSS @ up to 6–8 mL/kg/h Maintain UO > 0.5 mL/kg/h Warm iv fluids Goal-Directed Fluid Management (see Appendix K, p. K-4)	⇐
Monitoring	Standard monitors Foley catheter ± Arterial catheter ± CVP catheter	Consider invasive monitoring for patients with underlying cardiopulmonary disease, advanced age, or in cases of large anticipated or unanticipated blood loss.
Positioning	✓ and pad pressure points ✓ eyes SCD	Steep Trendelenburg may be necessary for laparoscopic and robotic-assisted procedures.
Complications	Injury to ureters Hypothermia	Watch for hematuria or ↓ UO. Warm iv fluids; use forced-air warmer.

POSTOPERATIVE

Complications	Atelectasis Hypothermia Bleeding VTE PONV	VTE prophylaxis See Appendix B, p. B-4 and p. B-6 .
Multimodal pain management	Multimodal approach ± PCA ± Epidural ± PNB/TAP	Consider the following multimodal options as appropriate: po/iv analgesics:

		acetaminophen 1000 mg po/iv q6h, gabapentin 300 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
2. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
3. Wright JD, Herzog TJ, Neugut AI, et al: Comparative effectiveness of minimally invasive radical hysterectomy for cervical cancer. *Gynecol Oncol* 2012; 127:11–17.

ROBOTIC-ASSISTED LAPAROSCOPY AND SURGERY IN GYNECOLOGIC ONCOLOGY

SURGICAL CONSIDERATIONS

Description

Laparoscopy and robotic-assisted laparoscopy are both widely used to treat gynecologic cancers. As reviewed previously, a recent Cochrane review has shown equivalent disease-free and overall survival when comparing laparoscopy to an open approach for the treatment of endometrial cancer. In addition, other advantages of a

laparoscopic approach include a shorter hospital stay, better pain control, faster recovery time, ↓ morbidity, and cosmetically appealing smaller incisions. Increasingly, more studies are being published to better answer questions regarding the safety of use in malignant diseases, differences in operative time, and cost, as well as survival and morbidity differences based on cancer type. The authors believe that there is a role for laparoscopy in select situations, such as:

- Assessment of adnexal masses and diagnosis of ovarian cancer
- Laparoscopic lymphadenectomy with laparoscopically assisted vaginal hysterectomy or laparoscopic hysterectomy in staging of endometrial cancer
- Laparoscopically assisted radical vaginal hysterectomy or laparoscopic radical hysterectomy with lymphadenectomy in early-stage cervical cancer
- Laparoscopically assisted application of interstitial brachytherapy implants
- Interval or secondary cytoreductive surgery for ovarian cancer

Details of each of these procedures are beyond the scope of this section; however, common principles of laparoscopic techniques (e.g., adhesiolysis, biopsies, oophorectomy, omentectomy) apply (see discussion in Laparoscopic Procedures for Gynecologic Surgery). The patient is placed in the dorsal lithotomy position, with the buttocks extended over the edge of the table to allow for instrument manipulation, and may need to be placed in steep Trendelenburg position during surgery. Various techniques are used for entry into the abdominal cavity, but the open laparoscopic technique is favored by many and may ↓ risk of vascular or visceral injuries. Other incision-related complications include dehiscence and hernia.

Usual preop diagnosis

Ovarian tumor and other gynecologic cancers/tumors

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, recovery plan, possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Dorsal lithotomy; legs in Allen universal stirrups; steep Trendelenburg
Incisions	Intraumbilical; bilateral suprapubic; midline suprapubic

Special instrumentation	CO ₂ laser; bipolar LigaSure® or other device; suction irrigator; may require Harmonic Scalpel®; various laparoscopic stapling instruments; beanbag or other support for robotic surgery
Unique considerations	Extended operative time and period of abdominal insufflation for radical procedures. Possible rapid need for laparotomy. Extensive use of electrocautery, ABC, CO ₂ laser
Antibiotics	Cefoxitin or cefotetan 2 g iv; redose as indicated by duration and blood loss
Surgical time	1.5–5 h
Closing considerations	Quicker than with laparotomy; but fascia of all 10- to 12-mm ports need to be closed. 5-mm trocar sites can be closed in a subcuticular fashion.
EBL	50–1500 mL, depending on extent of procedure
Postop care	Advance diet as tolerated, ambulate POD 0, and discharge POD 1–2.
Mortality	0.1%
Morbidity	Overall complication rate: 0.2–10.3% Conversion to laparotomy: 2.1–7% Port-site metastasis: <1% Subcutaneous emphysema: 0.3–2% Pneumomediastinum: 0.26% Brachial plexus neuropathy: 0.16% Incisional hernia: 0.5–2% Bowel injury: 1% Major vascular injury: <1% Bladder/ureteral injury: 1–3% Gas embolism: rare Wound infection: uncommon
Pain score	3–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive and postreproductive year
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Incidence	Depends on primary disease
Etiology	Numerous, depending on primary disease (see appropriate sections earlier in this chapter)
Associated conditions	See specific procedures in this chapter.

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations for Laparoscopic Hysterectomy.

Suggested Readings

1. Coronado PJ, Herrai MA, Magrina JF, et al: Comparison of perioperative outcome and cost of robotic-assisted laparoscopy, laparoscopy, and laparotomy for endometrial cancer. *Eur J Obstet Gynecol Reprod Biol* 2012; 165(2):289–94. doi: 10.1016/j.ejorgb.2012.07.006.
2. Walker JL, Piedmonte MR, Spirtos NM, et al: Recurrence and survival after random assignment to laparoscopy versus laparotomy for comprehensive surgical staging of uterine cancer: Gynecologic Oncology Group LAP2 Study. *J Clin Oncol* 2012; 30:695–700.
3. Wright JD, Herzog TJ, Neugut AI, et al: Comparative effectiveness of minimally invasive and abdominal radical hysterectomy for cervical cancer. *Gynecol Oncol* 2012; 127:11–7.
4. Zakhari A, Czuzoj-Shulman N, Spence AR, et al: Laparoscopic and robot-assisted hysterectomy for uterine cancer: a comparison of costs and complications. *Am J Obstet Gynecol* 2015; 213(5):665.e1-7.

INTERSTITIAL PERINEAL IMPLANTS

SURGICAL CONSIDERATIONS

Description

Radiotherapy is the treatment of choice for International Federation of Gynecology and Obstetrics (FIGO) Stage IIB-IVA carcinoma of the cervix; Stage I, II, III, and IV vaginal cancers; selected vulvar cancers; and pelvic recurrences of gynecologic cancers. In addition, radiotherapy is also commonly used as an adjuvant treatment with chemotherapy and surgery for advanced endometrial cancer. Often, **external beam therapy** is combined with **brachytherapy**, either in the form of intracavitary insertion

or as an interstitial perineal template implant. Intracavitary radiotherapy utilizes devices such as Fletcher-Suit tandem and ovoid applicators (Fig. 8.1-11) or cylinders that are fitted into the vagina and provide a therapeutic boost to the vaginal apex region after external beam irradiation. In general, these devices do not require a laparotomy or laparoscopy for guidance. For selected patients with distorted anatomy and/or bulky tumors, interstitial implants provide superior dose distribution and better local tumor control. The two most widely used systems are the **Martinez Universal Perineal Interstitial Template (MUPIT)** and the **Syed-Neblett applicator**. These systems have similar efficacy, and both are performed in the OR with the patient under MAC or GA. A **laparoscopy** or **laparotomy** is frequently performed at the time of interstitial implant placement to accurately guide the needles into their target tissues and to avoid radiation injury to bowel and/or other pelvic organs not involved with tumor. To minimize postop patient discomfort, some centers use laparoscopy instead of laparotomy for needle guidance. A two-team approach is used: a gynecologic oncology team performs the laparotomy or laparoscopy and guides the needles from above; a radiation oncology team inserts the implants from below. The implants are afterloaded with the appropriate radiation sources when the patient has returned to her shielded room. If a **hysterectomy** has been done previously, an **omental pelvic carpet**, or a **pelvic lid** made of delayed, absorbable mesh, is performed to provide additional space between the radiation source and bowel.

One important and recent advance in radiation therapy is the technique, intensity-modulated radiation therapy (IMRT). IMRT is a form of external beam radiation and allows creation of a gradient of radiation dose over a specified treatment area, in contrast to the previous method of delivering a constant dose to all tissues. As more IMRT is being used, fewer patients are needing an operative procedure to assist with delivery of the appropriate radiation dose.

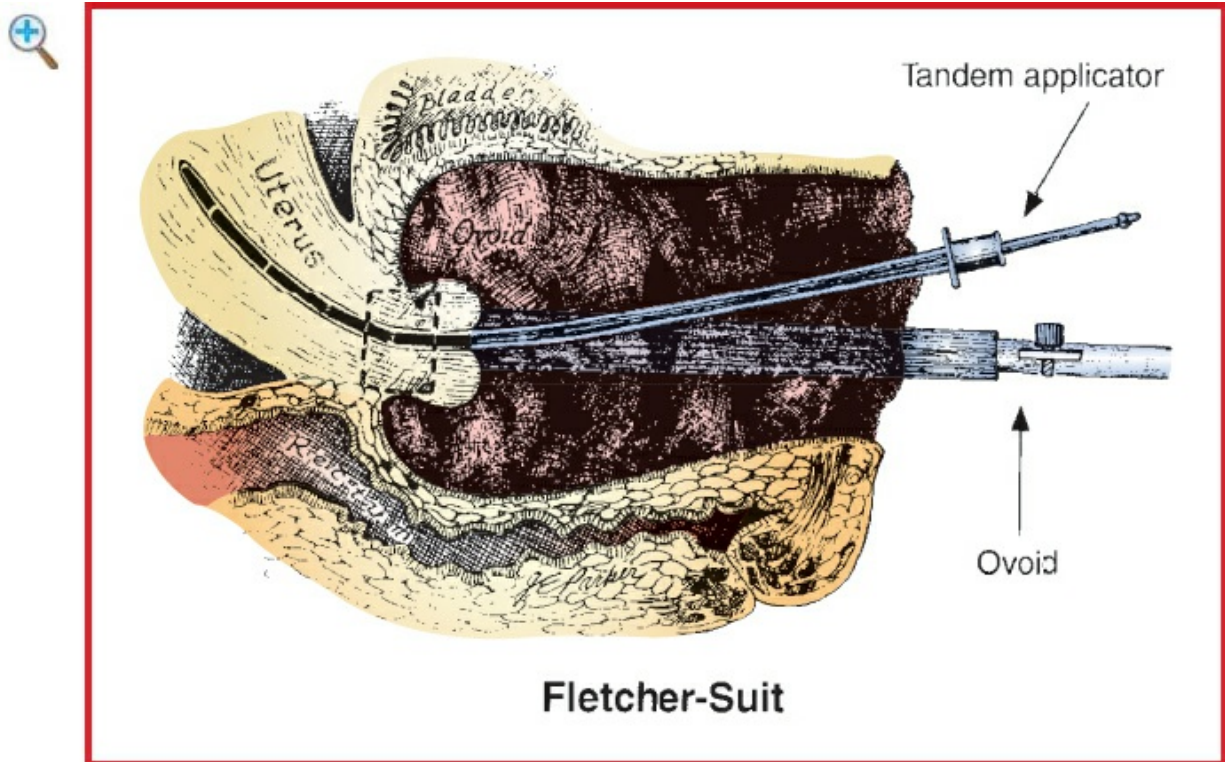


Figure 8.1-11 “Fletcher-Suit.” The tandem and ovoid applicators are packed into the vagina, leaving the maximum distance between the bladder and radium sources. (Reproduced with permission from Wheeler CR: Atlas of Pelvic Surgery, 3rd edition. Williams & Wilkins, Baltimore: 1997.)

Usual preop diagnosis

Cervical, vaginal, vulvar carcinomas; pelvic recurrence of gynecologic malignancies

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading VTE prophylaxis, postop pain management options, recovery plan, possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Laparotomy Guided	Laparoscopy Guided
Position	Modified dorsolithotomy; Allen stirrups	⇐
Incision	Midline longitudinal abdominal	Vertical infraumbilical and multiple small transverse incisions at the pubic hairline

Special instrumentation	Syed-Neblett or MUPIT systems or modifications thereof	⇐ + Videolaparoscopy equipment, preferably with a 3-chip camera
Unique considerations	MRI and/or CT scans + information from physical exam are used to preplan implant with a computer dosimetry program. Patients require thorough preop mechanical and antibiotic bowel prep. Preop epidural placement or postop PCA may prove helpful for pain control. Foley catheter needs to be inserted through an opening at the top of the clear plastic template before implant positioning.	⇐ + Adhesions of variable severity may be present from prior surgery or radiation. Care should be taken to avoid possible bowel injury at time of trocar insertion. Patient needs to be in steep Trendelenburg position for duration of procedure.
Antibiotics	Cefoxitin 2 g or cefotetan 2 g iv; then q12h × 3 doses	⇐
Surgical time	1.5–3 h	⇐
EBL	150–350 mL	Minimal
Closing considerations	Suture template to perineum. Perform rectal exam and adjust any needles that are too close to, or have protruded through, the rectal mucosa. Pack any space between template and perineum with Vaseline gauze. Obtain A-P and lateral orthogonal localization films with the patient in the supine bed	⇐ + Release the pneumoperitoneum completely. Consider insertion of 1 L heparinized LR to cause the bowel to float and remain mobile, thus minimizing the risk of radiation injury.

rest position. Insert Hypaque dye into Foley catheter balloon before localization films. Consider insertion of a large Foley into the rectum, and attach to a drainage bag.

Postop care

Patient is confined to bed ⇐ while interstitial implants are in place. SCDs and minidose heparin for VTE prophylaxis. Vigorous use of incentive spirometry. Consider constipating medications (e.g., Lomotil). Patient must be placed in a shielded room. Visitors and medical personnel should interact with patient from behind a lead shield until radiation sources have been removed.

Mortality

0.1–0.3% ⇐

Morbidity

Hemorrhagic proctitis with diarrhea and tenesmus: 7–18% ⇐

Radiation cystitis: 7–10% ⇐

Cervical necrosis: 5–6% ⇐

Rectovaginal fistula: <5% ⇐

Vesicovaginal fistula: <5% ⇐

Vaginal vault necrosis: 3–5% ⇐

	Rectal fibrosis: 2–5% Pelvic infection: 2–3%	≤ ≤
Pain score	8–9	8–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive and postreproductive age; childhood—rare
Etiology	Cervical, vaginal, or vulvar carcinomas; pelvic recurrence of gynecologic malignancies

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Radiation implants are used for palliation or cure in cervical, endometrial, and ovarian cancer. The implants concentrate the radiation close to the site of the tumor and may be supplemented by external beam radiation. The unique tolerance of the uterus and vagina to radiation permits large doses and accounts for the success in treating cervical lesions. The sigmoid, rectum, and large bowel are much more sensitive to radiation injury, limiting the dose of radiation that may be given to the pelvis. There is a trend toward the use of sophisticated external beam radiation (intensity modulated), resulting in fewer cases requiring radiation implants.

Respiratory	Usually not significant unless underlying lung disease is present
Cardiovascular	Many patients with pelvic tumors are elderly and prone to cardiovascular disease. Tests: ECG if age ≥ 65; others as indicated from H&P
Hematologic	Tests: CBC; consider PT/PTT if at risk for coagulopathy.
Laboratory	Renal panel
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 1000 mg po, gabapentin 300 mg po, and/or celecoxib 200 mg po
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Regional or GA, depending on site involved. A postop epidural is useful for pain management.

Induction	Standard induction	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. Consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion).	
Emergence	Consider PONV prophylaxis. See Appendix B, p. B-4 and p. B-6 . PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	Small blood loss IV: 18–20 ga × 1 BSS @ 2–4 mL/kg/h	Goal-Directed Fluid Management (see Appendix K, p. K-4)
Monitoring	Standard monitors	
Positioning	May be necessary to keep patient motionless during implant insertion. ✓ and pad pressure points ✓ eyes SCD	

▼ POSTOPERATIVE

Multimodal pain management	Multimodal approach ± PCA ± epidural ± PNB/RA	Consider the following multimodal options: po/iv analgesics: acetaminophen 1000 mg
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		po/iv q6–8h, gabapentin 200 mg po tid, ibuprofen 200–800 mg po/iv q6h, celecoxib 200 mg po q12h, oxycodone 5–10 mg po q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain.
ERAS	PONV protocol (see Appendix B, p. B-1)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider VTE prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Koh WJ, Sales LR: Principles of radiation therapy. In: Karlan BY, Bristow RE, IAJ, eds. *Gynecologic Oncology Clinical Practice and Surgical Atlas* 1st edition. McGraw-Hill Co., San Francisco: 2012.
2. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
3. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
4. Syed AM, Puthawala AA, Abdelaziz NN, et al: Long-term results of low-dose-rate interstitial-intracavitary brachytherapy in the treatment of carcinoma of the cervix. *Int J Radiat Oncol Biol Phys* 2002; 54(1):67–78.

CHAPTER 8.2

Gynecology/Infertility Surgery

SURGEONS

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ANESTHESIOLOGIST

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DILATATION AND CURETTAGE (D&C)

SURGICAL CONSIDERATIONS

Description

During **dilatation and curettage (D&C)**, the endometrial lining of the uterus and coexisting lesions (myoma, polyp) are sampled or removed. This procedure is performed to diagnose and treat bleeding from uterine and cervical lesions, to treat cervical stenosis, to complete an incomplete or missed spontaneous abortion, or as a method for pregnancy termination. An office endometrial biopsy and medical management are usually initially performed for bleeding problems.

With the patient in the dorsal lithotomy position, the surgeon initially performs a bimanual examination under anesthesia to obtain information about both the presence of adnexal pathology and anatomic detail of the uterus. A speculum is inserted into the vagina, and the cervix is grasped with a single-tooth tenaculum for manipulation. The cervix is pulled gently toward the operator, who then uses a uterine probe to delineate the length of the uterus and the angulation between the cervical canal and uterus. The uterine cavity is reached by dilating the cervical canal with progressively larger

dilators (Hegar or Pratt) to the desired diameter for the procedure. A polyp forceps may be used at this stage to remove existing polyps, and a sharp curette is used to remove the endometrial lining (Fig. 8.2-1). A suction curette is often used for abortion procedures.

Usual preop diagnosis

Uncontrolled uterine bleeding refractory to hormonal treatment; abnormal uterine bleeding; incomplete, missed, or induced abortion; pregnancy termination; cervical stenosis causing dysmenorrhea or obstruction of menstrual flow

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, carb-loading and postop pain management options. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy with stirrups
Incision	None
Special instrumentation	To prevent a peroneal nerve injury, the area of the leg leaning against the stirrup (lateral knee) should be well cushioned. To prevent a femoral nerve injury, prolonged hyperflexion of the hips should be avoided.
Unique considerations	Following induction, the perineum is positioned at the very end of the table to ensure optimal exposure. During the cervical dilatation, a vasovagal response can occur, with subsequent bradycardia and ↓ BP. Uterine perforations will often manifest as severe postop pain.
Antibiotics	None or doxycycline 100 mg orally or iv before the procedure
Surgical time	5–15 min
EBL	0–50 mL
Mortality	Minimal
Morbidity	Postop fever: 1.7% Uterine perforation: 0.63% Severe immediate postop bleeding (caused by either uterine artery perforation or cervical injury): <1%

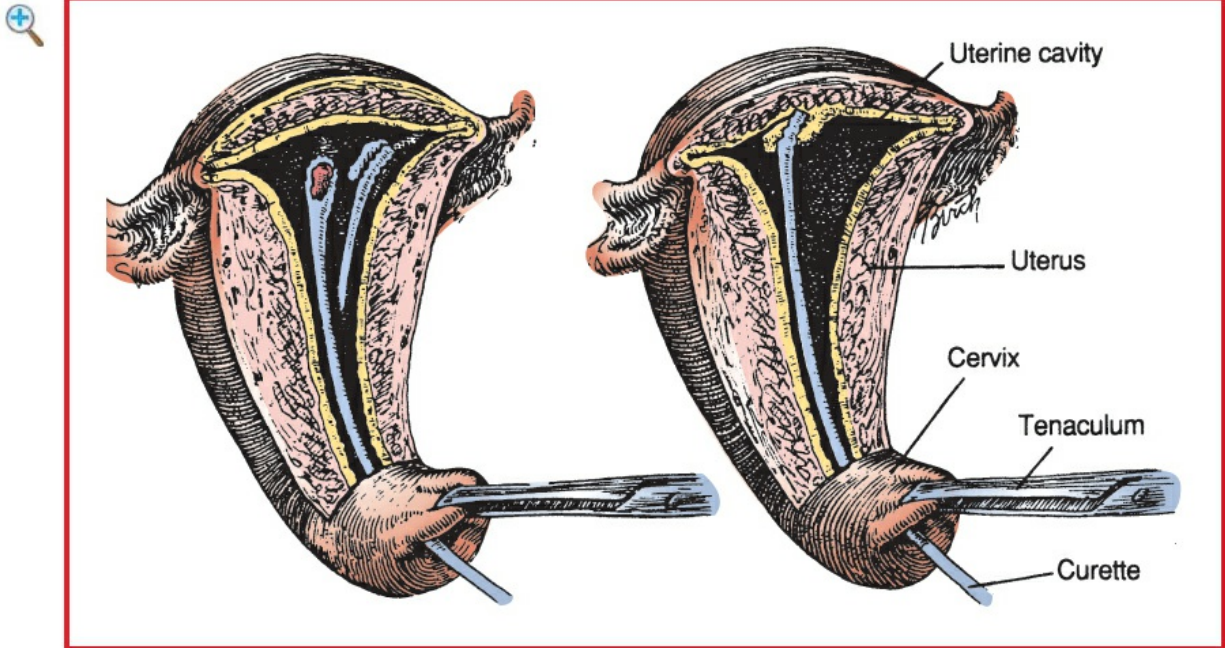


Figure 8.2-1 Curettage of endometrial lining. (Reproduced with permission from Rock JA, Jones HW, eds: TeLinde's Operative Gynecology, 10th edition. Wolters Kluwer Health/Lippincott Williams & Wilkins, Philadelphia: 2008.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–80 yr
Incidence	>1/50 females
Etiology	Dysfunctional uterine bleeding, endometrial pathology, pregnancy issues
Associated conditions	Obesity, HTN

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Therapeutic Abortion, Dilatation, and Evacuation.

Suggested Reading

1. Tangsirawatthana T, Sangkomkamhang US, Lumbiganon P, et al: Paracervical local anaesthesia for cervical dilatation and uterine intervention. Cochrane Database Syst Rev 2013; (9):CD005056.

THERAPEUTIC ABORTION, DILATATION, AND EVACUATION (D&E)

SURGICAL CONSIDERATIONS

Description

Therapeutic abortion (TAB) is the elective termination of a pregnancy before viability (usually considered to be 24 wk). In 2017, 879,000 abortions were reported in the United States, a rate of 14.6 abortions per 1000 women aged 14–44. **Suction curettage** is the most efficient method to terminate pregnancies during the 1st trimester (<12 wk), and the great majority of abortions are performed in this manner. Few (<5%) 1st trimester abortions in the United States are performed with a sharp curette due to increased operative time and blood loss, and also the risk of intrauterine adhesions potentially resulting in infertility. Very early termination can be performed without anesthesia via medical termination.

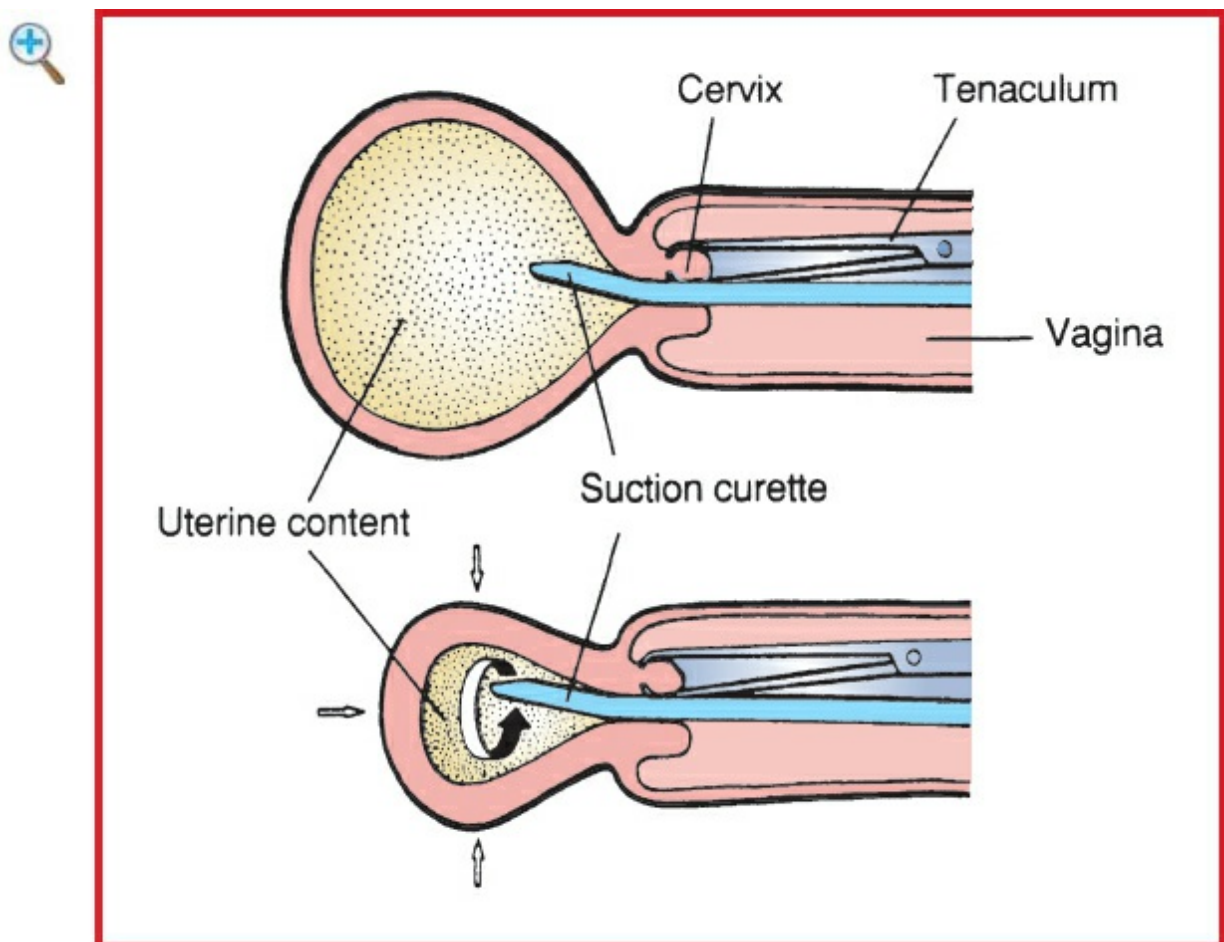


Figure 8.2-2 Uterine aspiration. Top, uterus at beginning of procedure; bottom, at conclusion of procedure. (Reproduced with permission from Rock JA, Thompson JD, eds. TeLinde's Operative

The TAB procedure consists of a standard cervical dilatation (required for gestations >6 wk), followed by vacuum aspiration of the uterine contents using a plastic suction curette ([Fig. 8.2-2](#)). Alternatively, the cervical canal can be dilated >6 h before the operation with laminaria or synthetic osmotic dilators, which, after insertion, swell to provide dilatation. A sharp curette is sometimes gently used at the very end to verify the emptiness of the cavity. Due to the risk of misdiagnosis and incorrect dating, physicians should confirm and date the intrauterine pregnancy via ultrasound before performing a TAB. Ergonovine maleate (0.2 mg im) and oxytocin (Pitocin) 20–30 U/1000 mL iv can be administered during the procedure to reduce bleeding, although the efficacy of oxytocin has been questioned. The uterus becomes increasingly responsive to oxytocin after 20 wk of gestation. The procedure is performed under local with or without iv sedation or GA; it is generally felt that local is safer. Regardless of the method used, the obtained products of conception (POC) are sent for histologic examination to exclude the presence of trophoblastic neoplasm or ectopic pregnancy.

Variant procedure

Dilatation and evacuation (D&E) remains the safest method for 2nd trimester pregnancy termination. It is performed similarly to 1st trimester suction curettage with the addition of using large-ring forceps to grasp and remove fetal parts intermittently. This procedure is often performed under GA.

Usual preop diagnosis

Pregnancy (viable or nonviable): 52% within 8-wk estimated gestational age; 90% within 12-wk estimated gestational age

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, carb-loading and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

1st Trimester Suction Curettage or 2nd Trimester D&E

Position	Dorsal lithotomy with patient in Allen stirrups. Following induction, the perineum is positioned at end of the table to ensure optimal exposure.		
Incision	None		
Special instrumentation	Suction; + large-ring forceps for 2nd trimester D&E		
Unique considerations	To prevent a peroneal nerve injury, the area of the leg leaning against the stirrup (lateral knee) should be well cushioned. To prevent a femoral nerve injury, prolonged hyperflexion of the hips should be avoided. During cervical dilatation, a vasovagal reaction can occur → ↓ HR + ↓ BP. During injection of lidocaine for paracervical block, seizure may occur. Uterine perforations often will result in excessive postop pain.		
Antibiotics	Doxycycline 100 mg iv		
Surgical time	5–15 min; 15–45 min (2nd trimester D&E)		
EBL	Number week gestation 1–4: 10 mL 5–8: 10–30 mL 9–10: 30–80 mL 11–12: 80–200 mL 13–14: 200–400 mL 300–500 mL (2nd trimester D&E)		
Postop care	PACU		
Mortality	0–3.1/100,000; 13/100,000 (2nd trimester D&E)		
Morbidity	Mild infection Resuctioned day of surgery Resuctioned subsequently Cervical stenosis causing amenorrhea Cervical incompetence Underestimation of gestational age Seizure (after local anesthesia)	1/216 1/553 1/596 1/6071 1/9444 1/15,454 1/25,086	For 2nd trimester D&E T > 38° for >1 d Cervical injury Cervical tear Retained POC Hemorrhage UTI
			13.4/1000 11.6/1000 10/1000 9/1000 7.1/1000 1.8/1000
Complications requiring hospitalization	Incomplete abortion Sepsis Uterine perforation Vaginal bleeding Inability to complete Coexisting tubal pregnancy Total	1/3617 1/4722 1/10,625 1/14,166 1/28,333 1/42,500 1/1405	For 2nd trimester D&E Endometritis Uterine perforation Need for laparotomy Need for blood transfusion
			8.5/1000 3.2/1000 2.1/1000 1.9/1000
Pain score	5		

■ PATIENT POPULATION CHARACTERISTICS

Age range	15–45 yr
Incidence	Abortions: 14.6/1000 women age 14–44 yr in the United States (2014)
Etiology	Desire for pregnancy termination

ANESTHETIC CONSIDERATIONS FOR D&C, TAB, D&E

PREOPERATIVE

These are among the most common procedures performed in gynecology. Patients presenting for these procedures are generally healthy; however, bleeding and sepsis may alter ASA status.

Cardiovascular	Hemodynamic status may be impaired 2° preop uterine bleeding, and patient may be septic from retained uterine products. Tests: As indicated from H&P
Laboratory	Hgb/Hct; consider T&S/T&C; other tests, as indicated from H&P.
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Local/MAC, regional, and GA all may be appropriate. In the younger patient population, spinal anesthesia may be less desirable because of increased incidence of post dural puncture headache (PDPH).

Local anesthesia

Some obstetricians/gynecologists perform these procedures under local anesthesia. Paracervical block has the potential for inadvertent intravascular administration, with

consequent local anesthetic toxicity.

Regional anesthesia

A T10 sensory level is sufficient to provide anesthesia for procedures on the uterus.

Spinal	0.75% bupivacaine 7.5–12 mg in 8.25% dextrose ± 10–20 mcg fentanyl (IT). (See Anesthetic Considerations for Cesarean Section) If spinal anesthesia is indicated, a pencil-point spinal needle (e.g., Whitacre or Sprotte) should be used to decrease the incidence of PDPH.
Epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL (1:200,000), 15–20 mL. Supplemental iv sedation. (See Anesthetic Considerations for Cesarean Section.)
CSE	Combined spinal/epidural (CSE): An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia. Apply monitors, administer fluid, and position patient as for spinal or epidural. The most common technique is the needle-through-needle. After the epidural space is located with a standard 17-ga Tuohy needle, insert a 4-inch 26- to 27-ga pencil-point spinal needle through it to administer 7.5–12 mg of spinal bupivacaine (0.75%) ± fentanyl 10–20 mcg and ± morphine sulfate 0.1–0.2 mg. Secure the epidural catheter and use if needed (a test dose is advisable).

General anesthesia

Induction	Consider full stomach precautions (see p. B-5 for RSI) at 20 wk gestation and beyond. First trimester and early 2nd trimester generally TAB low risk for aspiration based on pregnancy-related changes. Supraglottic airway often appropriate
ERAS	Verify preprocedure antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and

	normothermia.	
Maintenance	Standard maintenance. Consider keeping MAC < 1 to lower the risk of uterine atony + bleeding.	
Emergence	Be ready with suction in the event of vomiting on emergence. Consider NSAIDs iv. Consider ketorolac (15–30 mg) after bleeding is controlled. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18–20 ga × 1 (unless hypovolemic)	Usual replacement of maintenance fluids and overnight deficit with BSS to restore euvolemia. Blood transfusion rarely indicated. See Restricted Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Oxytocin 20–30 U/L iv Methylergonovine maleate 0.2 mg im	Oxytocin causes uterine contraction, with a consequent decrease in blood loss. Rapid iv bolus may lead to hypotension. 20–30 U of oxytocin is usually diluted in 1 L of crystalloid and then infused. Methylergonovine maleate also causes uterine contraction. Dose is usually 0.2 mg im. Side effects include HTN, myocardial ischemia, and dysrhythmias. Do not give iv.
Monitoring	Standard monitors	
Positioning	✓ hip, leg, hand position ✓ and pad pressure points ✓ eyes Shoulder abducted	Lithotomy position may worsen pulmonary function. Consider L uterine displacement if >20 wk.

	<90°	
Vagal stimulation	Bradycardia	When the cervix is grasped and dilated, the patient may have excessive vagal stimulation, which can be treated by prompt cessation of stimulation and glycopyrrolate 0.1–0.2 mg iv or, if preferred, atropine 0.2–0.4 mg.
Complications	Nerve injury	Common peroneal nerve palsy (e.g., foot drop) is possible if pressure on the nerve over the fibula is not prevented by adequate padding or positioning. Hyperflexion of the hip joint can cause femoral and lateral femoral cutaneous nerve palsy. Obturator and saphenous nerve injuries are also complications of the lithotomy position.
	Finger trauma	Take care to ensure safety of patient's fingers when manipulating foot of the bed. Avoid finger injury by placing patient's arms on arm boards or by wrapping her hands.
	Laryngospasm	Avoid light anesthesia.

POSTOPERATIVE

Complications	PONV Uterine perforation (rare) Severe hemorrhage,	→ Severe abdominal pain Severe postop bleeding may be caused by uterine atony,
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	necessitating blood transfusion (rare)	retained POCs, uterine perforation, or cervical injury.
Multimodal pain management	Oral analgesia IV opiates	Acetaminophen ± opiates, NSAIDs IV fentanyl. With extreme pain, consider uterine perforation.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization.
Tests	Hgb/Hct, if hemorrhage	

Suggested Readings

1. Chapman JS, Roddy E, Ueda S, et al: Enhanced recovery pathways for improving outcomes after minimally invasive gynecologic oncology surgery. *Obstet Gynecol* 2016; 128(1):138–44.
2. van den Berg MMJ, Dancet EAF, et al: Patient-centered early pregnancy care: a systematic review of quantitative and qualitative studies on the perspectives of women and their partners. *Hum Reprod Update* 2018; 24(1):106–18.

HYSTEROSCOPY

SURGICAL CONSIDERATIONS

Description

Hysteroscopy is a procedure in which the endometrial cavity is directly visualized via the hysteroscope. The procedure is used primarily to investigate abnormal uterine bleeding, often caused by intrauterine submucous myomata and polyps. After the diagnosis, these lesions can be removed through the operative hysteroscope using a variety of techniques.

Variant procedure or approaches

Diagnostic and operative hysteroscopies can be performed under either GA or local anesthesia with iv sedation. An examination under anesthesia is performed, followed by the insertion of an open-sided speculum and the placement of a tenaculum on the cervix. The cervical canal is dilated until the hysteroscope can be introduced with its sheath

(Fig. 8.2-3). A distension medium—usually a low-viscosity fluid—is then used to provide visualization of the uterine cavity. The most commonly used low-viscosity fluids are NS, sorbitol 3%, mannitol 0.5%, and mannitol 5% solutions. NS is most commonly used for diagnostic hysteroscopies, whereas the other nonconductive fluids are used for operative hysteroscopies. Certain bipolar operative hysteroscopies also allow the use of NS.

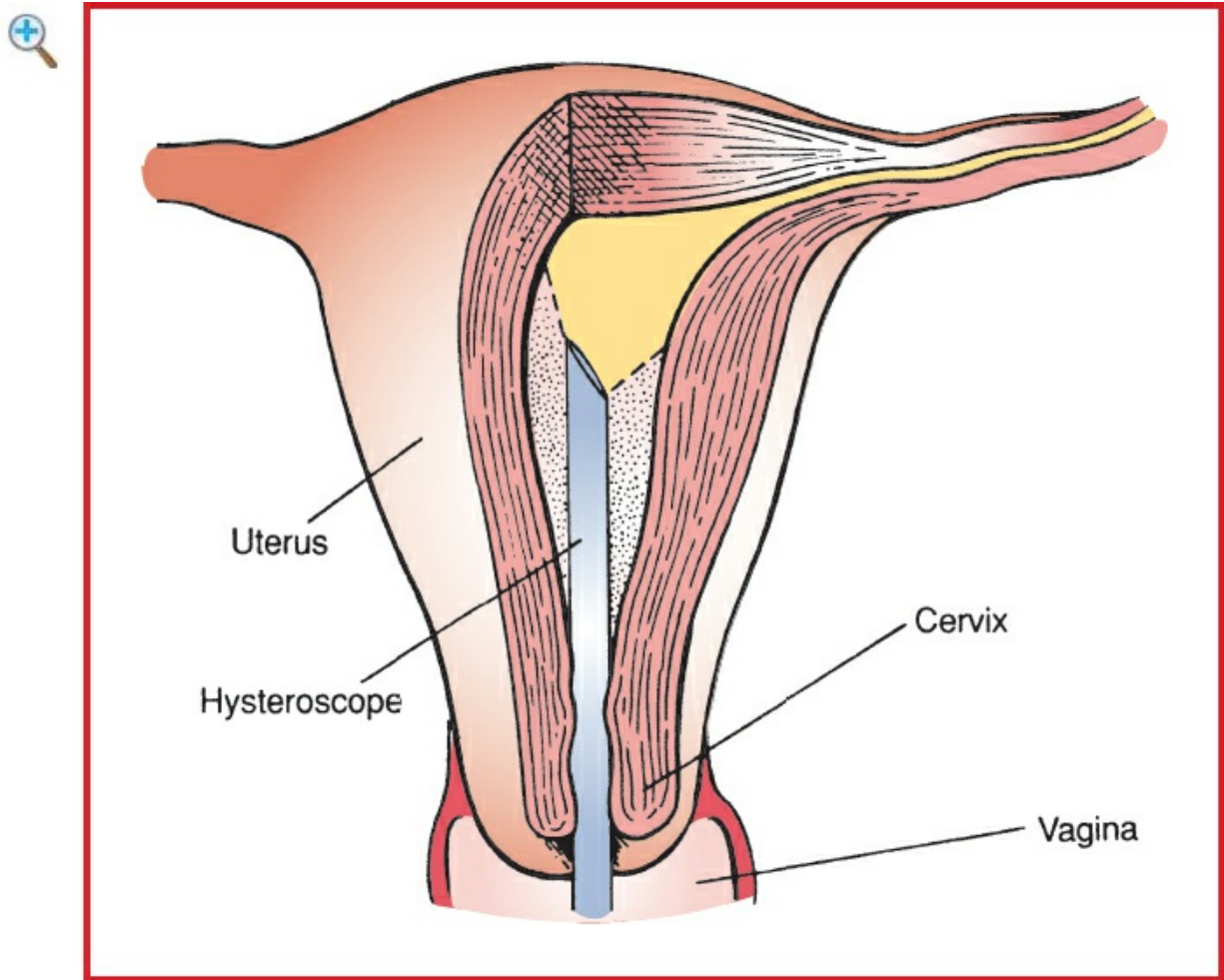


Figure 8.2-3 Hysteroscopy. (Reprinted from Baggish MS, Barbot S, Valle RF Diagnostic & Operative Hysteroscopy, 1st edition. Year Book Medical Publishers, Chicago: 1989. Copyright © 1989 Elsevier. With permission.)

It is important for the surgeon to be vigilant regarding the fluid uptake by closely monitoring the infused volume and the collected volume. Intracervical injection of a dilute vasopressin solution before the procedure reduces media absorption. At a fluid uptake of 500 mL of a hypo-osmolar solution (e.g., sorbitol/mannitol) or 1000 mL of an iso-osmolar solution (e.g., NS), consideration should be given to stopping the procedure given concerns for fluid overload. The maximum fluid deficit of 1000 mL is suggested when using hypotonic solutions and 2500 mL when using isotonic solutions. However,

decisions regarding when to terminate the procedure based on fluid deficit should be individualized based on the patient’s condition and comorbidities. Ongoing communication between the surgeon and anesthesiologist is essential, as iv fluids can add to increased circulatory volume and potential fluid overload. Fluid overload can lead to complications such as pulmonary edema and hyponatremia. A rapid increase in fluid deficit can also be a sign of an occult uterine perforation.

The distension medium is delivered to the hysteroscope by means of gravity or by a pump. Most systems use gravity, whereby the fluid is delivered via a wide-bore tubing, and the maximum intrauterine pressure (IUP) is determined by the height of the fluid container above the uterus. The maximum IUP is thus limited by gravity, making this system relatively safe. The lowest IUP necessary to provide adequate visualization should be used to decrease the rate of absorption of the distension medium. If using a pump, the pressure should be set at or just above the patient’s mean arterial pressure (MAP). A video camera is usually attached to the hysteroscope to allow for easier visualization. During operative cases, an accompanying laparoscope is sometimes inserted into the abdomen to evaluate the progress of the hysteroscopy and to safeguard against uterine perforation and potential bowel injury. GA is required if a laparoscopy is to be performed concurrently.

Usual preop diagnosis

Abnormal uterine bleeding; infertility; recurrent pregnancy loss

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, carb-loading and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy (Allen stirrups)
Incision	None
Special instrumentation	Fluid pump; laser; electrocautery equipment
Unique considerations	Accelerated fluid absorption with lengthy procedures or with resections may → pulmonary edema. Laparoscopy may accompany this procedure.
Surgical time	15 min–2 h

Antibiotics	None
EBL	0–100 mL
Postop care	Excessive postop bleeding can be controlled by using a 5-mL Foley balloon catheter in the uterus for several hours.
Mortality	Minimal
Morbidity	Pleural effusion can be seen with use of low-viscosity medium.
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	20–80 yr
Incidence	>1/50
Etiology	Unexplained uterine bleeding; infertility
Associated conditions	Obesity

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Hysteroscopy may be performed for diagnosis or treatment of intrauterine pathology.

Cardiovascular	As patient may be undergoing hysteroscopy for uterine bleeding, consider hypovolemia and anemia. Tests: As indicated from H&P
Laboratory	Hgb/Hct if bleeding Hx. Other tests, as indicated from H&P; consider pregnancy testing.
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg po. Consider aprepitant 40 mg po for GA patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Local, regional, or GA may be used. If spinal anesthesia is indicated, pencil-point spinal needles (e.g., Sprotte or Whitacre) should be used to decrease the incidence of PDPH.

Local anesthesia

Some procedures may be done under local, usually with supplemental iv sedation, especially if they are diagnostic. Paracervical block has the potential for an inadvertent intravenous administration and local anesthetic toxicity.

Regional anesthesia

A T10 sensory level is sufficient to provide anesthesia for these procedures.

Spinal	0.75% bupivacaine 7.5–10 mg in 8.25% dextrose ± 10–20 mcg fentanyl (See Anesthetic Considerations for Cesarean Section.)
Epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL (1:200,000), 15–20 mL; supplemental iv sedation (See Anesthetic Considerations for Cesarean Section.)
CSE	Combined spinal/epidural (CSE): An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia. Apply monitors, administer fluid, and position patient as for spinal or epidural. The most common technique is the needle-through-needle. After the epidural space is located with a standard 17-ga Tuohy needle, insert a 4-inch 26- to 27-ga pencil-point spinal needle through it to administer 7.5–12 mg of spinal bupivacaine (0.75%) ± fentanyl 10–20 mcg and ± morphine sulfate 0.1–0.2 mg. Secure the epidural catheter and use if needed (a test dose is advisable).

General anesthesia

Induction	Standard induction. Supraglottic airway may be appropriate.	
ERAS	Verify preprocedure antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	Standard emergence. PONV prophylaxis may include dexamethasone and ondansetron.	
Blood and fluid requirements	IV: 18–20 ga × 1 BSS @ 2–4 mL/kg/h	Minimal blood loss. See Restricted Fluid Management, Appendix K, p. K-4 .
Positioning	✓ hip, leg, and hand positions ✓ and pad pressure points ✓ eyes Shoulder abduction <90°	Lithotomy position can be deleterious to pulmonary function, as it may impair respiratory mechanics. Common peroneal nerve palsy is possible if pressure on the nerve over the fibula is not prevented by adequate padding or positioning. Hyperflexion of hip joint can cause femoral and lateral femoral cutaneous nerve palsy. Obturator and saphenous nerve injuries are also complications of the lithotomy position.
Monitoring	Standard monitors	
Surgical stimulation	When cervix is grasped and dilated, patient may have excessive vagal nerve stimulation.	Rx: prompt cessation of stimulation by surgeons and treatment with glycopyrrolate (0.1–0.2 mg) or atropine (0.2–0.4 mg)
Complications	Air embolism Pulmonary and cerebral edema	Air embolism can occur with the use of gas distension media; thus, most institutions

	(2° hypotonic fluid overload) Coagulopathy Anaphylactoid reactions	currently use low-viscosity fluids for distension. Low-viscosity media can be grouped as follows, based on their tonicity and electrolyte content: (a) hypotonic, electrolyte-free media that may cause hypotonic fluid overload, and (b) isotonic, electrolyte-containing media that may cause isotonic fluid overload. It is not uncommon to use 10–20 L of distension fluid during an operative hysteroscopy. Monitor for fluid balance. Adjust iv fluid to maintain euvolemia.
	Finger injury (2° positioning)	When manipulating the foot of the bed, avoid finger injury by placing patient's arms on arm boards or by wrapping her hands.

POSTOPERATIVE

Complications	PONV Fluid overload/pulmonary edema from excessive fluid absorption	
Multimodal pain management	Oral analgesics Acetaminophen ± NSAIDs	Patients usually do not require opiate medications.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early enteral nutrition and mobilization.
Tests	Consider CXR and	For patient with respiratory

Suggested Readings

1. AAGL Advancing Minimally Invasive Gynecology Worldwide; Munro MG, Storz K et al: AAGL practice report: practice guidelines for the management of hysteroscopic distending media: (Replaces Hysteroscopic Fluid Monitoring Guidelines. J An Assoc Gynecol Laparosc. 2000;7:167-168.). J Minim Invasive Gynecol 2013; 20(2):37–48.
2. Chapman JS, Roddy E, Ueda S, et al: Enhanced recovery pathways for improving outcomes after minimally invasive gynecologic oncology surgery. Obstet Gynecol 2016; 128(1):138–44.
3. Cooper JM, Brady RM: Intraoperative and early postoperative complications of operative hysteroscopy. Obstet Gynecol Clin North Am 2000; 27(2):247–66.

PELVIC LAPAROTOMY

SURGICAL CONSIDERATIONS

Description

Laparotomy and its variants are all common gynecologic procedures. Laparotomy for benign gynecologic procedures is most frequently performed via a Pfannenstiel incision, which is often used for a cesarean section, thus is familiar for the gynecologic surgeon. A vertical incision is used in oncologic surgery or in the presence of a large uterus or mass. A knife or electrocautery (Bovie) is used to cut through the skin and underlying tissue until the rectus fascia is reached. The fascia is incised and then sharply extended bilaterally 2–4 inches with scissors or Bovie. The rectus muscle is separated sharply in the midline down to the pubis, and the peritoneum is entered. The peritoneal incision is then extended vertically and/or transversely. The pelvis and entire abdominal cavity are explored first by palpation. Then the bowels are usually packed in a cephalad direction with surgical laparotomy sponges (laps) to prevent them from falling back into the pelvis. Good muscle relaxation is important during this stage to ensure optimal packing. A self-retaining retractor is frequently used to keep the lap pads in place and to enhance exposure. After the desired operation has been performed, the retractor and packs are removed. During the peritoneal closure, abdominal muscle relaxation is again important to minimize tension on this layer and risk of bowel injury with the needle. The rectus fascia, the subcutaneous tissue, and the skin are closed in succession.

Variant procedures

Myomectomies are performed to remove myomata that are causing abnormal bleeding, bulk symptoms, infertility, and sometimes pain. Myomata are heavily vascularized at the base, and the surgeon has several ways to minimize this bleeding. A tourniquet can be placed across the uterine vasculature to minimize blood flow to the uterus. A common alternative is the use of a vasoconstrictor, such as a diluted epinephrine (1:200,000) or vasopressin (1–5 U/10 mL NS) solution. The solution (2–10 mL) is injected around the myoma before incising the uterus, which invariably $\rightarrow \uparrow$ HR (epi) or $\downarrow$ HR (vasopressin) and $\uparrow$ BP. Gonadotropin-releasing hormone (GnRH) agonists may be used for a few months before the operation to render the patient hypoestrogenic and, thus, decrease the vascularity of the myomata. After the myomata have been removed, the uterine defects within the myometrium are closed with several layers of suture, and the uterine serosa is then closed.

Ovarian cystectomies are performed to alleviate related pain and associated bulk, also to provide a diagnosis of asymptomatic cysts. Small, single-functional ovarian cysts are found at different stages in a woman's menstrual cycle. At times, these cysts can increase in size and quantity, which may cause severe pain. The ovary can also contain various nonfunctional cysts (e.g., tumors), which should be removed, even if asymptomatic, to exclude malignancy and prevent ovarian torsion or rupture, which may require more urgent surgical intervention. After the pelvic structures are well visualized, the cystic ovary is stabilized with instruments or surgical laps. Sharp or blunt dissection is used to shell out the cyst intact. If there is any suspicion about the nature of the cyst, an intraoperative frozen section is obtained. The ovary is then reapproximated, and the abdomen closed. If the cyst is large and little healthy ovarian stroma remains, or if there is uncontrollable bleeding, an **oophorectomy** is performed.

Ectopic pregnancies are usually surgical emergencies. Laparoscopy is the most common approach, although a laparotomy may be utilized if the patient is unstable from internal hemorrhage. The abdomen is entered, and the pregnancy located quickly. An attempt is made to control the bleeding with the surgeon's hand, clamp, or suture. Frequently, large amounts of blood in the pelvis are suctioned, and the ectopic pregnancy can be removed via a **partial tubal resection**, **salpingostomy**, or **salpingectomy**; then the abdomen is closed. A **D&C** may be performed at the end to prevent late bleeding from the pregnancy-induced endometrial proliferation.

An **abdominal, laparoscopic, or robotic colpopexy** (fixation of vagina) is utilized to repair vaginal vault prolapse after hysterectomy. The patient is placed in the lithotomy position, usually with Allen stirrups, to perform an examination under anesthesia. A

urethral catheter is initially inserted. Typically, a synthetic graft is used. The bowel is packed, and the defect is examined from the abdominal perspective. Frequently, there is an accompanying enterocele that must be closed initially (see **Moschcowitz procedure**, below). The peritoneum over the vaginal apex is then entered, and the neighboring rectum and bladder are dissected a distance away from the vaginal apex. The peritoneum over the sacrum is incised, and the cephalad end of the graft is sutured to the anterior sacral ligament of the third vertebrae. Severe bleeding can occur if there is injury to the middle sacral artery. The caudal end of the graft is attached to the vaginal apex. The peritoneum and abdomen are closed, and the patient's legs are elevated to the dorsal lithotomy position for a high-posterior **colporrhaphy** (repair of vagina).

The **Moschcowitz procedure** is used to reduce enteroceles via an abdominal approach. The abdomen is entered in the usual fashion, the uterus held up with a traction suture, the bowel packed, and the patient placed in the Trendelenburg position. Multiple concentric purse-string sutures are used to close the defect in the pouch of Douglas, and the abdomen is closed. A McCall procedure can also be performed to prevent enterocele formation. Here, the uterosacral ligaments are plicated and sutured together in the midline, at the level of the pouch of Douglas, to obliterate that space.

A **presacral neurectomy** is an operation performed for women with severe dysmenorrhea and chronic midline pelvic pain. Many patients have had previous surgery and have a diagnosis of endometriosis. The procedure is usually performed laparoscopically. The rectosigmoid is brought over to the left to make a vertical, posterior peritoneal incision over the sacral area. The anatomy is examined closely, and the superior hypogastric plexus is dissected out and obliterated or excised; then the peritoneum and abdomen are closed. Severe bleeding can be seen intraop from the hemorrhoidal and sacral veins but can usually be controlled with pressure.

Usual preop diagnosis

Myomata (pelvic pain, menorrhagia, pelvic pressure symptoms, infertility); ovarian cysts and pelvic pain with undiagnosed ovarian mass; ectopic pregnancy; vaginal vault prolapse; enterocele; presacral neurectomy (chronic pelvic pain)

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Supine or lithotomy
Incision	Pfannenstiel or low midline abdominal laparoscopic or robotic incisions
Unique considerations	Muscle relaxation is important during bowel packing and abdominal closure. Vasoconstrictor solutions for controlling myomata bleeding (1:200,000 epinephrine and 1–5 U vasopressin/10 mL NS) are often used and can alter BP and HR.
Antibiotics	Cefoxitin 1–2 g iv
Surgical time	45 min–4 h Abdominal colpopexy: 4–5 h
EBL	50–1000 mL (maximum related to procedure)
Postop care	PACU
Mortality	Minimal
Morbidity	Gastric dilatation: 3% Thrombophlebitis: 3% PE: 2% Ureteral stenosis: 1%
Pain score	8

PATIENT POPULATION CHARACTERISTICS					
	Myomectomy	Ovarian Cystectomy, Oophorectomy	Ectopic Pregnancy	Abdominal Colpopexy, Moschcowitz	Presacral Neurectomy
Age range	20–45 yr	20–85 yr	15–45 yr	40–80 yr	20–50 yr
Incidence	>1/5 females	⇐	1/100 females	1/500 females	⇐
Etiology	Congenital	Endometriosis Anovulation Adenoma	Preexisting tubal disease	Multiparous Obesity Chronic cough Older age group	Endometriosis
Associated conditions	Menorrhagia and pelvic symptoms associated with bulk	Endometriosis	Pelvic adhesions	Pelvic relaxation	Endometriosis

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Infertility Operations.

Suggested Readings

1. Allen RH, Micks E, Edelman A: Pain relief for obstetric and gynecologic ambulatory procedures. *Obstet Gynecol Clin North Am* 2013; 40(4):625–45.
2. Nelson G, Dowdy SC, Lasala J, et al: Enhanced recovery after surgery (ERAS) in gynecologic oncology—practical considerations for program development. *Gynecol Oncol* 2017; 147(3):617–20.
3. Verheijen R, Zweemer R: Robotic surgery for gynaecologic cancer: an overview. *Curr Oncol Rep* 2012; 14(6):544–9.

TRANSVAGINAL OOCYTE RETRIEVAL

SURGICAL CONSIDERATIONS

Description

Transvaginal oocyte retrieval (TVOR) is performed on patients who have undergone ovarian stimulation using ovulation-inducing agents. This procedure is performed 35–36 h after the patient has been injected with human chorionic gonadotropin (HCG) and/or leuprolide acetate to induce oocyte maturation. It is crucial for the success of the procedure that the retrieval is performed within this time period. If a retrieval is

performed too late, ovulation will have occurred, and oocyte retrieval will no longer be possible.

The patient should empty her bladder just before the procedure to facilitate visualization with transvaginal ultrasound. In the procedure room, the patient is placed in the dorsal lithotomy position, and moderate to heavy iv sedation is started. A sterile speculum is inserted into the vagina, and a vaginal prep is performed with sterile saline irrigation as other bactericidal preparations may also be harmful to the oocytes. Next, an ultrasound probe with a 16- to 17-ga needle on a needle guide is placed into the patient’s vagina. One of the ovaries is identified and entered by inserting the needle through the vaginal fornix. Patients may experience a combination of pain and pressure at this point in the procedure. After the needle is in the ovary, the surgeon will then proceed with sequential aspiration of the ovarian follicles. It is important that the patient remain relaxed and motionless as movement may prevent aspiration of oocytes and increase the risk of injury to the surrounding organs and vessels. After retrieval is completed in the first ovary, the needle is withdrawn; the other ovary is identified, and a second puncture is made through the vaginal fornix. Procedure duration is 10–30 min, depending on the number of follicles present. After all of the follicles have been aspirated, the needle and ultrasound probe are removed. A sterile speculum is then reintroduced into the vagina, and the vaginal wall and cervix are inspected for hemostasis. Pressure is applied for 3–5 min if bleeding is noted. The patient is then recovered for 30–60 min before going home. If the patient is undergoing an embryo transfer within the same menstrual cycle (fresh embryo transfer), NSAIDs are contraindicated intra- or postoperatively. If the patient is planning to cryopreserve oocytes or embryos for later use, and hemostasis is adequate, NSAIDs are appropriate.

ERAS: Preop counseling by the surgeon/anesthesiologist should include limiting NPO duration, alcohol, and smoking cessation, carb-loading and postop pain management options. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy
Incision	Vaginal punctures (usually 2)
Special instrumentation	Ultrasound probe with needle guide, 16- to 17-ga needle
Unique considerations	Requires complete relaxation/sedation to prevent injury and loss of oocyte during follicle aspiration

Antibiotics	Cefotetan or cefazolin 2 g iv before incision
Procedure time	Setup: 5–10 min Procedure: 10–30 min (depending on number of follicles)
Postop care	PACU→home
Morbidity	Bleeding: rare Infection: rare
Pain score	1–3 (abdominal cramping)

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–50 yr (rare after age 43)
Incidence	It is estimated that at least 10–15% of American couples of reproductive age who desire pregnancy are unable to conceive within 1 yr.
Etiology	A male factor is responsible for 1/3 of cases and female factor 1/3, and 1/3 are a combination or unexplained.
Associated conditions	Thyroid disorders; polycystic ovarian syndrome; endometriosis; tubal disease; depression

Suggested Reading

1. Rose BI: Approaches to oocyte retrieval for advanced reproductive technology cycles planning to utilize in vitro maturation: a review of the many choices to be made. J Assist Reprod Genet 2014; 31(11):1409–19.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

This is generally a healthy patient population. Little is required beyond routine tests unless otherwise indicated.

Laboratory	Tests as indicated from H&P
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg. Consider aprepitant 40 mg po for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Moderate to heavy IV sedation with propofol effective and most commonly used technique for IVF. Local anesthetic and mild to moderate sedation with midazolam +/- fentanyl an alternative if propofol sedation contraindicated. The most stimulating parts of the procedure occur when the vaginal fornix is pierced on each side and when the ovarian follicles are entered for aspiration of the eggs.

MAC	Propofol infusion (50–200 mcg/kg/min), ± midazolam (1–2 mg iv), and ± fentanyl titrated to achieve the desired combination of moderate to heavy sedation and analgesia.	
Blood and fluid requirements	Minimal blood loss IV: 20–22 ga × 1	See Restricted Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ hip, leg, and hand positions ✓ and pad pressure points	

▼ POSTOPERATIVE

Complications	PONV Bleeding: rare	PONV rare with propofol-based TIVA. Minimal postop pain → limit intraop opiates.
Multimodal pain management	Pain: usually minimal	Oral analgesics (e.g., acetaminophen, NSAIDs) are usually sufficient occasionally supplemented with fentanyl (25–50 mcg iv).
ERAS	PONV protocol (see	Encourage early enteral

Suggested Readings

1. Chapman JS, Roddy E, Ueda S, et al: Enhanced recovery pathways for improving outcomes after minimally invasive gynecologic oncology surgery. *Obstet Gynecol* 2016; 128(1):138–44.
2. Hadimioglu N, Titiz TA, Dosemeci L, et al: Comparison of various sedation regimens for transvaginal oocyte retrieval. *Fertil Steril* 2002; 78(3):648–9.
3. Matsota P, Kaminioti E, Kostopanagiotou G: Anesthesia related toxic effects on in vitro fertilization outcome: burden of proof. *Biomed Res Int* 2015; 2015:475362.
4. Van de Velde M: Anesthesia for in-vitro fertilization. *Curr Opin Anaesthesiol* 2005; 18(4):428–30.

INFERTILITY OPERATIONS/ASSISTED REPRODUCTIVE TECHNOLOGIES

SURGICAL CONSIDERATIONS

Description

These operations all deal with reproductive problems. The general trend is to avoid laparotomies and to perform operations using outpatient laparoscopy and hysteroscopy techniques whenever possible.

Fimbrioplasty is used to repair distal fallopian tubal occlusion—a common cause for infertility—which is usually a consequence of pelvic inflammatory disease (PID). The operation may be performed by **pelvic laparotomy** or **laparoscopy**. A urethral catheter is inserted to empty the bladder, followed by the insertion of a transcervical uterine catheter for chromopertubation (dye injection). The abdomen is opened, and the pelvic structures are exposed. During the operation, microsurgical techniques are followed closely to minimize trauma. Meticulous hemostasis is important. A wound protector is often used instead of self-retaining retractors. The peritoneum and pelvic structures are kept moist with intermittent irrigation. **Salpingolysis** and **ovariolysis** may be performed microsurgically. After the adnexae have been freed, they are elevated by loosely packing the pouch of Douglas with insulated pads (plastic sheathed covered laps). **Chromopertubation** is then performed, and if occlusion is present, a new stoma is created using microsurgical instruments and sutures. The abdomen is then closed.

Tubal reanastomosis, performed to restore fertility, is very similar to fimbrioplasty, with microsurgical techniques followed diligently. After the tubal segments have been freed slightly from their underlying mesosalpinx, the occluded ends are cut, and chromopertubation is performed to ensure patency. After patency has been established, anastomosis is performed in two layers. The mesosalpinx is reapproximated to the tubal serosa and the abdomen closed.

The uterus is embryologically formed by the fusion of two paramesonephric tubes. At times, the fusion is incomplete, and a septated uterus or bicornuate uterus is formed. The malformed uterus is associated with an increased risk for miscarriages and preterm labor. **Metroplasty** is used to correct this condition. The **Strassman procedure** (extremely rare) for bicornuate uteri uses a standard pelvic laparotomy. Following uterine exposure, an incision is made on the medial side of each hemicorpus and carried down until the uterine cavity is entered. The edges are reapproximated to form a single uterus. Septated uteri are usually repaired via a hysteroscopic approach (see Hysteroscopy) with scissors or laser.

Proximal tubal cannulation is a procedure in which proximal tubal occlusion can be repaired through either fluoroscopic or hysteroscopic approach. The hysteroscopic approach, usually performed under GA, allows the surgeon to insert a small cannula to restore tube patency. This procedure is often done with laparoscopy to follow the progress of the cannulization and to visualize the chromopertubation (see Hysteroscopy and Laparoscopy).

Gamete intrafallopian transfer (GIFT), **zygote intrafallopian transfer (ZIFT)**, and **tubal embryo transfer (TET)** are methods of **assisted reproductive technology (ART)** but are **seldom used**. Couples who have undergone extensive infertility workups and treatment without success may become candidates for GIFT, ZIFT, or TET procedures. Ovarian follicles are stimulated to grow with the help of gonadotropins. These follicles are then punctured with a needle transvaginally to “harvest” the eggs. These eggs can be mixed with semen and placed directly into the distal end of the fallopian tube (GIFT) using laparoscopic techniques and a small tubal catheter (see Laparoscopy). In the ZIFT and TET procedures, the semen and eggs are allowed to incubate 1–2 d in vitro; embryos form and are transferred to the fallopian tubes like GIFT procedure. Most patients undergoing ART procedures utilize conventional in vitro fertilization (IVF) and do not need laparoscopy.

Usual preop diagnosis

Infertility; history of recurrent pregnancy loss and preterm labor

ERAS: For elective procedures, preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol, and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

(For summaries of specific procedures, see Laparoscopy; Hysteroscopy; or Pelvic Laparotomy.)

■ PATIENT POPULATION CHARACTERISTICS

Age range	18–50 yr
Incidence	1/20 women
Etiology	PID; endometriosis; idiopathic
Associated conditions	Obesity

ANESTHETIC CONSIDERATIONS

(Procedures covered: pelvic laparotomy for myomectomy; ovarian cystectomy; oophorectomy; ectopic pregnancy removal; abdominal colpopexy; Moschcowitz enterocele repair; presacral neurectomy; infertility operations.)

PREOPERATIVE

This is generally a healthy patient population; however, this procedure can be performed for a wide variety of pathologic conditions.

Cardiovascular	Patients undergoing myomectomy and, especially, ectopic pregnancy removal may have had a significant amount of preop bleeding; therefore, assess for hypovolemia. Preexisting comorbidities and preop status will vary considerably. Assess as appropriate. Tests: As indicated from H&P
Laboratory	Hgb/Hct. Consider T&S/T&C. Patients with ectopic

	pregnancy may have urine/serum pregnancy tests as well as pelvic ultrasound.
Premedication	Patients with ruptured ectopic pregnancies may come to the OR urgently and should be treated with full stomach precautions.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or epidural block/analgesia for postop pain management. Consider preop acetaminophen 1000 mg, ± gabapentin 300 mg, ± celecoxib 200 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA is indicated for patients undergoing laparoscopic surgery and in patients presenting for emergency surgery. Neuraxial anesthesia is best avoided in hemodynamically unstable patients (i.e., ectopic pregnancies) and for laparoscopy where breathing difficulty may develop 2° to pneumoperitoneum and Trendelenburg position. Regional anesthesia may be suitable for simple pelvic laparotomies. In the younger patient population, spinal anesthesia is less desirable because of an increased incidence of postdural puncture headache (PDPH). If spinal anesthesia is indicated, a pencil-point needle (e.g., Whitacre, Sprotte) should be used to decrease the incidence of PDPH.

General anesthesia

Induction	Standard induction: In a patient with a ruptured ectopic pregnancy, bleeding, and hemodynamic instability, consider ketamine 1–2 mg/kg or etomidate 0.1–0.4 mg/kg. These patients are generally considered unfasted and would require rapid-sequence induction. Use OGT/NGT to decompress the stomach after induction for laparoscopic surgery.
ERAS	Verify preincision antibiotics. Assess volume status

	(see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Consider PONV prophylaxis.
Emergence	Standard emergence. Discuss with surgeon local wound infiltration. Consider iv acetaminophen if po dose given >6 h ago. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

A T4-T6 sensory level is recommended for pelvic/lower abdominal surgery.

Spinal	0.75% bupivacaine 12–14 mg in 8.25% dextrose (see Anesthetic Considerations for Cesarean Section) + 10–20 fentanyl ± 0.1–0.2 mg morphine	
Epidural + narcotic	1.5–2.0% lidocaine with epinephrine 5 mcg/mL, 15–20 mL; redose with 3–5 mL as needed. Provide supplemental iv sedation. (See Anesthetic Considerations for Cesarean Section)	
CSE	Combined spinal/epidural (CSE): An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia. Apply monitors, administer fluid, and position patient as for spinal or epidural. The most common technique is the needle-through-needle. After the epidural space is located with a standard 17-ga Tuohy needle, insert a 4-inch 26- to 27-ga pencil-point spinal needle through it to administer 12–14 mL of spinal bupivacaine (0.75%) ± fentanyl 10–20 mcg ± morphine sulfate 0.1–0.2 mg. Secure the epidural catheter and use if needed (a test dose is advisable).	
Blood and fluid requirements	Blood loss variable IV: 16–18 ga × 1–2 BSS @ 4–7 mL/kg/h	Blood loss will vary by procedure. Generally limited unless ruptured ectopic pregnancy or vascular complication of laparoscopy.

		See Goal-Directed Fluid Management, p. K-4 .
Control of blood loss	Epinephrine Vasopressin	During myomectomies, surgeons may inject vasopressors into the area surrounding myomata before excision. This may cause HTN, ↑ HR (epic), or ↓ HR (vasopressin).
Monitoring	Standard monitors ± Foley catheter ± Arterial catheter	Patients with ectopic pregnancies may need intra-arterial monitoring if major hemorrhage occurs.
Positioning	✓ and pad pressure points ✓ eyes	During abdominal colpopexy, patient is placed intermittently in both the lithotomy and supine positions. (See p. 895 and p. 899 for concerns regarding the lithotomy position.)
Complications	Pneumoperitoneum ↑ PaCO ₂ , ↓ PaO ₂ ETT migration	Pneumoperitoneum with CO ₂ insufflation and steep Trendelenburg position cause cephalad displacement of diaphragm with ↓ FRC, ↓ pulmonary compliance, and ↑ airway closure/atelectasis. Hypercarbia and hypoxia, due to respiratory compromise, can result unless ventilation is controlled during GA. Check for endobronchial migration of ETT upon assumption of Trendelenburg position.
	Pneumothorax	Pneumothorax due to retroperitoneal dissection of

	insufflated gas into the mediastinum can cause hypoxemia, ↑ airway pressure, subcutaneous emphysema, and ↓ BP.
↓BP	↓BP can result from ↓ venous return caused by pneumoperitoneum.
Hemorrhage	Hypotension may result from hemorrhage or rapid reversal of head-down position.
Dysrhythmias	
Peripheral nerve injury	
Venous CO ₂ embolism	Unintended intravascular injection of CO ₂ gas can → ↓ BP and dysrhythmias.

▼ POSTOPERATIVE

Complications	PONV Anemia Shoulder pain	
Multimodal pain management	Regional analgesia PCA	Emphasize multimodal approach combining nonopioid ± bolus/PCA opioid analgesics with epidural analgesia and/or TAP blocks (see Appendix K, p. K-2).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.
Tests	Hgb/Hct	If hemorrhage occurs

■ ANESTHETIC CONSIDERATIONS FOR ASSISTED

REPRODUCTIVE TECHNOLOGIES (ART)

PREOPERATIVE

This is generally a fit, healthy patient population. Little is required beyond routine tests unless otherwise indicated.

Laboratory	Tests as indicated from H&P
Premedication	Standard premedication
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider aprepitant 40 mg po for GA patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Conscious sedation, local anesthesia, and regional (e.g., spinal and epidural) and GA all have been employed for ART. The standard IVF technique consists of transvaginal egg retrieval and embryo transfer (not laparoscopic technique), and one study suggests higher pregnancy and delivery rates if iv sedation or epidural anesthesia is used as opposed to GA. Laparoscopy is only performed for GIFT, ZIFT, or TET, but not fo IVF. For TVOR, use moderate → heavy iv sedation. For laparoscopic procedures breathing difficulty can develop due to pneumoperitoneum and Trendelenburg position. Therefore, GETA with controlled ventilation is most commonly used. No difference in pregnancy rates with the use of isoflurane, propofol, N₂O, or midazolam has been demonstrated.

Induction	Standard induction. Avoidance of succinylcholine may decrease postop myalgia.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. N ₂ O does not appear to adversely affect success of fertilization.
Emergence	No special considerations. PONV prophylaxis should include dexamethasone and ondansetron.

Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1 BSS @ 2 mL/kg/h	Blood loss minimal, unless trauma to vasculature. Rarely, trauma to blood vessels or organs following laparoscopy may necessitate laparotomy. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Ventilatory	Pneumoperitoneum with CO ₂ and steep Trendelenburg position cause cephalad displacement of the diaphragm with ↓ FRC, ↓ pulmonary compliance, and ↑ airway closure/atelectasis. Hypercarbia and hypoxia, due to respiratory compromise, can result unless ventilation is controlled during GA. Check for endobronchial migration of ETT upon assumption of Trendelenburg position.
	Pneumothorax	Pneumothorax due to retroperitoneal dissection of insufflated gas into the mediastinum can cause hypoxemia, ↑ airway pressure, subcutaneous emphysema, and ↓ BP.
	Cardiovascular	↓ BP may result from ↓ venous return 2° pneumoperitoneum.
	Hemorrhage	Hemorrhage may result from

Venous CO₂ embolism
Dysrhythmias
Nerve injury
Bowel injury

blood vessel injury or rapid reversal of head-down position.
Unintended intravascular injection of CO₂ gas → ↓ BP and dysrhythmias.
Use of Trendelenburg position incurs risk of nerve injury.
Hyperextension of the arm may result in brachial plexus injury, and careful padding of vulnerable points is necessary.

▼ POSTOPERATIVE

Complications	Shoulder pain	Postop pain may be referred to the shoulder, due to irritation of the diaphragm by residual pneumoperitoneum or bleeding.
Multimodal pain management	Oral analgesics are usually sufficient.	NSAIDs or acetaminophen
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis.

Suggested Readings

1. Allen RH, Micks E, Edelman A: Pain relief for obstetric and gynecologic ambulatory procedures. *Obstet Gynecol Clin North Am* 2013; 40(4):625–45.
2. Beilin Y, Bodian C, Eisenkraft J, et al: The use of propofol, nitrous oxide, or isoflurane does not affect the reproductive success rate following gamete intrafallopian transfer (GIFT). *Anesthesiology* 1999; 90:36–41.
3. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018;

132(2):281–90.

4. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
5. Nelson G, Altman AD, Nick A, et al: Guidelines for pre- and intra-operative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part I. *Gynecol Oncol* 2016; 140(2):313–22.
5. Nelson G, Altman AD, Nick A, et al: Guidelines for postoperative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part II. *Gynecol Oncol* 2016; 140(2):323–32.
7. Nelson G, Dowdy SC, Lasala J, et al: Enhanced recovery after surgery (ERAS®) in gynecologic oncology—practical considerations for program development. *Gynecol Oncol* 2017; 147(3):617–20.

HYSTERECTOMY—VAGINAL, ABDOMINAL, LAPAROSCOPIC, OR ROBOTIC

SURGICAL CONSIDERATIONS

Description

Hysterectomy is one of the most commonly performed operations in the United States (650,000/yr). Several approaches are possible: vaginal, abdominal, laparoscopic, and robotic. The **vaginal approach**, performed with the patient in a dorsal lithotomy position, is preferred because it offers significantly less morbidity and mortality. Use of the vaginal approach may be limited by such factors as the pelvic bony architecture, excessive uterine size, pelvic adhesions, or the presence of gynecologic cancers, thus requiring an **abdominal or laparoscopic/robotic approach**. The approach may be changed in the OR, where a pelvic examination under anesthesia will determine the actual uterine size, degree of prolapse, and the presence of pelvic pathology. More recently, minimally invasive approaches with laparoscopy and robotic surgery have allowed for decreased morbidity and recovery time when compared to an abdominal approach. For a **total laparoscopic, laparoscopic-assisted, or robotic-assisted hysterectomy**, patients are placed in a low lithotomy position, and abdominal entry is confirmed with the laparoscope once the abdomen has been insufflated. The procedural steps are identical to the abdominal approach. For a laparoscopic or robotic-assisted hysterectomy, the patient is placed in a Trendelenburg position. Pelvic organ prolapse (POP) is the most frequent preoperative diagnosis in patients having a vaginal

hysterectomy. POP includes one or more of the following: prolapse of the uterus, intestine into the pouch of Douglas (enterocele), bladder into the anterior vaginal wall (cystocele), urethra into the anterior vaginal wall (urethrocele), and rectum into the posterior vaginal wall (rectocele). In these cases, the hysterectomy is often accompanied by an anterior/posterior **colporrhaphy**, **vault suspension**, and **perineoplasty**.

Variant approaches

Abdominal hysterectomy is performed through a Pfannenstiel or vertical midline incision, depending on the uterine size and the need to perform a **lymph node dissection** for cancer. A Pfannenstiel incision often can be extended with two types of muscle-splitting steps: the **Maylard**, in which the rectus muscles are cut, or a **Cherney rectus muscle detachment** performed at the pubic insertion. After entering the abdomen, a self-retaining retractor is placed, and the round, ovarian, and broad ligaments are clamped, cut, and tied, in that order. The uterine vessels are identified and ligated, then ligation of the uterosacral and cardinal ligaments is performed. The vagina is entered and the cervix removed. Then, the vaginal cuff is closed in a way that should incorporate the uterosacral ligaments for pelvic support. The visceral peritoneum is reapproximated, the retractors are removed, and the abdominal layers are closed. During a **supracervical hysterectomy**, the uterine corpus is detached from the cervix, and the vagina is not entered.

In a **vaginal hysterectomy**, the cervix is retracted, a paracervical incision is made, and the anterior and posterior cul-de-sacs are entered ([Fig. 8.2-4](#)). The uterosacral and cardinal ligaments and the uterine vessels are cut and ligated. With steady downward traction, the broad ligament is ligated in a stepwise manner until either the ovarian or infundibulopelvic ligament is reached, and one of the two is ligated, depending on whether the ovaries are to be removed or not. After the uterus has been removed, closure of the vaginal cuff follows, which often includes the uterosacral and cardinal ligaments for support. A vaginal pack often is left in place. **Laparoscopic-assisted vaginal hysterectomy (LAVH)** is sometimes performed to facilitate a unilateral or bilateral salpingo-oophorectomy (USO or BSO) or perform any adnexal surgery by ensuring good visualization of the infundibulopelvic and utero-ovarian ligaments as well as ovarian vessels. Once the abdomen is entered, the ureters are identified as they cross the iliac vessels at the level of the pelvic brim. The infundibulopelvic or utero-ovarian ligaments are then coagulated and cut. The remainder of the procedure is performed vaginally, including entry into the peritoneum, ligation of the ovarian vessels, and suturing of the vaginal cuff. Once the vaginal cuff is closed, and pneumoperitoneum achieved, the surgeon can look into the abdomen through the laparoscope to ensure

hemostasis at the end of the case.

For a **total laparoscopic hysterectomy (TLH)**, with or without robotic assistance, the entire procedure is performed laparoscopically. Some surgeons may prefer to suture the cuff vaginally. The uterine specimen is removed vaginally, either intact or after morcellation.

Usual preop diagnosis

Uterine myoma; pelvic organ prolapse; pelvic pain 2° endometriosis or adhesions; uncontrolled uterine bleeding/dysmenorrhea; endometrial hyperplasia; gynecologic cancers; pelvic masses

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol, and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

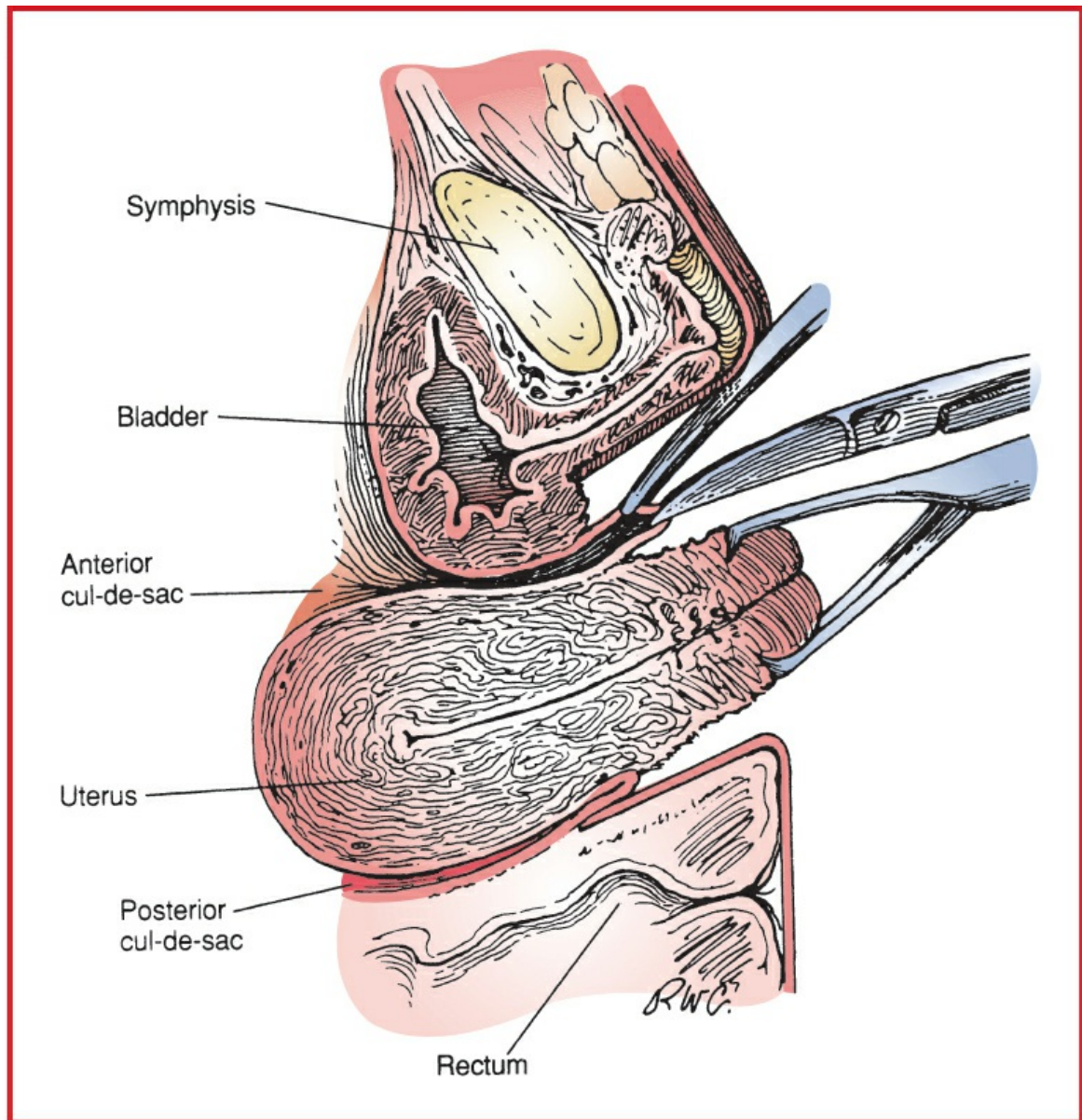


Figure 8.2-4 Surgical anatomy for vaginal hysterectomy. (Reproduced with permission from Rock JA, Thompson JD, eds: TeLinde's Operative Gynecology, 8th edition. Lippincott-Raven, Philadelphia: 1997.)

■ Summary of Procedures

	Abdominal Approach	Vaginal Approach
Position	Supine	Lithotomy. Following induction, position patient so that the perineum is at the end of the operating table to ensure optimal surgical exposure.
Incision	Pfannenstiel, low vertical midline, or	Pericervical, vaginal

	laparoscopic. The Pfannenstiel incision can be extended with a Maylard muscle-splitting procedure or a Cherney rectus muscle detachment at pubic insertion.	
Special instrumentation	None	Stirrups
Unique considerations	None	To prevent peroneal nerve injury, the area of leg leaning against stirrup should be well cushioned. Often, vasoconstriction agents (1:200,000 epinephrine and vasopressin) are used to cut down perioperative vaginal cuff bleeding.
Antibiotics	Cefoxitin or cefotetan 2 g iv + clindamycin 600 mg iv	⇐
Surgical time	1–2 h	45 min–1.5 h
EBL	200–300 mL	100–200 mL
Postop care	PACU	⇐
Mortality	Overall (10,000 patients): 14.6	20
	<25 yr: 8.9	0
	25–34 yr: 4.7	0.9
	35–44 yr: 3.8	0.5
	45–54 yr: 6.5	2.7
	55–64 yr: 41.3	1.9
	65–74 yr: 93.0	18.3

	>75 yr: 255.8	56.8
Morbidity	Infection	⇐
	Unexplained fever: 10–20%	5–8%
	Pelvic infection: 3.2–10%	3.9–10%
	Wound infection: 4–8%	–
	Urinary tract infection: 1.1–5%	1.7–5%
	Femoral nerve injury: 11.6%	–
	Hemorrhage	⇐
	Intraop: 1–2%	0.7–2–5%
	Requiring transfusion: 2–12%	2–8.3%
	Unintended major procedures: 1.7%	5.1%
	Injury	⇐
	Bladder: 1–2%	0.5–1.5%
	Bowel: 0.1–0.5%	0.1–0.8%
	Ureter: 0.1–0.5%	0.05–0.1%
	Vesicovaginal fistula: 0.1–0.2%	⇐
	Thromboembolic events: 0.4–1.3%	0.62–1.7%
Pain score	5–8	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	30–80 yr
Incidence	>1/5 females; 650,000/yr in the United States
Etiology	Uterine myomata; endometriosis; uterine prolapse; uterine cancer; pelvic mass
Associated conditions	Stress urinary incontinence; obesity

Suggested Reading

1. Moen M: Hysterectomy for benign conditions of the uterus: total abdominal hysterectomy. *Obstet Gynecol Clin North Am* 2016; 43(3):431–40.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Although many patients presenting for this procedure are otherwise healthy, others may have advanced cancer.

Respiratory	CXR may be indicated to r/o pleural effusion or other lung pathology in cancer patients. Also, ABG, $\pm$ PFTs, may be indicated preop in patients with significant pulmonary involvement. Tests: As indicated from H&P
Cardiovascular	Patient may have blood loss from the primary problem. Bowel prep may cause dehydration and electrolyte abnormalities. Tests: As indicated from H&P
Hematologic	Hgb/Hct. Consider T&S/T&C.
Laboratory	Consider pregnancy testing. Other tests, as indicated from H&P
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of a transversus abdominis plane or epidural block/analgesia for postop pain management. Consider preop

acetaminophen 1000 mg, $\pm$ gabapentin 600 mg, $\pm$ celecoxib 200 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

◆ INTRAOPERATIVE

Anesthetic technique

Increasingly, these procedures are done laparoscopically or robotically. Thus, GA would be the preferred anesthetic technique. Spinal, epidural, or combined spinal-epidural (CSE) anesthesia may be appropriate for normovolemic patients undergoing simple hysterectomy through a Pfannenstiel incision or vaginal hysterectomy. If spinal anesthetic is indicated, a pencil-point needle (e.g., Sprotte or Whitacre) should be used to decrease the incidence of PDPH.

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance. Muscle relaxation is necessary if the procedure is performed abdominally. These patients have a high incidence of PONV; consider ketamine (0.5 mg/kg bolus + 5 mcg/kg/min infusion).
Emergence	Standard emergence. PONV prophylaxis should include dexamethasone and ondansetron.

Regional anesthesia

A T4-T6 sensory level is sufficient to provide anesthesia for procedures on the uterus.

Spinal	0.75% bupivacaine 10–15 mg in 8.25% dextrose $\pm$ fentanyl 10–20 mcg; $\pm$ morphine 0.1–0.2 mcg. (See Anesthetic Considerations for Cesarean Section.) Supplement with iv sedation.
Epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL (1:200,000), 15–20 mL; supplement with 3–5 mL as

	needed. Supplemental iv sedation. (See Anesthetic Considerations for Cesarean Section.) Supplement with iv sedation.
CSE	Combined spinal/epidural (CSE): An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia. Apply monitors, administer fluid, and position patient as for spinal or epidural. The most common technique is the needle-through-needle. After the epidural space is located with a standard 17-ga Tuohy needle, insert a 4-inch 26- to 27-ga pencil-point spinal needle through it to administer 0.6 mL of spinal bupivacaine (0.75%) ± fentanyl 10 mcg ± morphine sulfate 0.1–0.2 mg. Secure the epidural catheter and use if needed (a test dose is advisable).

General considerations

Blood and fluid requirements	IV: 18 ga × 1–2 BSS; warm fluids T&S/T&C patients at ↑ risk of bleeding	Consider ↑ blood loss in cancer patients, prior XRT, and prior abdominal surgery. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Vaginal, laparoscopic, and robotic hysterectomy: Minimal blood loss Abdominal hysterectomy: Moderate to heavy blood loss	See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Vaginal hysterectomy: Standard monitors	

	± Foley catheter Abdominal hysterectomy: Standard monitors Foley catheter ± Arterial line ± CVP line	Consider in patients with anticipated ↑ blood loss or major medical comorbidities.
Positioning	✓ and pad pressure points Shoulder abduction <90°	The lithotomy position has several considerations for safety. (See p. 895 and p. 899 for details.)
Complications	Vaginal hysterectomy: Cervical stimulation Epinephrine/vasopressin injection Abdominal hysterectomy: Blood loss Epinephrine/vasopressin injection	Vagal stimulation and ↓↓ HR may occur with cervical/peritoneal manipulation. Surgeon may inject epinephrine or vasopressin to decrease local bleeding → HTN and brady- or tachydysrhythmias.

▼ POSTOPERATIVE

Complications	PONV VTE Anemia	
Multimodal pain management	Regional analgesia PCA	Emphasize nonopiate analgesia. Epidural opiates if catheter to be used postop
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Tests	Hct CXR (if CVP catheter placed intraop)	CXR to evaluate central line placement and rule out pneumothorax
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Suggested Readings

1. Bauchat JR, Habib AS: Evidence-based anesthesia for major gynecologic surgery. *Anesthesiol Clin* 2015; 33(1):173–207.
2. Meyer LA, Lasala J, Iniesta MD, et al: Effect of an enhanced recovery after surgery program on opioid use and patient-reported outcomes. *Obstet Gynecol* 2018; 132(2):281–90.
3. Modesitt SC, Sarosiek BM, Trowbridge ER, et al: Enhanced recovery implementation in major gynecologic surgeries: effect of care standardization. *Obstet Gynecol* 2016; 128(3):457–66.
4. Nelson G, Altman AD, Nick A, et al: Guidelines for pre- and intra-operative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part I. *Gynecol Oncol* 2016; 140(2):313–22.
5. Nelson G, Altman AD, Nick A, et al: Guidelines for postoperative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part II. *Gynecol Oncol* 2016; 140(2):323–32.
5. Nelson G, Dowdy SC, Lasala J, et al: Enhanced recovery after surgery (ERAS®) in gynecologic oncology—practical considerations for program development. *Gynecol Oncol* 2017; 147(3):617–20.

OPERATIONS FOR PELVIC ORGAN PROLAPSE: ANTERIOR AND POSTERIOR COLPORRHAPHY, ENTEROCELE REPAIR, SACROSPINOUS LIGAMENT VAGINAL VAULT SUSPENSION, UTEROSACRAL LIGAMENT VAGINAL VAULT SUSPENSION, AND SACROCOLPOPEXY

SURGICAL CONSIDERATIONS

Description

Pelvic organ prolapse often includes coexisting anatomical hernias that can be reported

as symptomatic vaginal pressure or bulges (e.g., enterocele [Fig. 8.2-5C] and uterine prolapse). Cystocele (Fig. 8.2-5A) and rectocele (Fig. 8.2-5B) are prolapses (relaxations) of the anterior and posterior vaginal wall, respectively. Cystoceles are often symptomatic due to bladder protrusion past the introitus during straining. Often, this relaxation will allow the bladder neck to lose its important anatomical relationship to the urethra and the rest of the bladder. The result can be bothersome stress urinary incontinence and/or incomplete emptying for the patient (see Operations for Stress Urinary Incontinence). The rectocele often is experienced as a vaginal bulge during straining and tends to cause incomplete evacuation of stool. The goal of the colporrhaphy is to restore the original anatomy. Due to the frequent coexisting relaxations, a **posterior colporrhaphy** (vaginal repair), **enterocele repair**, and **vaginal hysterectomy** are frequently performed at the same time.

In an **anterior colporrhaphy**, the patient is placed in a high dorsal lithotomy position with the perineum at the end of the operating table for surgical access. The bladder is emptied, and a weighted speculum is inserted into the vagina. A **vaginal hysterectomy** is performed at this point if indicated (see Vaginal Hysterectomy). The extent of the urethrocytocele is determined manually, and the vaginal mucosa is grasped at its cephalic border with two clamps. From this point to the external urethral meatus, the mucosa is undermined with a vasoconstrictive solution epinephrine (5–10 mL, 1:200,000), phenylephrine (1:200,000), or vasopressin (1–5 U/10 mL NS). This decreases blood loss significantly and helps to determine the depth of the vaginal mucosa. The mucosa is cut over this undermined area, and, with the help of sharp and blunt dissection, the mucosa is dissected laterally from its underlying fascia. A series of fascial plication sutures are placed to reduce the cystourethrocele. The redundant mucosa is excised, and the edges are reapproximated.

Paravaginal repair is another procedure for repair of cystocele and can be performed via the abdomen or the vagina. The space between the bladder and pubic bone is entered, and the bladder is dissected off the pelvic sidewall, taking care to avoid injury to the obturator nerve and vessels. The arcus tendineus fascia pelvis is visualized, running from the inferior margin of the pubic symphysis posteriorly to the ischial spine. The surgeon places a hand in the vagina to elevate the lateral vaginal sulcus to the arcus tendineus fascia. Multiple fine sutures are placed to secure the paravaginal tissues to the arcus tendineus fascia pelvis. The same is done for the opposite pelvic sidewall if bilateral defects are present.

If necessary, a **posterior colporrhaphy** may be performed. A small portion of the perineum posterior to the introitus is removed initially. The vaginal mucosa over the

rectocele is undermined with vasoconstrictor fluid before incision, followed by dissection of the overlying mucosa in a manner nearly identical to anterior colporrhaphy. One or several layers of stitches are placed to plicate the pararectal fascia, allowing for reduction of the rectocele. Redundant mucosa is excised, and the edges are reapproximated. A vaginal pack is usually placed to minimize bleeding.

An **enterocele** often is first noticed during a posterior colporrhaphy procedure and is repaired during the posterior repair. The enterocele is well identified and dissected away from the surrounding tissue. Two or more parallel purse-string stitches are used to close the enterocele. The enterocele tissue distal to the purse-string closure is excised. To reduce the enterocele in an optimal fashion the intra-abdominal pressure has to be at a minimum.

Abdominal approach: An **abdominal sacrocolpopexy** or **sacrocolpexy** can be performed for vaginal vault or cervical stump prolapse. This procedure can be performed through a Pfannenstiel or midline incision or via the laparoscopic/robotic approach. After entering the abdomen, a self-retaining retractor is used to retract the bowels. The peritoneum over the sacral promontory is carefully incised longitudinally, and the sacral promontory is exposed. The peritoneum is opened along the right pararectal space down to the right uterosacral ligament and the vaginal cuff or cervical stump (**cervicopexy**). Careful attention is paid to visualization of the right ureter. An EEA sizer is introduced into the vagina, and the peritoneum is dissected off the anterior and posterior vagina. The bladder and rectum are also dissected from the vaginal vault. A “Y”-shaped polypropylene mesh graft is then introduced into the abdomen. One end of the mesh is attached to the vaginal cuff (cervix) and the opposite end attached to the sacral promontory, thereby elevating the vaginal apex. The peritoneum is then closed over the mesh.

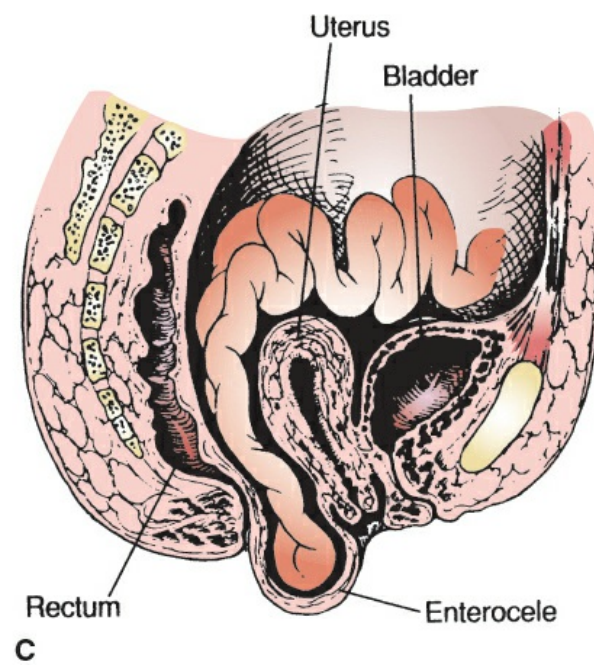
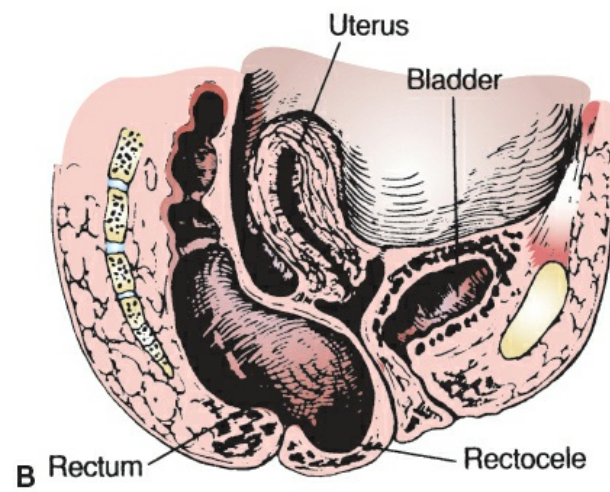
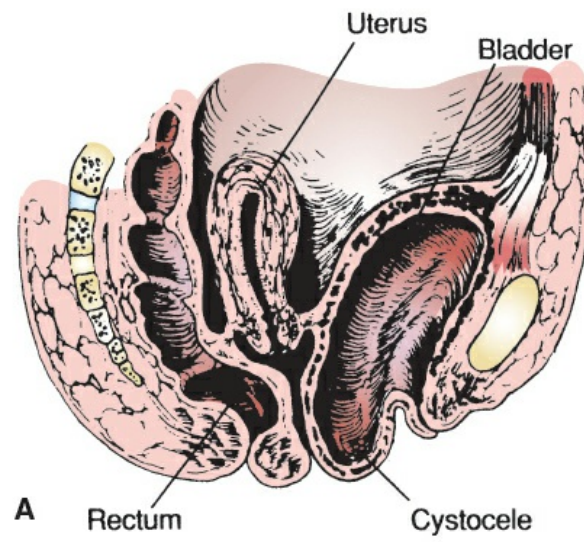


Figure 8.2-5 **A:** Anatomy of cystocele. **B:** Anatomy of rectocele. **C:** Anatomy of enterocele. (Republished with permission of McGraw-Hill Education from Pernoll ML, ed *Current Obstetrics and Gynecological Diagnosis and Treatment*, 7th edition. Appleton & Lange, Norwalk, Connecticut: 1991; permission conveyed through Copyright Clearance Center, Inc.)

During laparoscopy or a robotic-assisted abdominal sacrocolpopexy or cervicopexy, the patient is placed in steep Trendelenburg to aid in bowel retraction. The procedure is performed in the same fashion as an abdominal sacrocolpopexy. **Cystourethroscopy** is performed after the procedure to ensure ureteral and bladder integrity. A **midurethral urinary incontinence sling** is often performed after completion of the sacrocolpopexy to prevent stress urinary incontinence.

Variant procedure or approaches

Vaginal sacrospinous ligament suspension is an alternative to sacrocolpopexy. The patient is placed in a dorsal lithotomy position, and an examination under anesthesia is performed. A vasoconstrictive solution is injected (usually epinephrine 3–5 mL, 1:200,000) in the posterior vaginal wall. A vertical incision is made, and the mucosa is bluntly dissected off the rectum in an anterolateral direction. An enterocele, if found, is repaired at this time. The pararectal tissue is then bluntly pierced to enter the pararectal space. The anatomy surrounding the sacrospinous ligament is well palpated and, with the help of the special Miya hook, a large suture (see Operations for Pelvic Organ Prolapse) is placed into the sacrospinous ligament. The other end of the suture is placed at the apex of the vagina, which, after tying, is pulled in a lateral cephalad direction. The mucosa is finally closed.

Uterosacral ligament suspension, or **high McCall culdoplasty**, is an alternate procedure for vaginal apex support. After the uterus is removed by a **vaginal hysterectomy**, the uterosacral ligaments are identified. Two separate permanent sutures are placed along the uterosacral ligament as far cephalad (toward the sacrum) as possible. The same is done on the opposite side. An additional stitch of absorbable suture is placed at the vaginal corners bilaterally and carried through the uterosacral ligaments as well. A cystoscopy is then carried out with all four stitches placed on tension to r/o ureteral obstruction due to ligation or kinking of the ureters. The stitches are replaced if obstruction is observed. After ureteral patency has been confirmed, these stitches are sutured to the fibromuscular layer of the vaginal cuff and tied to bring the apex as cephalad as possible. The vaginal cuff is closed with absorbable sutures.

The **LeFort procedures**, **colpectomy**, and **colpocleisis**, are highly efficacious, minimally invasive operations performed in women with complete prolapse of the uterus and/or vagina, respectively, who do not desire to remain sexually active. With

the patient in a dorsal lithotomy position, the anterior and posterior vaginal mucosae are dissected from the underlying fascia, and then the anterior and posterior vaginal fasciae are plicated together, resulting in near-complete closure of the vagina. Advantages of these techniques are that they can be performed rapidly, they do not require general anesthesia, and most patients can be discharged from the hospital the following day. Closure procedures are ideal for patients who are elderly and/or have multiple medical conditions that make more invasive surgery risky.

Use of vaginal mesh: In 2008 and 2011, the U.S. Food and Drug Administration (FDA) issued safety communications warning doctors and consumers about an increase in adverse event reports related to mesh used for urogynecologic procedures. In 2016, the U.S. FDA reclassified surgical mesh for transvaginal POP repair as a class III (i.e. high-risk) device. While transvaginal synthetic mesh slings have become the standard of care for surgical treatment of stress urinary incontinence, concerns about the safety of synthetic mesh in treating POP have led to an overall decline in use of vaginal mesh.

Cystourethroscopy is performed after the procedure to ensure ureteral and bladder integrity.

Usual preop diagnosis

Symptomatic cystocele; symptomatic uterine prolapse; enterocele; rectocele

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol, and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, physiotherapy plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy
Incision	Vaginal mucosal; peritoneum for enterocele repair
Unique considerations	To prevent peroneal nerve injury, the area of the leg leaning against the stirrup should be well cushioned. Putting the legs in a high position increases venous return to the heart. Infiltration with epinephrine and vasopressin often is used to reduce intraop bleeding. This causes changes in CO, HR, and BP. Low intra-abdominal pressure (good muscle relaxation) is needed during the enterocele reduction.

Antibiotics	Cefoxitin or cefotetan 2 g iv	
Surgical time	45 min (anterior colporrhaphy) 30 min (posterior colporrhaphy/enterocele repair) 45 min (uterosacral ligament suspension) 1 h (paravaginal repair)	
Closing considerations	Suprapubic catheter (anterior colporrhaphy); vaginal pack (enterocele repair)	
EBL	20–500 mL	
Postop care	PACU	
Mortality	1%	
Morbidity	Anterior colporrhaphy Delayed voiding 1–7 d: 30% Foul vaginal discharge: 14% Bacteriuria: 13% Pyrexia (>100.4° F): 11% Atelectasis: 3% Delayed voiding >7 d: 0.6% Need for blood transfusion: 0.6% Urethrovaginal fistula: 0.4% Acute gastric dilatation: 0.2% Uterosacral ligament suspension Ureteral obstruction Perirectal abscess due to stitch in the	Vaginal sacrospinous suspension (Miya hook) Bleeding in gluteal and pudendal vessels Pararectal burning pain × 2 mo Sciatic pain (indicating misplaced sutures, which must be removed) Peritoneal tear (during Miya hook insertion) Paravaginal repair Obturator vessel laceration Obturator nerve injury Ureteral obstruction Bladder laceration

rectum

Pain score 5–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80 yr
Incidence	>1/5 women
Etiology	Multiparous state; obesity; chronic cough; previous pelvic surgery
Associated conditions	Pelvic relaxation syndrome

■ ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Operations for Stress Urinary Incontinence.

Suggested Readings

1. Iglesia CB, Smithling KR: Pelvic organ prolapse. *Am Fam Physician* 2017; 96(3):179–85.
2. Mowat A, Maher D, Baessler K, et al: Surgery for women with posterior compartment prolapse. *Cochrane Database Syst Rev* 2018; (3):CD012975.
3. Zwain O, Aoun J, Eisenstein D: Minimally invasive surgery in pelvic floor repair. *Curr Opin Obstet Gynecol* 2017; 29(4):276–81.

OPERATIONS FOR STRESS URINARY INCONTINENCE

■ SURGICAL CONSIDERATIONS

Description

Stress urinary incontinence is a common condition affecting mostly older and multiparous women. It is a disorder of the musculofascial support to the bladder neck, urethra, and pelvic floor. Many of these patients have had extensive preop workup to exclude urge or other types of urinary incontinence, and many have been treated with pelvic floor exercises (Kegel) before surgery. Many surgical approaches exist: **abdominal suspension** procedures, **suburethral slings**, and **transurethral injections of**

bulking agents. Evidence suggests that **midurethral slings** are the most successful procedures. Given that they can be performed with regional or even local anesthesia as outpatient procedures, midurethral slings are the most commonly used procedures to treat stress urinary incontinence. **Bladder neck slings** and injection of urethral bulking agents can also be used for those with contraindication to a midurethral sling. Patient positioning is crucial for all vaginal surgery.

Vaginal approaches

The **Kelly urethral plication** was historically used as the primary surgical treatment, especially when other vaginal surgeries are to be performed. The patient initially is placed in a high dorsal lithotomy position with the perineum at the end of the operating table for surgical exposure. The bladder is emptied, and a weighted speculum is inserted into the vagina. The extent of the cystourethrocele is determined and the vaginal mucosa is grasped at its cephalad border with two clamps. From this point to the external urethral meatus, the mucosa usually is undermined with 5–10 mL of a vasoconstrictive solution (epinephrine 1:200,000, phenylephrine 1:200,000, or vasopressin 1–5 U/10 mL NS). This decreases blood loss significantly and helps to determine the depth of the mucosa. With the help of sharp and blunt dissection, the mucosa is freed laterally from its underlying adherent fascia. Series of vertical mattress sutures are placed in the mobilized paraurethral and paravesical fascia to reduce the cystourethrocele and elevate the posterior urethra to a high retropubic position. The redundant mucosa is excised, and the edges are reapproximated. A suprapubic catheter is often inserted at the end of the surgery to prevent bladder overdistension. The Kelley urethral plication is rarely performed in modern practice and has been supplanted by the midurethral sling.

Two anterior bladder neck suspension techniques—Stamey and modified Pereyra—are very similar procedures wherein the vaginal mucosa is incised and dissected off the underlying paraurethral fascia, much the same way as in the Kelly plication. Instead of using a layer of mattress sutures, both suspension methods use two lateral sutures that suspend the vesical neck on each side (see [Fig. 9-22](#)). The ends of the sutures are tied over the rectus fascia to provide support. The Stamey method uses a small Dacron cuff to prevent the suture from tearing through the paravesical fascia, whereas in the modified Pereyra method, the posterior loop is attached firmly to the pubourethral ligament. One or two small suprapubic abdominal incisions must be made to allow for the tying of the sutures. Specialized long needles are used to help the placement of these sutures, and a cystoscope often is used to verify their placement. Perforation of the bladder is a common complication found upon cystoscopy. Finally, a suprapubic catheter is placed at the end of the operation. Stamey and Pereyra

suspensions are rarely performed and are mostly of historical significance as they have been replaced by midurethral urinary incontinence slings.

Abdominal approaches

Burch (urethropexy) and the **Marshall-Marchetti-Krantz (MMK)** are both abdominal suspension procedures, though most clinicians have abandoned the MMK procedure due to higher complication rates. The patient is placed in Allen stirrups in the frog-leg position with a urethral catheter in place. A Pfannenstiel incision is used to enter the space of Retzius, which lies between the parietal peritoneum and the rectus fascia under the pubic bone. Blunt dissection is used to open and extend this space. The surgeon then inserts two fingers into the vagina to raise the anterior vagina and bladder neck. This enables the surgeon to place two or more sutures in the tissue just lateral to the urethra and attach them to Cooper ligament (Burch) or to the periosteum of the pubic symphysis (MMK).

The **pubovaginal sling** and highlight sling are often reserved for women in whom other incontinence operations have failed. It can also be used as primary treatment for stress urinary incontinence, though it is more invasive than a midurethral sling. The goal of the sling procedure is to produce extrinsic compression of the urethrovesical junction with the help of a strip anchored to the rectus fascia or pubic bone (see [Fig. 9-22](#)). With the patient in the dorsal lithotomy position, a urethral catheter is placed and the vaginal mucosa incised and dissected off the underlying paravesical and paraurethral fascia, similar to the Kelly plication. The retropubic space is entered through a Pfannenstiel incision, and a strip of rectus fascia is obtained. The strip is then brought through the vagina, around the urethra, and back to the abdomen, where it is fastened to the rectus fascia, creating a sling under the urethra at the junction of the bladder neck. The vaginal and abdominal incisions are closed, and a suprapubic catheter is often placed. If donor fascia lata or synthetic material is used, a small (~2 inches) horizontal incision is made above the pubic bone for attachment of the sling. The patient's own fascia lata (from the lateral thigh) can also be used.

Midurethral sling placement may be performed under local, regional, or general anesthesia. Local anesthesia may be combined with conscious sedation. If local anesthesia is used, it will also be administered to the vaginal incision site as well as the groin areas or suprapubic areas where the trocar and mesh will exit.

Tension-free vaginal tape (TVT) sling is a newer sling procedure for the treatment of stress urinary incontinence. A polypropylene mesh sling is used to support the urethra, although it is not attached to the fascia. The mesh is woven in such a way that the sling does slide out over time. Fibroblasts grow into the sling to anchor it throughout the

endopelvic fascia and around the dependent surface of the urethra. The patient is placed in a high dorsal lithotomy position with the perineum at the end of the operating table for surgical exposure. The bladder is emptied and a weighted speculum or retractor is inserted into the vagina. The vaginal mucosa is grasped near the middle of the urethra with two clamps. The vaginal mucosa is hydrodissected with a small amount of vasoconstrictive solution (epinephrine 1:200,000, phenylephrine 1:200,000, vasopressin 1–5 U/10 mL NS). This decreases blood loss significantly and helps to determine the depth of the mucosa. With the help of sharp and blunt dissection, the mucosa is freed laterally from its underlying adherent fascia, and two midurethral submucosal tunnels are made toward the underside of the pubic symphysis. The TVT curved needle is introduced from the vagina through the space of Retzius on each side of the urethra and is brought out to the abdomen above the pubic bone. The sling mesh is attached to the ends of each needle; thus, when both needles are pulled through the abdominal sites, the mesh will rest under the urethra. A cystoscopy is done before pulling the mesh through to verify that neither needle is in the bladder or the urethra. The mesh is then adjusted so that the urethra is resting on the sling, under no tension. The **transobturator tape (TOT) sling** is a commonly used alternative to the TVT sling that does not enter the retropubic space. The sling is inserted under the midurethra and exits through the obturator membrane, inferior to the adductor longus tendon, through a poke incision lateral to the vagina in the groin. There are two variations: **inside-out**, where the trocars are passed from a midurethral vaginal incision to exit through bilateral groin incisions, and **outside-in**, where the trocars are passed from bilateral groin incisions to exit through a midurethral vaginal incision.

Usual preop diagnosis

Stress urinary incontinence; intrinsic sphincter deficiency

ERAS: Preop counseling by the surgeon/anesthesiologist should include no bowel prep, limiting NPO duration, alcohol, and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, postop recovery plan, and possibility of urinary catheter postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURES

	Kelly Plication	Midurethral Sling	MMK/Burch	Pubovaginal Sling
Position	Dorsal lithotomy	Lithotomy	Frog-leg	⇐
Incision	Vaginal mucosa	2 suprapubic and a vaginal incision	Pfannenstiel	Vaginal and suprapubic incision
Unique considerations	Cushion area of leg against stirrup to prevent peroneal injury. Epinephrine (1:200,000) or vasopressin is often infiltrated to reduce intraop bleeding for vaginal approaches. Injections may cause ↑ BP and tachydysrhythmias (epi) or bradydysrhythmias (vasopressin). The urethral catheter is frequently removed and reinserted during surgery.	⇐	⇐	⇐
Antibiotics	Cefoxitin 1–2 g iv	⇐	⇐	⇐
Surgical time	1 h	⇐	⇐	⇐
Closing considerations	Suprapubic catheter	⇐	⇐	⇐
EBL	50 mL	Roughly 50 mL	100–200 mL	50–100 mL
Postop care	Inpatient	Outpatient	⇐	± Outpatient
Mortality	Minimal	Minimal	⇐	⇐
Morbidity	Detrusor instability: Rare (due to vaginal approach) Enterocoele: Rare Incisional hernia: Rare Osteitis pubis: Rare Voiding difficulties: Rare Wound infection: Rare	<7% Rare ⇐ Rare <2% Rare	14% 15% 0.09% ⇐ 2% 8.7%	⇐ N/A Rare ⇐ ~40% Rare Prolonged catheter time: >50% Outlet obstruction: 10–20% Urethral perforation: 7% Bladder perforation: 1%
Pain score	4–6	1–3	6–8	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	40–80 yr
Incidence	>1/5 women
Etiology	Multiparous state; obesity; chronic cough
Associated conditions	Other components of pelvic relaxation syndrome (rectocele, uterine prolapse, enterocele)

Suggested Reading

1. Lukacz ES, Santiago-Lastra Y, Albo ME, et al: Urinary Incontinence in women: a review. JAMA 2017; 318(16):1592–604.

■ ANESTHETIC CONSIDERATIONS

(Procedures covered: anterior and posterior colporrhaphy; enterocele repair; vaginal sacrospinous suspension; operations for stress urinary incontinence.)

■ PREOPERATIVE

This is usually an older patient population, past childbearing age. Assess for medical comorbidities.

Laboratory	Hgb/Hct, as indicated from H&P
Premedication	Consider midazolam 1–2 mg iv.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of spinal or epidural block/analgesia for postop pain management. Consider preop acetaminophen 1000 mg. Consider aprepitant 40 mg po for GA patients with h/o severe PONV.

■ INTRAOPERATIVE

Anesthetic technique

Regional or GA may be used.

Regional anesthesia

A T10 sensory level is sufficient to provide anesthesia for procedures using a perineal approach, but a T4 level is recommended for abdominal/combined approaches.

Spinal	0.75% bupivacaine 10–15 mg in 8.25% dextrose (see Anesthetic Considerations for Cesarean Section) $\pm$ 10–20 mg fentanyl $\pm$ 0.1–0.2 mg morphine
Epidural	1.5–2.0% lidocaine with epinephrine 5 mcg/mL, 15–20 mL; supplement with 3–5 mL as needed. Supplemental iv sedation. (See Anesthetic Considerations for Cesarean Section) Spinal may provide better perineal block.
CSE	Combined spinal/epidural (CSE): An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia. Apply monitors, administer fluid, and position patient as for spinal or epidural. The most common technique is the needle-through-needle. After the epidural space is located with a standard 17-ga Tuohy needle, insert a 4-inch 26- to 27-ga pencil-point spinal needle through it to administer 0.6 mL of spinal bupivacaine (0.75%) $\pm$ fentanyl 10 mcg $\pm$ morphine sulfate 0.1–0.2 mg. Secure the epidural catheter and use if needed (a test dose is advisable).

General anesthesia

Induction	Standard induction
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance
Emergence	No special considerations. PONV prophylaxis should include dexamethasone and ondansetron.
Blood and fluid	IV: 18–20 ga $\times$ 1 Normally minimal blood loss.

requirements	BSS @ 2–4 mL/kg/h	See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Control of blood loss	Epinephrine, vasopressin, phenylephrine used by surgeons	Surgeons may inject vasopressors into the submucosa to minimize blood loss, which may cause ↑ BP and tachydysrhythmias (epi) or bradydysrhythmias (phenylephrine/vasopressin).
Monitoring	Standard monitors	Although bladder catheterization before incision is normal, the catheter is not left in place throughout surgery in those patients undergoing anterior and posterior colporrhaphy, enterocele repair, and Kelly urethral plication. Suprapubic bladder catheters are placed toward the end of surgery in these procedures, as well as in the Stamey, Pereyra, and urethral sling procedures.
Positioning	✓ and pad pressure points ✓ eyes	See p. 895 and p. 899 for concerns regarding the lithotomy position.

POSTOPERATIVE

Complications	PONV	
Multimodal pain management	Regional analgesia, IV opiates, ketorolac	Usually rapid conversion to po pain medications
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6)	Encourage early (within 24 h) enteral nutrition, Foley catheter removal, and

mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Tests	None indicated
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Suggested Readings

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CHAPTER 8.3

Obstetric Surgery

SURGEONS

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CESAREAN SECTION—LOWER SEGMENT AND CLASSIC

SURGICAL CONSIDERATIONS

Description

Cesarean section (C-section) is the delivery of the fetus through a horizontal or, less commonly, through a vertical incision in the lower uterine segment. The skin incision is made either as a Pfannenstiel's (transverse in the crease above the pubis), a Maylard (in extremely obese patients), or vertical midline from umbilicus to pubis. The peritoneal cavity is entered as in any laparotomy. A retractor is placed inferiorly, and the reflection of visceral peritoneum from the bladder dome to the anterior lower segment of the uterus (bladder flap) is incised and may be displaced inferiorly, along with the bladder. The uterus is entered sharply and the incision extended with digital pressure

and/or bandage scissors. The fetal head is elevated out of the pelvis and delivered through the uterine incision. In cases of nonvertex lie, the infant's breech or foot is grasped and brought out of the incision. After the delivery of the fetus, the cord is double-clamped and cut, and cord blood is obtained for analysis. The placenta is removed manually and the uterine cavity cleared of all debris and clots. The uterine incision is closed with a running, interlocking stitch, followed by a second imbricating layer. The bladder flap and parietal peritoneum do not require closure, and the rectus muscles do not routinely require reapproximation. Finally, the fascia is closed and the skin reapproximated with staples. **Classic C-section** usually involves a fundal vertical uterine incision ([Fig. 8.3-1](#)). Patients with a history of prior classical C-section should be delivered abdominally via a repeat C-section because their risk of uterine rupture with labor and vaginal delivery is ~12%.

Usual preop diagnosis

Failure to progress in labor; elective repeat C-section; fetal distress; malpresentation

ERAS: For elective procedures, preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, carb loading, DVT prophylaxis, postop pain management options, and postdelivery plan. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

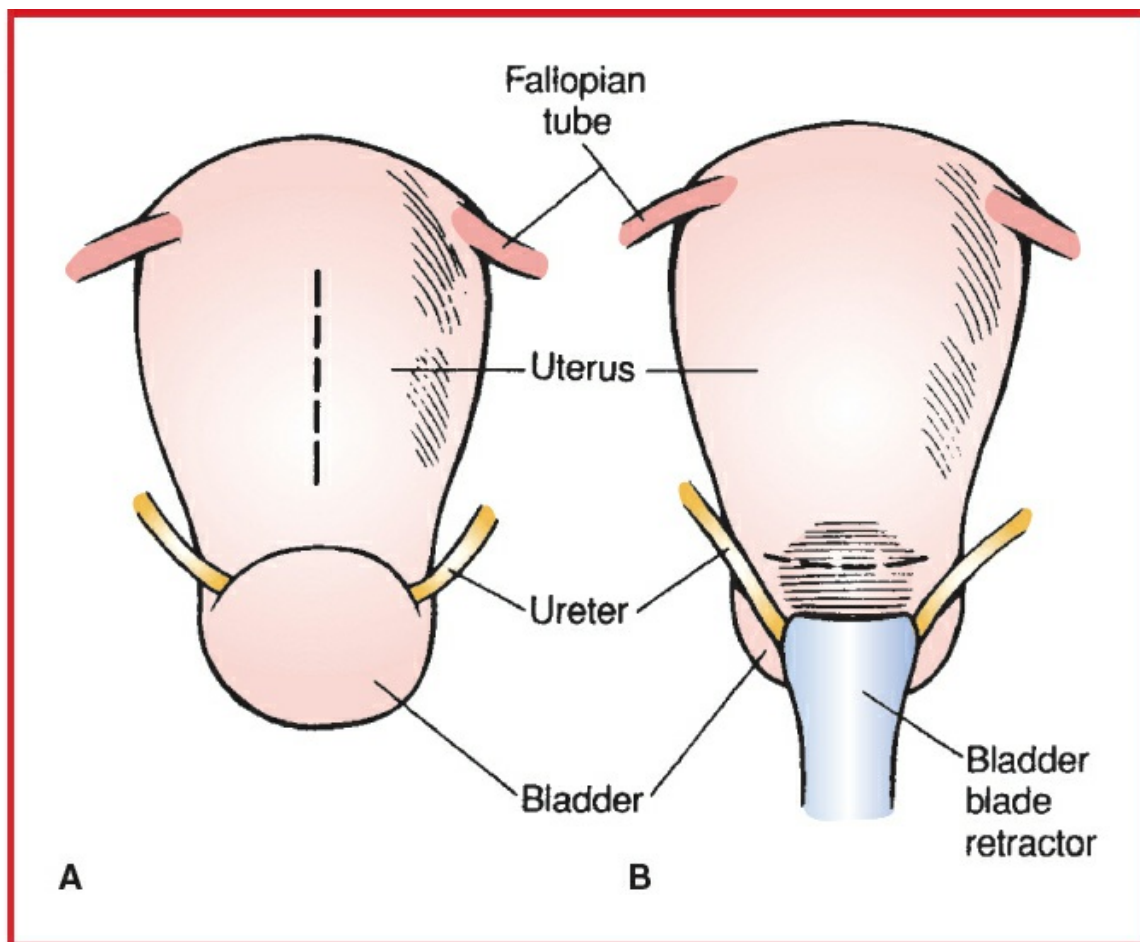


Figure 8.3-1 Typical C-section incisions. **A:** Classic incision, in the upper uterine segment. **B:** Low transverse incision.

■ SUMMARY OF PROCEDURES

	Lower-Segment C-Section	Classic C-Section
Position	Supine with left lateral tilt. (In obese patients, the pannus may be lifted superiorly by tape or towel clips or inferiorly for a supraumbilical incision.)	⇐
Incision	Skin: transverse low abdominal (Pfannenstiel's) or repeat vertical. Uterus: transverse (Kerr) or low vertical (for premature infants or nonvertex lie)	Skin: Pfannenstiel's or, more commonly, vertical midline Uterus: vertical fundal

Special instrumentation	Bladder blade retractor; small ring forceps; bandage scissors; suction bulb; DeLee suction trap (if meconium)	⇐
Unique considerations	✓ fetal heart tones before the procedure. If for CPD: vaginal exam within the last 15 min before the procedure. If for fetal distress: continuous monitoring until skin incision	⇐
Antibiotics	If in labor or membranes ruptured: continue prophylactic antibiotics or cefazolin 2 g iv (before incision).	⇐
Surgical time	20–90 min	40–90 min
Closing considerations	Low transverse: closed in two layers. Low vertical: two layers; may require additional operative time for control of incision bleeding or repair of incision extension into the cervix	Three-layer closure requires additional time.
EBL	750–1000 mL	1000–2000 mL
Postop care	Observation for bleeding and ↓ BP	Special attention to VS is needed due to additional blood loss.
Mortality	<0.1%	⇐
Morbidity	Infection: Not in labor: <5% In labor/ruptured membranes: 50% (antibiotics reduce incidence to 15%) Small bowel obstruction (SBO): rare	⇐ ⇐ ⇐
Pain score	1–5 (treated) 6–10 (untreated)	

■ PATIENT POPULATION CHARACTERISTICS

Age range	14–45 yr
Incidence	10–30%
Etiology	Failure to progress (30%); repeat C-section (30%); fetal anomaly/other (20%); abnormal presentation (10%); fetal distress (10%)
Associated conditions	Preeclampsia/eclampsia; DIC; hemolysis, elevated liver enzymes, low Plt count (HELLP) syndrome; obstetrical hemorrhage/shock; chorioamnionitis

■ ANESTHETIC CONSIDERATIONS

(Procedures covered: C-section; emergent obstetrical hysterectomy; repair of uterine rupture.)

■ PREOPERATIVE

In general, patients are young and healthy, although the pregnant patient has undergone profound physiologic changes that affect the administration of anesthesia. Women present for C-section for a multitude of reasons, including failure to progress in labor, repeat cesarean delivery, fetal indications including breech presentation, and certain medical indications, and for emergency cesarean delivery for nonreassuring fetal heart rate tracing and/or hemorrhage (placenta previa, abruptio placenta, and, rarely, uterine rupture).

Respiratory

The pregnant patient has a compensated respiratory alkalosis ($\text{PaCO}_2 = 32\text{--}34$), $\uparrow$ minute ventilation (MV; $\uparrow 50\%$), and $\downarrow$ FRC ($\downarrow 20\%$). $\uparrow$ O_2 consumption ($\uparrow 20\%$) with $\downarrow$ FRC results in rapid onset of hypoxemia if ventilation is compromised. Small airway closure due to elevation of the diaphragm (exaggerated by obesity and supine position) can $\rightarrow$ VQ mismatch and $\downarrow$ PaO_2 . $\uparrow$ MV and $\downarrow$ FRC enhance uptake of inhalational anesthetics. Mucosal capillary engorgement in the upper airways may necessitate a smaller ETT and mandates careful airway suctioning

	to avoid bleeding. Avoid any nasal airways. Airway changes can occur in labor and necessitate an airway evaluation just before administration of anesthesia. Tests: As indicated from H&P
Cardiovascular	Typically, there is a ↓ SVR (↓ 15%), ↓ diastolic pressure, and ↓ MAP (↓ 15%) with ↑ HR (↑ 20%) and ↑ CO (↑ 40%, higher in multifetal pregnancy, higher in labor). Use left uterine tilt (15°) to minimize aortocaval compression and supine hypotension. Immediately postpartum, 600–800 mL blood enters the central circulation, due to placental autotransfusion, with further ↑ in CO. Tests: As indicated from H&P
Hematologic	These patients have ↑ red cell mass (↑ 300–400 mL), ↑ plasma volume (↑ 1200–1300 mL), ↑ blood volume (↑ 1500–1600 mL), ↑ more with multifetal pregnancy. WBC count may ↑ to 15,000/mm³. Iron deficiency anemia often is superimposed on the dilutional anemia of pregnancy (Hct 33%). The typical blood loss of 700–1000 mL is usually well tolerated. Excessive blood loss is possible with uterine atony, placenta previa, and abruptio placenta. Repeat C-section associated with placenta previa poses high risk for hemorrhage because of increased risk for associated abnormal placental implantation. Tests: Hgb/Hct
Gastrointestinal	Abnormalities, including ↑ intragastric pressure, ↓ esophageal sphincter tone, and ↓ gastric motility (after onset of labor), predispose to aspiration pneumonitis. All parturients should be considered to have full stomachs and should receive nonparticulate antacid (e.g., 0.3 M Na citrate 30 mL) iv metoclopramide 10 mg and ranitidine 150 mg or famotidine 20 mg before general anesthesia.
Hepatic	Liver enzymes can be mildly elevated, alkaline phosphatase is elevated from placental source, and

	plasma protein concentration is diminished (↑ unbound drug levels).
Renal	Tests: As indicated from H&P These patients have ↑ renal blood flow (↑ 50%), ↑ GFR, and ↑ creatinine clearance and ↓ serum creatinine and ↓ BUN. Dependent edema results from increased water and Na⁺ retention 2° resetting of the osmotic threshold for thirst and vasopressin secretions. Tests: As indicated from H&P
Laboratory	T&S maternal blood if risk factors for blood loss are present. Routine cross-match unnecessary unless significant blood loss is anticipated. Routine autologous blood donation is not recommended. Coagulation studies and Plt count are recommended with PIH, abnormal placentation, and heavy maternal bleeding. BUN, Cr, UA, and fasting blood glucose as indicated from H&P.
Premedication	Agents to decrease risk of aspiration pneumonitis include a nonparticulate antacid (e.g., 0.3 M Na citrate 30 mL po), ranitidine 150 mg iv, and metoclopramide 10 mg iv. Sedatives are not routinely administered. In extremely anxious patients, however, 0.5–1.0 mg midazolam iv is an excellent anxiolytic, without apparent effect on maternal memory, alertness, or neonatal condition.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status. Discuss with surgeon the possibility of an epidural block for anesthesia and postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients with h/o severe PONV.

SPECIAL CONSIDERATIONS

Pregnancy-induced	PIH is characterized by generalized vasoconstriction
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hypertension (PIH)

with relative intravascular volume depletion and, occasionally, diffuse capillary leak. Cautious hydration before regional anesthesia is necessary to prevent hypotension or pulmonary edema. Hepatic dysfunction may be present (HELLP syndrome). Epidural, spinal, and combined spinal-epidural (CSE) are all considered safe techniques in PIH assuming normal Plt count. Cardiovascular stability is better with regional than GA, provided intravascular volume is adequate. Abnormal coagulation ($\downarrow$ Plt count or dysfunctional Plt) contraindicates regional anesthesia. If GA is necessary, it is vital that precautions are taken to $\downarrow$ laryngoscopic hypertensive response. Use supplementation along with RSI technique; esmolol (1.5 mg/kg), NTG (2 mcg/kg), lidocaine (1.5 mg/kg), and/or remifentanyl (0.5–1 mg/kg). Labetalol (20 mg) or hydralazine (5–10 mg) can be used, but beware of onset times (<5 min and 5–15 min, respectively) and longer duration of action, so choose drugs appropriate to urgency. There is a potential for difficult intubation in PIH due to airway edema; therefore, a small ETT (6.0 mm) should be available. MgSO_4 potentiates neuromuscular blocking agents; avoid defasciculating dose of muscle relaxant before induction, use smaller than normal doses of nondepolarizing agents, and monitor neuromuscular function. MgSO_4 may potentiate hypotension and increase vasopressor requirements.

Tests: PT; PTT; Plt; TEG or bleeding time; LFTs

Eclampsia

Treat eclamptic Sz with adequate oxygenation and a small dose of either midazolam (1–2 mg) or propofol (20–40 mg). Intubate if necessary to protect airway. Initiate MgSO_4 therapy (loading dose: 4–6 g iv over 20 min; then, infuse at 1–2 g/h). If 2nd seizure occurs during maintenance dose (rare): administer 2nd bolus dose of 2 g over 5 min.

Massive maternal hemorrhage:

- Placenta previa
- Abruptio placenta
- Ruptured uterus

Activate massive transfusion protocol (MTP) if available at institution. Insert two large-bore iv catheters (14–16 ga) and consider sending Hg/HCT and coagulation studies. Ensure immediate availability of cross-matched blood and if unavailable consider uncross-matched or O-negative blood. Rapidly restore intravascular volume with crystalloid, colloid, or both until blood is available. If bleeding continues, administer MTP, consisting of a transfusion of PRBC, plasma, and platelets. Induction with either ketamine (1–1.5 mg/kg) or etomidate (0.2–0.6 mg/kg) is preferred in hypovolemic or unstable patients. DIC and dilutional coagulopathy may follow abruptio placenta, amniotic fluid embolism, massive blood loss, and resuscitation. A ratio of RBC:FFP:PLT of 6:4:1 or RBC:FFP of 1:1 is currently recommended during massive transfusion. Consider tranexamic acid (e.g., 10 mg/kg) if EBL > 1000 mL. Fibrinogen concentrate or cryoprecipitate empirically or as directed by PT, PTT, fibrinogen levels, and TEG.

Uterine atony is treated with oxytocin 30 U/L in NS at a rate sufficient to control atony (250–500 mL/h). Small boluses 1–2 U are appropriate, but avoid larger doses (risk of ↓ BP); alternatively, methylergonovine, 0.2 mg im (risk of HTN); carboprost tromethamine (Hemabate®; a synthetic analog of PGF_{2α}), 0.25 mg im or intramyometrially (risk of pulmonary HTN, bronchospasm); or misoprostol (600 mg buccal or 1000 mg PR). Bakri balloon, B-Lynch sutures or uterine artery embolization may be effective in controlling continued postpartum bleeding and may be considered before surgical artery ligation or hysterectomy. Emergency hysterectomy, however, may be the only solution to continued bleeding. Conversion to GA may be necessary if massive bleeding occurs during regional anesthesia.

Diabetes

Diabetic patients may have an increased propensity to ↓ BP following regional anesthesia. Determine

	blood glucose hourly and maintain at 80–100 mg/dL. Insulin requirements decrease drastically after delivery, and insulin dosage must be reduced to prevent maternal hypoglycemia. Tests: Fasting blood glucose; UA
Response to anesthetic drugs	In pregnant patients, MAC is ↓ 30% for inhaled agents; combined with more rapid uptake, this predisposes to anesthetic overdose. Sensitivity to local anesthetics also is increased. Epidural space capacity is decreased 2° engorgement of epidural veins; this decreases requirements for local anesthetics and increases possibility of intravascular injection of drugs. Increased sensitivity to nondepolarizing muscle relaxants (especially in patients receiving $MgSO_4$) mandates careful TOF monitoring and use of reduced doses. Decreased protein binding may increase toxicity of highly protein-bound drugs such as bupivacaine.

INTRAOPERATIVE

Anesthetic technique

Regional (neuraxial) anesthesia has many maternofetal advantages (↓ maternal mortality, ↓ risks of aspiration, participation in childbirth, better pain control) over GA. Anesthetic choice in specific circumstances depends on maternal and fetal conditions and degree of urgency.

Spinal anesthesia is preferred for elective or unplanned C-section (unless patient has an existing epidural) when no contraindications to regional anesthesia exist (e.g., patient refusal, coagulopathy, active neurologic disease, elevated ICP, hypovolemia, infection at site of insertion). With the use of small-gauge (e.g., 25–27 ga), pencil-point needles (e.g., Sprotte, Whitacre), the risk of headache is low (0.5–1%). Advantages of spinal over epidural anesthesia include rapid onset of block and more reliable anesthesia. ↓ BP, however, is more common with spinal anesthesia. Fluid (1–1.5 L crystalloid or 500 mL colloid) preloading or co-loading (at the time of the spinal), leg wrapping (e.g., compression stockings), and vasopressors infusions (phenylephrine 25–50 mcg/min) reduce the incidence and severity of ↓ BP, but do not eliminate it. Additional boluses of vasopressors (e.g., phenylephrine [50–100 mcg] or ephedrine [5–10 mg]) should be

used as appropriate to treat ↓ BP. Phenylephrine should be considered the vasopressor of choice due to its rapid onset, efficacy in treating ↓ SVR, and limited placental transfer. Ephedrine may be associated with ↑ fetal acidosis. Atropine (0.4 mg iv) should be used to treat ↓ HR. Consider using epinephrine (50–100 mcg iv) if other vasopressors are unsuccessful in reversing severe ↓ BP or profound ↓ HR.

General anesthesia is used when regional anesthesia is contraindicated or when there is inadequate time to institute regional blockade. Obstetric emergencies for which rapid induction of GA may be indicated include severe maternal hemorrhage, prolapsed umbilical cord, severe fetal bradycardia, severe persistent fetal decelerations, or the need for intrauterine manipulation. Less dire situations often permit the performance of a quickly placed spinal or extension of a functioning epidural block with an agent having a rapid onset (e.g., 15–20 mL 3% 2-chloroprocaine or 2% lidocaine with epinephrine 1:200,000 and bicarbonate). Continuous monitoring of the fetal heart rate (FHR) in the OR may allow use of regional anesthesia if the FHR tracing is reassuring. Constant communication with the obstetrician regarding maternal and fetal condition is essential. Although situations exist in which a GA is preferable to regional, the risks must be weighed against the benefits for patients with greater potential for complications. If difficult intubation is anticipated, rapid-sequence induction of GA with cricoid pressure should not be undertaken. Alternative approaches include awake intubation, regional anesthesia, or local infiltration by the obstetrician. Sometimes, a nonreassuring FHR pattern is diagnosed as “fetal distress.” Fetal distress is an imprecise and nonspecific term with little positive predictive value. The duration and severity of any FHR abnormality should be considered when the type of anesthesia is determined. C-section performed for a nonreassuring FHR pattern does not necessarily preclude the use of regional anesthesia.

Regional anesthesia

Epidural

Apply monitors, fluid load, and place the patient in the sitting or lateral decubitus position. Locate the epidural space with a 17–18-ga Tuohy needle and LOR to saline (preferable) or air technique. A 3-mL test dose of 1.5–2% lidocaine (45–60 mg) with 1:200,000 epinephrine (15–20 mcg) is given through the epidural catheter to exclude intravascular injection (Sx: tachycardia, palpitations, dizziness, tinnitus, new taste in the mouth) or subarachnoid placement (motor/sensory block in lower extremities,

rapid onset of pain relief). After 3–5 min, inject 15–20 mL 2% (300–400 mg) lidocaine with 1:200,000 epinephrine (75–100 mcg) incrementally over 5–10 min. Sodium bicarbonate, 1-mEq/10-mL lidocaine, hastens onset of block and improves the block quality, but increases risk of ↓ BP. Fentanyl (50–100 mcg) can improve block quality and decrease intraoperative nausea and vomiting. 2-Chloroprocaine (450–600 mg) or 2% lidocaine with epinephrine should produce adequate surgical anesthesia within 5–10 min. Bupivacaine, levobupivacaine, and ropivacaine 0.5%, 15–20 mL (75–100 mg), with or without epinephrine 1:200,000 and/or fentanyl (50–100 mcg) are alternative options but with slower onset of blockade.

Left uterine displacement is best achieved by elevating the right hip with a wedge or towel. Administer O₂ by mask or nasal cannula if needed, and continue to monitor FHR before abdominal prep. Monitor BP every min until stable and then every 3–5 min. Treat ↓ in BP with further uterine displacement, additional fluids, and vasopressors as outlined earlier. Maintain BP at baseline levels to optimize placental perfusion and minimize maternal nausea. For inadequate anesthesia, give additional epidural local anesthetic, 50–100 mcg fentanyl iv or epidurally, 50% N₂O/O₂, ketamine 10–20 mg iv, midazolam 1–2 mg iv, and/or infiltrate with local anesthetic. The patient must remain conscious to avoid risk of aspiration. If anesthesia is still inadequate, induce GA (see below).

After delivery of infant and placenta, bolus (1–2 U) and infuse (30 U in 1 L at 250–500 mL/h) oxytocin. Monitor for excessive blood loss, recognizing that significant blood loss can occur quickly and outside of direct observation and is often underestimated. Chest pain, mild oxyhemoglobin desaturation, and SOB after delivery may be due to irritation of the

diaphragm by blood or packs, too high or inadequate level of anesthesia, or venous air or amniotic fluid embolization. Transient ST-segment changes on ECG may occur, but do not usually signify myocardial ischemia.

Spinal

Apply monitors, administer fluid, and position as for epidural anesthesia. Insert 25- to 27-ga pencil-point needle and verify free flow of CSF. If technical difficulties or urgent situations are encountered, a larger pencil-point needle (e.g., 22-ga Sprotte) may be considered. Inject hyperbaric 0.75% spinal bupivacaine 11–12 mg (1.5–1.6 mL) $\pm$ fentanyl 10–20 mcg and preservative-free morphine 0.1–0.15 mg, and position the patient with left uterine displacement. Monitor and treat $\downarrow$ BP as for epidural. Adjust operating table position to insure a T4 level of anesthesia. If anesthesia is inadequate and time permits, consider repeating block with CSE (reduced intrathecal dose) or epidural. Incrementally dose the epidural catheter as needed. Treat persistent inadequate anesthesia as for epidural. Induce GA if these measures fail.

Combined spinal-epidural (CSE)

An alternative technique combining the rapid onset and density of spinal anesthesia with the flexibility of continuous epidural anesthesia (e.g., if necessary to extend the duration or intensity of the block). Apply monitors, administer fluid, and position as for spinal/epidural. The recommended technique is the needle-through-needle. When the epidural space is located with a standard 17–18-ga Tuohy needle, insert a 25- to 27-ga pencil-point spinal needle through it, and administer either a standard C-section spinal dose of bupivacaine and narcotics as described for spinal or reduced dose if indicated (e.g., maternal cardiac disease or short stature). Secure the epidural catheter and use if needed. Appreciate that the epidural catheter is untested and an epidural test dose is advisable before dosing.

General anesthesia

Induction	Elevate the right hip (wedge or towel) to provide left uterine displacement. Preoxygenation for 3–5 min is optimal; however, 4–8 maximal inspiratory breaths in 30 sec is a satisfactory substitute in an emergency. Head up and CPAP may improve oxygenation in obese parturients. Place patient in maximal “sniff” position (ramp patient if necessary), to optimize position for intubation. After patient is prepped and draped and obstetric team is ready to begin, perform rapid-sequence induction with cricoid pressure. Administer propofol 2–3 mg/kg (or ketamine 1–1.5 mg/kg or etomidate 0.2–0.6 mg/kg in hypovolemic or cardiovascularly unstable patients) and succinylcholine 1–1.5 mg/kg to induce GA and facilitate intubation. Inflate cuff of ETT and verify tracheal placement by ETCO ₂ waveform and auscultation of bilateral breath sounds.
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Failed intubation	If tracheal intubation is unsuccessful, oxygenation and mask ventilation must become a priority, and numerous attempts to intubate should be avoided. Summon experienced help and quickly decide whether surgery must proceed. The risks of continuing with mask GA and cricoid pressure must be weighed against the risk of allowing the mother to awaken. If mask ventilation is impossible, quickly attempt ventilation with an LMA. If this succeeds, either continue to use throughout the case or place an ETT (6 mm ID) through LMA blindly or with a fiberoptic scope; alternatively, use an intubating LMA. A ProSeal LMA has been used successfully as a rescue supraglottic airway device in this situation

and may negate the necessity for ET intubation. Cricoid pressure may interfere with mask ventilation or LMA placement and should be released if necessary. If LMA fails to allow ventilation, attempt emergency transtracheal ventilation using a 12–14-ga iv catheter (or specific cricothyroid airway device) and appropriate tubing to connect to a high-pressure O₂ source (e.g., jet ventilator). If these measures are unsuccessful, an emergency cricothyrotomy or tracheostomy should be performed by experienced personnel. **Planning for a failed intubation must occur before it happens.** A difficult-intubation tray, including video laryngoscopes, supraglottic airways, and equipment for emergency jet ventilation, must be immediately accessible in or very near the delivery room.

Maintenance

50% N₂O/O₂ with isoflurane, sevoflurane, or desflurane (limit MAC < 1.5 to reduce uterine atony). Propofol infusion for maintenance negates the uterine relaxant effect of the halogenated agents but may lead to more neonatal depression. Avoid hypocapnia (ETCO₂ 32–36 mm Hg) during controlled ventilation, which may ↓ umbilical blood flow. After delivery, NDMR is rarely required; give additional opioid (e.g., fentanyl 150–250 mcg) and increase N₂O concentrations if necessary, or consider converting to TIVA (with propofol and remifentanyl infusions). Administer small doses of muscle relaxants (e.g., vecuronium 2–3 mg or rocuronium 10–20 mg) if needed. Reverse with neostigmine 0.05 mg/kg and glycopyrrolate 0.01 mg/kg or atropine 0.02 mg/kg or sugammadex 2–4 mg/kg. Consider titrating longer-acting opioids (e.g., hydromorphone 1–2 mg) for postpartum analgesia. Midazolam (1–2 mg), given after delivery, helps avoid maternal awareness, which occasionally occurs with this anesthetic technique. Consider TAP (transversus abdominis plane) block pre-emergence if consented.

Emergence	Delay extubation until patient is fully awake and muscle strength has returned to normal. Discuss with surgeon local wound infiltration or nerve block. Consider iv acetaminophen if po dose is not given. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Moderate blood loss IV: 16–18 ga × 1 (additional larger iv access if hemorrhage) BSS: 1–3 L typical replacement	Infuse 1–2 L BSS (± colloid 500 mL) immediately before (preload) or after (coload) regional anesthesia. Typical blood loss = 700–1000 mL. A rapid-fluid infuser and blood warmer should be available in the event that large-volume blood transfusion is required. See Goal-Directed Fluid Management, Appendix K, p. K-4.
Monitoring	Standard monitors FHR monitor ±arterial line ±CVP ±TTE/TEE	Arterial BP monitoring via automated BP device or arterial line (e.g., severe or labile HTN, hemorrhage). TTE (awake) or TEE (asleep) should be considered to evaluate volume status and cardiac function. Occasionally, a CVP is needed (e.g., pulmonary edema, vasopressor requirements, unresponsive oliguria).
Positioning	Left uterine displacement (blanket under the right hip and/or table tilt)	Minimizes aortocaval compression
Complications	Amniotic fluid embolism	Rare cause of hemodynamic collapse hypoxemia and DIC. Often fatal. Rx: supportive,

100% O₂, PEEP, and vasopressors. Correct Plt, clotting factors, and metabolic disturbances. CPB has been used successfully.

POSTOPERATIVE

Complications	VTE	Pregnancy and C-section increase risk of venous thromboembolism. DX: pleuritic chest pain, cough, hypoxemia, ↑ RR, ↑ HR, ↑ A-a gradient. Rx: supportive, 100% O ₂ , volume expansion, vasopressors, or thrombectomy
	Postpartum hemorrhage	See Anesthetic Considerations for Removal of Retained Placenta.
Multimodal pain management	Epidural: 3–4 mg (preservative-free) morphine after delivery Intrathecal: Morphine (preservative-free) 0.1–0.15 mg given with spinal local anesthetic	Common side effects include pruritus (70%), nausea (30%), and, rarely, respiratory depression. Nalbuphine (2.5–5 mg) and naloxone (0.1–0.4 mg) are used for reversal of these side effects. Metoclopramide (10 mg iv) and/or ondansetron (4 mg iv) may be needed for persistent nausea. Risk of delayed respiratory depression in healthy patients is small; however, adequately trained nursing staff and a protocol for treatment of complications are mandatory

if neuraxial opioids are used. These patients should not routinely receive sedatives or other systemic opioids for 12 h, and close monitoring of RR and level of consciousness every hour is necessary for 12 h and then every 2 h for the next 12 h. Pulse oximetry and monitors of ventilation (e.g., ETCO₂) should be considered in high-risk patients, including those with obstructive sleep apnea, those with super-morbid obesity, and patients receiving additional sedatives.

Parenteral opioids:

iv opioids (e.g., fentanyl 25–50 mcg, hydromorphone 0.2–0.4 mg) boluses prn or PCA instituted in recovery room.

NSAIDs (e.g., ketorolac): iv either in OR at the end of case or in PACU or if unable to take orally after surgery. Acetaminophen: iv 1000 mg in OR or PACU

TAP block is a beneficial rescue analgesic option in select patients.

Oral analgesics:

NSAIDs (e.g., ibuprofen) and acetaminophen around-the-clock for 48–72 postcesarean. Oxycodone (or equivalent oral opioid), for breakthrough pain management.

ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6.)	Encourage early enteral nutrition and mobilization.
Tests	As indicated	Consider DVT prophylaxis.

Suggested Readings

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MEDICAL AND SURGICAL MANAGEMENT OF POSTPARTUM HEMORRHAGE

SURGICAL CONSIDERATIONS

Description

The most common indication for **postpartum uterine devascularization and hysterectomy** is **intractable postpartum hemorrhage (PPH)**. PPH is clinically defined as any uncompensated postpartum blood loss → tissue hypoperfusion. There are four major causes of PPH: uterine atony, retained products of conception (POC) laceration of the genital tract, and coagulopathies. Inherited coagulopathies include von Willebrand disease, hemophilia, and factor XI deficiency. Acquired coagulopathies are most often related to thrombocytopenia 2° preeclampsia/eclampsia, hypofibrinogenemia 2° long-standing fetal demise, placental abruption, and DIC related to massive blood loss.

Postpartum blood loss can be reduced by prophylactic use of oxytocin, methylergonovine, or prostaglandins, and these same agents are used as the first line of treatment for PPH. Methylergonovine should be given im only (0.2 mg q2–4 h up to 1 mg) because iv infusion has been reported to cause acute HTN, stroke, and seizure. Ergot derivatives are contraindicated in patients with Hx of HTN, asthma, Raynaud syndrome, or migraine. PGF_{2α} (Hemabate®) may be injected im (intramyometrial) at a dose of 0.25 mg, up to a total of 2 mg. Misoprostol, a PGE, may be given rectally (up to 1000 mcg).

Simultaneously, the surgeon should explore the cause of PPH and apply a specific treatment. If PPH is not controlled with treatment of uterine atony, and after volume replacement and correction of any coagulopathy, temporizing measures should be applied while preparing the patient for definitive invasive treatments. Temporizing measures include packing of uterine cavity with a long gauze, use of balloon tamponade (Bakri balloon), or B-Lynch suture. Nonpneumatic antishock garment (ASG) on nonpregnant patients is applied to postpartum patients with some success in temporizing hypovolemic shock from abdominal and pelvic bleeding. ASG can be applied quickly and results in an immediate 1500–2000 mL autotransfusion. ASG should not be used with fetus in situ or thoracic site of hemorrhage. After stabilization, patient should be transferred to radiology for uterine artery embolization under fluoroscopic control where uterine arteries are selected, and absorbable Gelfoam pledgets are introduced. Treatment may be repeated until bleeding is stopped. In known cases of placenta accreta, in anticipation of PPH, catheters have been placed in uterine arteries before C-section.

If selective embolization is not available or fails to stop hemorrhage, more invasive surgical intervention should be employed, including uterine compression sutures, iliac artery ligation, **uterine devascularization**, and **hysterectomy**. The decision for surgical intervention is made when other options (i.e., medical, interventional radiology) have not been successful in decreasing the hemorrhage. PRBC, Plt, FFP, and cryoprecipitate replacement should continue while proceeding with surgery. Tranexamic acid and fibrinogen concentrates should be considered.

The technique for an **emergent obstetrical hysterectomy** is largely similar to a hysterectomy for other indications. Of note is the engorged and prominent nature of the vessels supplying the gravid uterus. The edematous tissues surrounding the uterus are very friable and may bleed profusely if improperly manipulated. A **supracervical** or **total hysterectomy** may be performed. Through a midline or Pfannenstiel incision, the uterus is elevated out of the abdominal cavity. The round ligaments are clamped,

transected, and ligated, and the anterior leaf of the broad ligament is incised bilaterally from the transected round ligaments to the vesicouterine reflection. The posterior leaf of the broad ligament adjacent to the uterus is entered at a level just below that of the fallopian tubes and utero-ovarian ligaments. These are then clamped, transected, and ligated. Next, incision of the posterior leaf of the broad ligament toward the cardinal ligaments is performed. With gentle blunt dissection, the bladder and attached vesicouterine peritoneal flap are dissected off the lower uterine segment. The ascending uterine arteries and veins are identified bilaterally, then clamped, transected, and ligated. If a **subtotal hysterectomy** is planned, the body of the uterus is amputated at this level, and the cervical stump is closed with interrupted sutures. If a total hysterectomy is planned, dissection of the bladder off the cervix is continued until the cervicovaginal margin is identified. The cardinal and uterosacral ligaments are clamped, transected, and ligated, with clamps placed as close to the cervix as possible without including cervical tissue. After the level of the lateral vaginal fornix is reached, a clamp is swung below the cervix, across the lateral vaginal fornix. The cervix is then amputated off the vaginal cuff. Throughout the procedure, it is vital to clamp and ligate any bleeding vessels and to take extra care to avoid damage to the ureter or bladder. Following removal of the uterus and cervix, the vaginal cuff angles are sutured to the ipsilateral cardinal ligament stumps and the vaginal cuff is closed with a running locked stitch. The abdominal wall is closed in layers.

Usual preop diagnosis

Intractable postpartum bleeding; rupture of gravid uterus

ERAS: Preop enhanced recovery options are limited by the emergent nature of this surgery.

■ SUMMARY OF PROCEDURE	
Position	Supine, with left lateral tilt
Incision	Pfannenstiel's or midline longitudinal
Unique considerations	Monitoring of coagulation parameters and correction of DIC. Consider central venous hemodynamic monitoring. Pediatrics team present, if indicated.
Antibiotics	Cefazolin 2 g iv q8h; total 3 doses
Surgical time	2–3 h
Closing	Subcutaneous intraperitoneal drains, if indicated

considerations	
EBL	3000–4000 mL
Postop care	ICU if blood loss severe; patient may require continued intubation and mechanical ventilatory support. Monitor for infectious morbidity and acute renal failure.
Mortality	<1%
Morbidity	Hemorrhage Postop febrile morbidity DIC Wound infection Sheehan syndrome Bladder injury Intraperitoneal bleeding requiring reoperation Vesicovaginal fistula Ureterovaginal fistula Transfusion-related complications
Pain score	5–10 (untreated) 0–5 (treated)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	0.11% of obstetric patients
Etiology	Unknown
Associated conditions	Placenta accreta; uterine atony nonresponsive to medical or other surgical intervention; extension of cervical tear to lower uterine segment; placenta previa; uterine rupture; uterine inversion

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Cesarean Section.

Suggested Readings

1. Baird E: Identification and management of obstetric hemorrhage. *Anesthesiol Clin* 2017; 35:15–34.

2. Collis R, Guasch E: Managing major obstetric haemorrhage: pharmacotherapy and transfusion. *Best Pract Res Clin Anaesthesiol* 2017; 31:107–24.
3. Crespo A, et al: Postpartum hemorrhage with pelvic arterial embolization, study of 33 cases. *J Matern Fetal Neonatal Med* 2017; 16:1–6.
4. El-Refaeey A, Gibreel A, Fawzy M: Novel modification of B-Lynch uterine compression sutures for management of atonic postpartum hemorrhage: VV uterine compression sutures. *J Obstet Gynaecol Res* 2014; 40(2):387–91.
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5. The American College of Obstetricians and Gynecologists: ACOG practice bulletin Postpartum hemorrhage. *Obstet Gynecol* 2017; 130:e168–86.

REPAIR OF UTERINE RUPTURE

SURGICAL CONSIDERATIONS

Description

Rupture of the gravid uterus is considered a true obstetric emergency and can be catastrophic, with significant maternal and fetal mortality. The classic symptoms are “shearing” pain, cessation of uterine contractions, loss of fetal heart tones, and the onset of vaginal bleeding. Unfortunately, these warning symptoms occur only in a minority of uterine rupture cases. Extrusion of the placenta through the uterine rupture may result in late decelerations due to uteroplacental insufficiency. Extrusion of the umbilical cord may be manifested by recurrent variable decelerations. Suprapubic pain as the only symptom has not been associated with uterine rupture. Causes of uterine rupture include breakdown of a previous uterine scar, obstructed labor, or uterine trauma. In cases where the uterine rupture occurs at the site of a prior uterine scar, the clinical course is usually less severe and the blood loss less than in cases of primary rupture of an intact uterus. The incidence of uterine rupture at the site of the old scar is 0.5% for lower-uterine transverse C-sections and 12% for classic C-sections.

Total abdominal hysterectomy, or **supracervical hysterectomy**, is the definitive therapy; however, depending on the clinical situation and the patient’s wishes for future fertility, a **uterine repair** may be undertaken. This consists of a two- to three-layered closure of the defect, using synthetic absorbable sutures. A transverse abdominal incision is made ~3 cm above the symphysis pubis and carried to the anterior rectus fascia. The fascia is incised, and the muscles of the anterior abdominal wall are separated sharply and bluntly from the midline. The peritoneum is elevated and entered

sharply. Because of the emergent nature of this condition and the possible massive blood loss associated with rupture of a gravid uterus, the anesthesiologist must act quickly. Prompt O₂ administration, together with aggressive iv fluid resuscitation, followed by transfusion of PRBC, Plt, and FFP. Cryoprecipitate, fibrinogen concentrate, and/or tranexamic acid should be considered. The use of unmatched O(–) or type-specific blood until cross-matched blood becomes available. Intraop **hypogastric or uterine artery ligation** may help minimize blood loss. Patient's coagulation parameters must be monitored because hypoxia and massive blood loss are associated with DIC.

Usual preop diagnosis

Uterine rupture

ERAS: Preop enhanced recovery options are limited by the emergent nature of this surgery.

■ SUMMARY OF PROCEDURE	
Position	Supine with left-lateral tilt
Incision	Pfannenstiel's (low, transverse abdominal) or midline longitudinal
Unique considerations	Pediatrics team present for infant resuscitation, if necessary. Thorough surgical exploration of the urinary tract (bladder and ureters), because ~10% of cases are associated with bladder lacerations. Cell-Saver may be helpful.
Antibiotics	Cefazolin 2 g iv q8h × 3 doses
Surgical time	1–2 h
EBL	500–3000 mL
Postop care	ICU if blood loss severe; MTP for continued hemorrhage; intubation and mechanical ventilatory support if aggressive fluid resuscitation results in pulmonary edema. Acute renal failure may occur 2° hypoxic and hypovolemic renal injury at time of acute uterine rupture with massive bleeding; monitor UO and serial renal function tests.
Mortality	Fetal: 35–45%

Morbidity	Maternal: 5% Blood transfusion >5 U: 58% Postop wound infection: 33% Pelvic abscess: 8% Repeat uterine rupture with subsequent pregnancies: 5%
Pain score	6–10 (untreated) 0–5 (treated)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	1/1400 deliveries
Etiology	Prior uterine surgery; grand multiparity; obesity; manual removal of the placenta; injury from tools of abortion; direct or indirect violence; oxytocin use; intra-amniotic or vaginal prostaglandins; breech extractions; internal or external version; forceps rotation; shoulder dystocia; fundal pressure; neglect (cephalopelvic disproportion, etc.); congenital uterine anomaly; cornual pregnancy; gestational trophoblastic neoplasia; placenta percreta; abruptio placenta

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Cesarean Section.

Suggested Readings

1. Al-Zirqi I, et al: Risk factors for complete uterine rupture. *Am J Obstet Gynecol* 2017; 216:165e1–8.
2. Markou G, Muray J, Poncelet C: Risk factors and symptoms associated with maternal and neonatal complications in women with uterine rupture: a 16 years multicentric experience. *Eur J Obstet Gynecol Reprod Biol* 2017; 217:126–30.
3. Stattmiller S, et al: Trial of labor after cesarean in the low-risk obstetric population: a retrospective nationwide cohort study. *J Perinatol* 2016; 36:808–13.
4. Vilchez G, et al: Contemporary epidemiology and novel predictors of uterine rupture: a nationwide population-based study. *Arch Gynecol Obstet* 2017; 296:869–75.
5. Yang Y, Ye X, Sun X: Maternal and neonatal morbidity: repeat cesarean versus a trial of labour after previous cesarean delivery. *Clin Invest Med* 2017; 40:e135–45.

POSTPARTUM TUBAL LIGATION

SURGICAL CONSIDERATIONS

Description

Postpartum tubal ligation (PPTL) is female surgical sterilization performed at the time of **cesarean section** (C-section) after delivery of the infant and repair of the uterine incision or within the first several days after a vaginal delivery. Although PPTL can be performed immediately postpartum, problems in the neonate may not be immediately evident, and a delay in surgery may be appropriate. If performed after a vaginal delivery, a small infraumbilical incision is made in the skin and carried down through the parietal peritoneum. The fallopian tubes are identified and brought out of the incision. It is important to identify the fimbriated end of the tube to ensure that the structure ligated is not the round ligament. A midsegment portion of the tube over an avascular portion of the mesosalpinx is selected, and tubal patency is disrupted by a variety of methods (**Pomeroy, Parkland, Irving, Uchida**, etc.). The Pomeroy, or a modification of it, is the most common technique used. The segment of tube grasped is ligated with absorbable suture, and the knuckle of tube formed is excised. The cut ends of the tubes should be hemostatic before replacing the tubes into the abdomen. The wound is closed in layers in the usual fashion.

The consent for sterilization requires special consideration. The procedure is strictly elective and voluntary and must be considered permanent, even though reversal may be possible. Some patients will eventually regret the decision to undergo permanent sterilization. The risk of sterilization failure and an increased risk of ectopic pregnancy in case of failure must be reviewed. The full range of alternatives to PPTL, including an interval sterilization procedure (sterilization performed remote from pregnancy), must also be considered.

Usual preop diagnosis

Desire for permanent sterilization

ERAS: For elective procedures, preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, carb loading, DV1 prophylaxis, postop pain management options, and recovery plan. See [Appendix K, p. K-1](#).

SUMMARY OF PROCEDURE

Position	Supine; steep Trendelenburg often required to allow the bowel to fall away for exposure
Incision	Infraumbilical
Special instrumentation	Small Richardson and Army/Navy retractors; Babcock clamps; vein retractor
Unique considerations	A special consent form for sterilization must be signed by the patient in advance of the surgery. The bladder must be drained before the procedure.
Antibiotics	None recommended
Surgical time	15–25 min (Uchida technique may ↑ operative time)
EBL	10 mL
Postop care	Routine postpartum care after recovery from anesthesia
Mortality	3/100,000
Morbidity	Hemorrhage Infection Incidental damage to bowel or bladder
Pain score	3–5 (untreated) 0–3 (treated)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	The most common contraceptive procedure in the United States

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Optimal timing of **tubal ligation** remains controversial. Often, it is more convenient for both the surgeon and the parturient with a functioning epidural catheter to have surgery immediately after delivery. Some obstetricians, however, favor waiting 8–24 h, when adequate assessment of the neonate should be complete and risk of postpartum complications, including maternal hemorrhage, is lessened. Alternatively, the patient's existing labor epidural catheter can be left in place and reinjected at a later time

(successful epidural reactivation within 12–24 h is possible in the majority of patients and drops significantly after 24 h). Because pulmonary aspiration remains a theoretical risk, initiation of GA or spinal anesthesia often is delayed 8–24 h until the acute GI changes of pregnancy have regressed and the patient is NPO for 8 h. Shorter hospital stays after vaginal delivery are encouraging more tubal ligations during the first 12 h after delivery, and it is unknown whether this affects morbidity or mortality.

Respiratory	FRC returns to normal almost immediately after delivery. Laryngeal edema may persist in postpartum patients after protracted expulsive efforts during labor and in the pre-eclamptic → requirement for a small ETT. Laryngeal edema can also turn an expected easy airway into a difficult intubation. Tests: As indicated from H&P
Cardiovascular	The physiologic changes of pregnancy return to normal at varying intervals after delivery. For example, risk of aortocaval compression disappears immediately. Blood volume and hemodynamic changes return to prepregnant values over several days to weeks. Postpartum hemorrhage can occur without warning. Tests: As indicated from H&P
Gastrointestinal	Postpartum patients continue to be at risk for acid aspiration, although it is not known exactly when normal GI function returns. If elective PPTL is planned within 8 h of delivery, patient should have no oral intake of solid foods during labor and the postpartum period. Precautions for prevention of acid aspiration should be followed as discussed in Cesarean Section.
Neurologic	Local anesthetic requirements for spinal and epidural anesthesia remain decreased compared to nonpregnant patients for 36–48 h after delivery.
Laboratory	Hct; other tests as indicated from H&P
Premedication	Precautions should be taken to ↓ risk of aspiration pneumonitis, as discussed in Cesarean Section.
ERAS	Check compliance with presurgical ERAS

recommendations (see above) including NPO status. Discuss the possibility of spinal or epidural block for anesthesia and postop pain management. Consider preop acetaminophen 1000 mg po with sip of water. Consider aprepitant 40 mg po with sip of water for patients undergoing GA with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

Spinal anesthesia is preferred if a functioning labor epidural catheter is not in place. Epidural catheters frequently become dislodged after a patient becomes ambulatory. GA is acceptable if patient has a strong preference or failed regional anesthesia or if contraindications to regional anesthesia exist. These patients may be at risk for aspiration of gastric contents at least 8–24 h postdelivery.

Regional anesthesia

Spinal	For technique and monitoring for spinal anesthesia, see Cesarean Section. Hyperbaric bupivacaine (7.5–10 mg) ± fentanyl (10–20 mcg), with goal of obtaining a T6 level of anesthesia. Sedate patient as necessary with small doses of iv midazolam 0.5–1.0 mg.
Epidural	A 3-mL epidural test dose, followed after 3–5 min by 15–20 mL of either 2% lidocaine with 1:200,000 epinephrine or 3% chloroprocaine injected incrementally. Titrate local anesthetic as needed to ensure adequate level of anesthesia. Consider adding fentanyl 100 mcg epidurally to reduce local anesthetic requirements and extend the duration of analgesia for several hours postoperatively.

General anesthesia

Induction	Rapid-sequence induction with propofol (2–3 mg/kg) and succinylcholine (1–1.5 mg/kg) for ET intubation. In selected patients >24 h after delivery, supraglottic
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	airway devices (e.g., ProSeal LMA) may be appropriate.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance	
Emergence	Extubation should be delayed until patient is fully awake and protective airway reflexes have returned. Consider iv acetaminophen if po dose is given >4–6 h ago. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Minimal blood loss IV: 18–20 ga × 1	1–1.5-L BSS. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Positioning	✓ and pad pressure points ✓ eyes	
Complications	None specific	

▼ POSTOPERATIVE

Complications	Minimal bleeding	
Multimodal pain management	Neuraxial opioids: Fentanyl intrathecally 10–20 mcg or epidurally 50–100 mcg. Some authors propose small doses of intrathecal or epidural morphine to improve duration of analgesia, but patients then require 24-h monitoring. Oral or parenteral opioids: Oral opioids	Emphasize multimodal approach combining nonopioid and bolus/PCA opioid analgesics with spinal or epidural analgesia (see Appendix K, p. K-6).

	(e.g., oxycodone 5–10 mg), iv opioids (e.g., morphine 2–4 mg iv, up to 20 mg), or hydromorphone 0.2–0.4 mg iv instituted in the recovery room NSAIDs: Intravenous (e.g., ketorolac 15–30 mg) or oral (e.g., ibuprofen 600 mg) as necessary may be effective for pain of a cramping nature. Acetaminophen iv or oral as needed	
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6.)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	None routinely indicated	

Suggested Readings

1. American College of Obstetricians and Gynecologists: ACOG Committee Opinion Committee on Ethics: Sterilization of women, including those with mental disabilities. No 371, 2007 (replaces No 63, 1988, No 73, 1989, and No 216 1999) Obstet Gynecol 2007; 110(1): 217–20.
2. Aaronson J, Goodman S: Obstetric anesthesia: not just for cesareans and labor Semin Perinatol 2014; 38:378–85.
3. American College of Obstetricians and Gynecologists: ACOG Practice bulletin Benefits and Risks of Sterilization. No 133, 2013 (replaces No 46, 2003, No 222 1996 and No 113, 1988). Obstet Gynecol 2013; 121(2):392–404.
4. Corso E, Hind D, Beever D, et al: Enhanced recovery after elective caesarean: a rapid review of clinical protocols, and an umbrella review of systematic reviews. BMC Pregnancy Childbirth. 2017; 17(1):91.
5. McKenzie C, Seden A, Gillian A, et al: Postpartum tubal ligation: a retrospective review of anesthetic management at a single institution and a practice survey of academic institutions. J Clin Anesth 2017; 43:39–46.
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postpartum tubal ligation: a prospective, observational study. J Clin Anesth 2016; 35:221–24.

REPAIR OF VAGINAL/CERVICAL LACERATIONS

SURGICAL CONSIDERATIONS

Description

Vaginal and cervical lacerations may occur 2° trauma of spontaneous and operative vaginal delivery. Adequate repair requires optimal surgical assistance, exposure, and patient comfort. Repair may be performed in a birthing bed or may require patient positioning, lighting, anesthesia, or monitoring capabilities available only in an OR. Vaginal and cervical lacerations can extend into the perineum, rectum, urethra, bladder, lower uterine segment, broad ligament, or peritoneal cavity.

Lacerations of the lower vagina generally are easy to identify and repair. Small, superficial lacerations that do not bleed usually do not need repair, whereas larger ones should be approximated. Deep lacerations may cause profuse bleeding; if it persists despite placement of multiple stitches, brief tamponade may be adequate to achieve hemostasis or vaginal packing may be required. Lacerations involving the perineum are classified as follows: first-degree—involves break in the mucosa and skin; second-degree—involves deeper tissue (bulbocavernosus and levator ani fascia and muscle); third-degree—involves anal sphincter; and fourth-degree—extends into the rectal mucosa. First- and second-degree lacerations are repaired in layers with continuous or interrupted stitches. The skin usually is closed with a subcuticular stitch. When the anal sphincter is lacerated, it often retracts. The ends are grasped with Allis clamps and approximated with multiple stitches. When the laceration extends into the rectum, the rectal mucosa usually is closed in two layers, with the second layer imbricating the first. With periurethral lacerations, a catheter may need to be placed in the urethra to prevent passing a stitch through it. A laceration involving the urethra or bladder should be closed in multiple layers, followed by bladder drainage for several days.

Lacerations of the upper vagina are often difficult to visualize. Uterine bleeding and the umbilical cord of an undelivered placenta can obscure the field, and it can be difficult to determine if bleeding is vaginal or uterine. It is helpful to deliver the placenta and control uterine bleeding before proceeding. After visualization is adequate, it is important to place the first stitch above the apex of the laceration to control bleeding from vessels that may have retracted. Again, vaginal packing may be

required if oozing of blood persists.

Superficial **lacerations of the cervix** occur with most deliveries but usually do not require treatment. Deep lacerations can cause significant blood loss, especially when they involve larger branches from the uterine artery or extend into the lower uterine segment. Again, the first stitch must be placed above the apex of the laceration to control bleeding from vessels that may have retracted. A **laparotomy** may be necessary if a laceration extends into the lower uterine segment or broad ligament and is causing significant bleeding that cannot be controlled otherwise. Alternatively, uterine artery embolization may be considered.

Usual preop diagnosis

Vaginal or cervical laceration

ERAS: Preop enhanced recovery options are limited by the urgent nature of this surgery.

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy
Incision	None (unless exploratory laparotomy is performed)
Special instrumentation	Right-angle retractors; ring forceps; Allis clamps; Gelpi retractor; vaginal packing
Antibiotics	May be used for lacerations involving entry into the peritoneal cavity or the rectal mucosa
Surgical time	10–45 min (possibly longer if exploratory laparotomy is performed)
EBL	Variable. Possible need for transfusion. Areas that persistently ooze after repeated placement of suture may be managed with vaginal packing.
Postop care	PACU → ward
Mortality	Rare
Morbidity	Hemorrhage Hematoma Infection Rectovaginal fistula

	Vesicovaginal fistula
Pain score	3–5 (untreated) 0–3 (treated)

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	Not uncommon
Etiology	Trauma 2° spontaneous or operative vaginal delivery (98%); other vaginal/pelvic trauma (2%)
Associated conditions	Major blood loss possible; with nonobstetric etiology, the possibility of sexual assault needs to be explored.

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Vaginal and cervical lacerations are reported in up to 11% of women during vaginal delivery. These lacerations may be associated with severe postpartum hemorrhage and can extend into the lower uterine segment leading to considerable blood loss that may go undetected. Patients should be examined carefully for Sx of hypovolemia with appropriate volume resuscitation before anesthesia. Evaluation and exploration of all but the most superficial of lacerations need to be done in the operating room to optimize anesthesia options, hemodynamic monitoring, and surgical exposure.

Respiratory	FRC returns to normal soon after delivery. Tests: As indicated from H&P
Cardiovascular	The physiologic changes of pregnancy return to normal at varying intervals after delivery. For example, risk of aortocaval compression disappears immediately. Blood volume and hemodynamic changes return to prepregnant values over several days to weeks. Postpartum hemorrhage can occur without warning. Ensure adequate fluid resuscitation before induction of GA or regional anesthesia. Tests: As indicated from H&P
Gastrointestinal	Postpartum patients continue to be at risk for acid aspiration, although it is not known exactly when

	normal GI function returns. Precautions for prevention of acid aspiration should be followed as discussed in Cesarean Section.
Neurologic	Local anesthetic requirements for spinal and epidural anesthesia remain decreased compared to nonpregnant patients for 36–48 h after delivery.
Laboratory	Hct; consider coagulation studies with excessive blood loss; other tests as indicated from H&P
Premedication	Precautions should be taken to ↓ risk of aspiration pneumonitis, as discussed in Cesarean Section.
ERAS	Discuss with surgeon the possibility of a nerve block for anesthesia and postop pain management.

INTRAOPERATIVE

Anesthetic technique

In many patients, a functioning epidural catheter will be in place, and supplemental doses of anesthetic may be given to provide adequate analgesia for the surgery. If no epidural is in place and the patient is hemodynamically stable, a spinal anesthetic may be satisfactory. GA may be required when regional is contraindicated.

Regional anesthesia

Epidural	Supplemental doses of local anesthetic (1.5–2% lidocaine 10–15 mL) with or without fentanyl (50–100 mcg) injected incrementally with patient in sitting position (if tolerated) to promote perineal anesthesia
Spinal	Hyperbaric bupivacaine 0.75% 7.5 mg combined with fentanyl 10–20 mcg with patient in sitting position, if tolerated. 25- to 27-ga pencil-point needle to ↓ incidence of spinal headache. Anesthesia to T10 is usually adequate. Repair of more extensive lacerations may require a higher anesthetic dermatomal level and, consequently, a higher dose of hyperbaric bupivacaine (e.g., 10 mg).
Combined spinal-	An alternative technique combining the rapid onset

epidural (CSE)	and density of spinal anesthesia with the flexibility of continuous epidural anesthesia (e.g., if necessary to extend the duration or intensity of the block). Position and prepare patient as for any neuraxial anesthetic. The recommended technique is the needle-through-needle, which entails after locating the epidural space with a 17-ga Tuohy needle, a 25- to 27-ga pencil-point needle is then inserted through the Tuohy needle into the intrathecal space. An intrathecal dose of 7.5 mg of bupivacaine is administered, and after removal of the spinal needle, the epidural catheter is threaded through the Tuohy needle. Secure the epidural catheter and use if needed. (An epidural test dose is advisable.)
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General anesthesia

Induction	Rapid-sequence induction with propofol (2–3 mg/kg) and succinylcholine (1–1.5 mg/kg) for ET intubation. If significant blood loss, ketamine 1.5 mg/kg or etomidate 0.3 mg/kg is preferred for induction.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia. Consider preincision placement of nerve blocks.	
Maintenance	Standard maintenance	
Emergence	Extubation should be delayed until patient is fully awake and protective airway reflexes have returned. Discuss with surgeon local wound infiltration or nerve block.	
Blood and fluid requirements	Potential large blood loss IV: 16–18 ga × 1	1–1.5 L BSS ± PRBC as per blood loss. Blood loss may be extensive until laceration is repaired. See Goal-Directed Fluid Management, Appendix K , p. K-4.

Monitoring	Standard monitors	
Complications	Bleeding	
Positioning	✓ pad pressure points ✓ eyes	To avoid peroneal nerve compression at lateral fibular head → foot drop

POSTOPERATIVE

Complications	Bleeding Peroneal nerve injury (2° lithotomy position)	Nerve injury manifests as foot drop and loss of sensation over the dorsum of the foot
Multimodal pain management	Neuraxial opioids: 10–20 mcg fentanyl intrathecally or 50–100 mg fentanyl epidurally enhances intraop anesthesia and provides 3–4 h postop analgesia. Oral or parenteral opioids: Oral opioids (e.g., oxycodone 5–10 mg) or iv opioids (e.g., morphine 2–4 mg iv up to 20 mg or hydromorphone 0.2–0.4 mg iv) instituted in the recovery room	Emphasize multimodal approach combining nonopioid and epidural analgesia. Avoid NSAIDs if significant bleeding.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6.)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.
Tests	Hct	

Suggested Readings

1. American College of Obstetricians and Gynecologists' Committee on Practice

- Bulletins—Obstetrics: Practice Bulletin No. 165: prevention and management of obstetric lacerations at vaginal delivery. *Obstet Gynecol* 2016; 128(1):e1–15.
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CERVICAL CERCLAGE—ELECTIVE AND EMERGENT

SURGICAL CONSIDERATIONS

Description

Cervical cerclage is the reinforcement of the cervix to prevent premature cervical dilation in a patient with an incompetent cervix. With cervical incompetence, there is painless dilation of the cervix in the midtrimester of pregnancy. The membranes bulge through the cervix and rupture, followed by delivery of a severely premature infant.

An elective cerclage is performed prophylactically before pregnancy or usually after the first trimester of pregnancy on a patient with a Hx of cervical incompetence. If cerclage is performed before pregnancy, it may need to be removed because of spontaneous abortion or fetal anomalies. It generally is performed between 14- and 16-wk gestation but may be performed as early as 10-wk gestation. An **emergent (rescue) cerclage** is performed in a patient who presents in the second trimester with painless cervical dilation and/or effacement. Ultrasound is performed before the procedure to confirm viability and to r/o major congenital anomalies. An emergent cerclage should not be performed if there is advanced cervical dilation or any evidence of infection, contractions, or uterine bleeding.

There are two types of cerclage procedures generally performed: the McDonald and the Shirodkar. The **McDonald cerclage** is technically easier, and the one most commonly performed. A purse-string stitch with nonabsorbable monofilament suture is placed high

around the cervix near the level of the internal os and tied at the 12-o'clock position. The end of the suture is cut long to facilitate removal. The cerclage is removed electively at term or earlier if there is rupture of membranes, persistent contractions, bleeding, or evidence of infection. The **Shirodkar cerclage** involves incising the cervix transversely, anteriorly, and posteriorly and advancing the bladder off the cervix. A nonabsorbable monofilament suture is placed submucosally between the incisions, and the mucosa is closed, burying the stitch. A Shirodkar cerclage may be left for future pregnancies if abdominal delivery is performed. If the cervix cannot be adequately accessed through the vagina, cerclage may be attempted through laparotomy or laparoscopy.

Usual preop diagnosis

Cervical incompetence

ERAS: For elective procedures, preop counseling by the surgeon and anesthesiologist should include limiting NPO duration and carb loading. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy, with the use of cane stirrups. Left lateral pelvic tilt (if performed during pregnancy); Trendelenburg
Incision	None with McDonald cerclage; transverse cervical with Shirodkar cerclage
Special instrumentation	Right-angle retractors; monofilament, nonabsorbable stitch
Unique considerations	For emergent cerclage, when prolapsing membranes are present, they may be reduced by filling the bladder and/or possibly removing amniotic fluid transabdominally.
Antibiotics	None recommended
Surgical time	30 min to 1 h (may be longer for Shirodkar cerclage)
EBL	25–50 mL (may be higher with the Shirodkar cerclage)
Postop care	PACU → ward; tocolysis with indomethacin or other agent may be considered.

Mortality	Rare
Morbidity	Morbidity is increased for emergent cerclage, especially when performed later in the 2nd trimester. The McDonald cerclage is associated with less trauma and bleeding than the Shirodkar cerclage. Cervical trauma Rupture of membranes Chorioamnionitis Preterm labor Spontaneous abortion
Pain score	McDonald, 0–3; Shirodkar, 0–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	Not uncommon
Etiology	Cervical trauma from previous vaginal delivery; cervical trauma at time of previous D&C; previous treatment for cervical dysplasia (laser therapy, cryotherapy, loop electrosurgical excision procedure [LEEP]/large loop excision of the transformation zone [LLETZ], cone biopsy); congenital anomalies; idiopathic

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Cervical insufficiency affects ~1% of pregnant women, and ~40,000 cerclage placement procedures are performed annually in the United States. This patient population is generally healthy, and little workup is needed unless otherwise indicated. Cerclage is usually performed between 14 and 24 wk of pregnancy. When performed after 20 wk, relevant physiologic changes are as discussed under Cesarean Section. Women requiring a cerclage may also have uterine irritability and potentially receive drugs such as β -sympathomimetics (e.g., terbutaline), nifedipine, or indomethacin.

Laboratory	Hct; other tests as indicated from H&P

Premedication	None usually, but the significantly anxious patient may benefit from midazolam (0.5–2 mg). If >18-wk gestation and GA required, precautions should be taken to decrease risk of aspiration pneumonitis, as discussed in Cesarean Section.
ERAS	Check compliance with presurgical ERAS recommendations (see above) including NPO status.

INTRAOPERATIVE

Anesthetic technique

Drug exposure during the critical period of organogenesis (15–56 d) should be minimized, although no particular anesthetic techniques or agents have proven teratogenic in humans. Although N₂O is teratogenic in rodents, there is no evidence of human teratogenicity when used for cervical cerclage or other operations. Ensure adequate uteroplacental perfusion and fetal oxygenation by maintaining normal maternal BP and oxyhemoglobin saturation. Use left uterine displacement after 20-wk gestation. Maternal hyperventilation and IPPV may diminish uteroplacental and umbilical blood flow. Monitoring FHR may permit optimization of fetal well-being by adjustment of maternal blood pressure, oxygenation, or position. Spinal anesthesia is ideal as it minimizes fetal drug exposure and provides good operating conditions. Risk of headache is low with the use of 25- to 27-ga pencil-point needles (e.g., Sprotte, Whitacre). Epidural anesthesia is an appropriate alternative for this procedure. GA may be required if regional anesthesia is contraindicated or if uterine relaxation is desired to facilitate cerclage placement (e.g., increased uterine tone with bulging fetal membranes).

Regional anesthesia

Spinal	Hyperbaric bupivacaine 7.5–10 mg ± fentanyl 10–20 mcg. Position patient to obtain T8–T10 block. Monitor BP every min until stable and then every 3–5 min. Treat ↓ in BP below baseline with fluids and a vasopressor (e.g., phenylephrine 50–100 mcg or ephedrine 5–10 mg iv).
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General anesthesia

Induction	Standard induction. If >16- to 18-wk gestation, rapid-sequence induction is indicated.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Standard maintenance. If <16- to 18-wk gestation, use of LMA or mask anesthesia with either O ₂ /N ₂ O/volatile agent/opioid or propofol infusion is appropriate. If >16- to 18-wk gestation, ET intubation will be necessary.	
Emergence	If >16- to 18-wk gestation, extubate patient when fully awake and protective airway reflexes have returned.	
Blood and fluid requirements	Minimal blood loss IV: 18 ga × 1	Fluid preloading (before regional anesthesia) or coload (at the time of spinal) with 1- to 1.5-L BSS or 500 mL of colloid may be beneficial in decreasing spinal hypotension. See Restrictive Fluid Management, Appendix K , p. K-2 and p. K-4.
Monitoring	Standard monitors ±FHR monitor	Consider FHR monitoring for viable fetuses and in earlier gestation, as the FHR may indicate inadequate placental perfusion and can guide BP management.
Positioning	Left uterine displacement, if >20-wk gestation ✓ and pad pressure points ✓ eyes	Lithotomy positioning: NB: peroneal nerve compression at lateral fibular head → foot drop.

Complications	Preterm labor Hypotension Peroneal nerve injury	Observe for preterm labor in recovery area. Tocolytic agents (beta-adrenergic agents) given to inhibit uterine contractions can cause maternal dysrhythmias or ↓ BP. Manifests as foot drop and loss of sensation over the dorsum of the foot
Multimodal pain management	Intraspinal opioids: 10 - to 20-mcg fentanyl Parenteral opioids: iv opioids (e.g., fentanyl 25–50 mcg iv, morphine 2–4 mg iv, or hydromorphone 0.2–0.4 mg) instituted in the recovery room Oral analgesics: NSAIDs (e.g., ibuprofen) should not be prescribed for analgesia during pregnancy (cardiopulmonary and renal effects in developing fetus). Acetaminophen ± oxycodone (or equivalent oral narcotics), once patient can tolerate oral medication.	Intrathecal fentanyl given with spinal improves intraop analgesia and provides several hours of postop analgesia. Emphasize multimodal approach combining nonopioid analgesics with spinal/epidural analgesia.
ERAS	PONV protocol (see Appendix B, p. B-4)	Encourage early enteral nutrition and mobilization.

Suggested Readings

1. American College of Obstetricians and Gynecologists: ACOG Practice Bulletin No. 142: Cerclage for the management of cervical insufficiency. Obstet Gynecol 2014; 123:372–9.
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REMOVAL OF RETAINED PLACENTA

SURGICAL CONSIDERATIONS

Description

In most deliveries, the placenta is easily removed with gentle cord traction and uterine massage. If, after 30 min, the placenta remains undelivered, **manual removal**, following either parenteral analgesia, sedation, or GA, must be initiated. NTG (100–200 mcg iv or 0.4–0.8 mg sublingually) may induce uterine relaxation during manual removal. A possible alternative to manual removal involves injection of 10 mL of oxytocin (10 U/mL) into the umbilical vein; however, the success of this procedure is unpredictable. A retained placental fragment may cause immediate or late postpartum hemorrhage. An ultrasound evaluation of the uterus may help in the detection of a retained fragment. If retained products are found, **curettage** is recommended. Frequently, the retained product will already have been flushed out of the uterus by brisk bleeding. In such

cases, iv oxytocin, rectal misoprostol, IM prostaglandins, or methylergonovine may be administered to contract the uterus before curettage.

Bleeding from a retained placenta or fragment is frequently brisk, so the anesthesiologist must be ready to administer iv fluids or blood, provide supplemental O₂, and correct any coagulopathy. Cross-matched blood must be available. Placenta accreta, if extensive, can cause profuse bleeding at delivery, and a hysterectomy is often necessary.

Oxytocin boluses (1–2 U) and infusion of 30 U in 1000 mL of LR at 250–500 mL/h should be administered at a rate sufficient to maintain uterine tone after manual removal of the placenta or after sharp/suction curettage of a retained placental fragment.

Usual preop diagnosis

Retained placenta

ERAS: Preop enhanced recovery options are limited by the urgent nature of this surgery.

■ SUMMARY OF PROCEDURE	
Position	Dorsal lithotomy
Incision	None
Special instrumentation	Banjo curette/suction cannula
Unique considerations	IV fluids; PRBC and blood products, as needed; monitoring of VS
Antibiotics	Cefazolin 2 g iv q8h; 3 total doses
Surgical time	30 min
EBL	300–900 mL
Postop care	PACU → ward. Monitor for infection and further bleeding.
Mortality	Rare
Morbidity	Hemorrhage Endometritis Uterine perforation 2° curettage

	Asherman syndrome Transfusion-related morbidity (hepatitis, HIV, transfusion reactions)
Pain score	0–4

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	0.25–0.8% of vaginal deliveries
Etiology	Unknown
Associated conditions	Placenta accreta; avulsed cotyledon; succenturiate lobe

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

The incidence of retained placenta is about 1–4%. The degree of urgency associated with these patients may vary dramatically. Some patients may be hemodynamically unstable as a result of heavy and/or persistent bleeding in the postpartum period; others may have a retained placenta with minimal bleeding. Delays in delivery of the placenta increase the risk of significant postpartum hemorrhage. Evaluation of the patient in the OR to optimize monitoring, anesthetic options, and surgical exposure is essential.

Respiratory	Laryngeal edema may persist in postpartum patients after protracted expulsive efforts during labor and increases the likelihood of difficult intubation. Tests: As indicated from H&P
Cardiovascular	Volume status should be carefully assessed, and resuscitation started early. Restore intravascular volume before institution of analgesia or anesthesia. Neuraxial anesthesia (either spinal or extension of existing lumbar epidural blockade) may aggravate hypovolemia. Consider possibility of placenta accreta (placental villi are attached to the myometrium), especially if the placenta is difficult to remove, and expect greater blood loss.

Gastrointestinal	Postpartum patients continue to be at risk for acid aspiration, although it is not known exactly when normal GI function returns. Precautions for prevention of acid aspiration should be followed as discussed in Cesarean Section.
Neurologic	Local anesthetic requirements for neuraxial anesthesia remain decreased compared to nonpregnant patients for 36–48 h after delivery.
Hematologic	Coagulopathy can develop rapidly with retained placenta, especially if bleeding is severe and persistent. Tests: Hct, PT, PTT, Plt, fibrinogen, as indicated
Laboratory	Other tests as indicated from H&P. T&C for 2–4 U+ if time permits. Emergency transfusion with type O(–) or type-specific blood may be necessary. If available, consider activation of MTP if severe hemorrhage, significant coagulopathy, unstable hemodynamics, and/or if there are significant delays in receiving cross-matched blood. Consider tranexamic acid if >1000-mL blood loss. PRBC, Plt, FFP, cryoprecipitate/fibrinogen concentrates as needed.
Premedication	Precaution should be taken to decrease risk of aspiration, as discussed in Cesarean Section. Anxiolysis (e.g., midazolam 0.5–2 mg iv) may be indicated.
ERAS	Discuss with surgeon the possibility of an epidural/spinal block for anesthesia (if MAC is not an option) and postop pain management. Consider preop acetaminophen 1000 mg po with sip of water.

INTRAOPERATIVE

Anesthetic technique

There are several anesthesia options (MAC/GA/neuraxial technique) for the removal of a retained placenta depending on the urgency and severity. Sedation (MAC) may be

sufficient to enable the obstetrician to manually extract the placenta. If better analgesia and additional uterine relaxation are needed, however, then GA may be required. If intravascular volume has been restored and an existing labor epidural catheter is in place, the sensory block can be extended to provide adequate anesthesia. Initiating spinal anesthesia is also an option if intravascular volume status is adequate, there is no significant active bleeding, and time permits. Small doses of iv opioids, midazolam, and/or ketamine sometimes provide sufficient analgesia and sedation to allow removal of a retained placenta without compromising maternal safety. If this proves inadequate or hemorrhage is severe, however, GA with ET intubation is required. NTG in 100-mcg iv boluses or sublingual NTG 400 mcg provides uterine relaxation and may aid delivery of a retained placenta in normovolemic patients.

Regional anesthesia

Spinal	For technique and monitoring of spinal anesthesia, see Cesarean Section. Hyperbaric bupivacaine 7.5–10 mg, with or without fentanyl 10–20 mg, to obtain T8 level of anesthesia.
Epidural	For technique and monitoring, see Cesarean Section. Administer increments of 2% lidocaine (with 1:200,000 epinephrine and bicarbonate) or 3% 2-chloroprocaine until block level is adequate. Additional local anesthetic and fentanyl 50–100 mcg can be administered epidurally as needed to ensure adequate level of anesthesia.
MAC	Titrate small doses of opioids (e.g., fentanyl 25–50 mcg) and midazolam 0.5–2.0 mg or ketamine 0.1–0.2 mg/kg. Sedation and analgesia should be titrated carefully, due to the potential risk of pulmonary aspiration in the parturient with an unprotected airway. Ensure patient is awake and responsive throughout. Consider NTG 100 mcg iv for uterine relaxation, repeated as necessary to obtain the desired effect. Transient ↓ BP may follow vasodilation due to NTG and should be treated with volume and vasopressors if necessary.

General anesthesia

Induction	Preoxygenation, rapid-sequence induction with cricoid pressure, and hydration, as discussed in Cesarean Section. Ketamine (1 mg/kg) is preferred for induction of hypotensive patient, but, in larger doses (>1.5 mg/kg), it theoretically may increase uterine tone and make removal of the placenta more difficult. Etomidate 0.2–0.4 mg/kg is also a suitable induction agent in this setting.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.	
Maintenance	Anesthesia with N ₂ O/O ₂ + volatile + opioid often permits delivery of placenta. If uterine relaxation is necessary, administer volatile agent (>1.5 MAC) or NTG (see above) until uterine tone decreases.	
Emergence	Extubation should be delayed until patient is fully awake and protective airway reflexes have returned.	
Blood and fluid requirements	Anticipate large blood loss IV: 16–18 ga × 1–2	1–1.5 L BSS or colloid 500 mL or greater volumes ± PRBC based on blood loss. Treat ↓ BP with fluids and vasopressors and by decreasing concentration of volatile agent. Surgery is usually brief; additional muscle relaxation is not usually necessary. See Goal-Directed Fluid Management, Appendix K, p. K-4 .
Monitoring	Standard monitors	
Complications	Bleeding Uterine atony	Oxytocin is the first-line uterotonic agent. Consider small boluses 1–2 U in addition to an infusion (e.g., 30 U in 1000 mL NS at 250–500 mL/h). Second-line

	uterotonics include methylergonovine 0.2 mg im (may → ↑ BP), PGF _{2α} (Hemabate) 250 mcg im/intrauterine (may → cardiovascular collapse/bronchospasm), and/or misoprostol (Cytotec) 600–1200 mg rectal/sublingual.
Positioning	NB: peroneal nerve compression at lateral fibular head → foot drop.

▼ POSTOPERATIVE

Complications	Peroneal nerve injury (2° lithotomy position) Bleeding	Nerve injury manifested as footdrop and loss of sensation over the dorsum of the foot. See uterine atony, above.
Multimodal pain management	Parenteral: Opioids (e.g., fentanyl 25–50 mcg iv; titrated to effect) instituted in recovery room. Avoid NSAIDs in patients with persistent or significant bleeding. Administer acetaminophen and oral opioids (e.g., oxycodone 5–10 mg) for postoperative pain management.	Emphasize multimodal approach combining nonopioid analgesics with epidural/spinal analgesia blocks.
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6.)	Encourage early enteral nutrition and mobilization. Consider DVT prophylaxis.

Suggested Readings

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MANAGEMENT OF UTERINE INVERSION

SURGICAL CONSIDERATIONS

Description

Uterine inversion is associated with fundal implantation of the placenta whereby a thinning of the uterine wall, together with placental separation, causes an invagination of the myometrium, resulting in inversion. Vigorous fundal pressure or cord traction also can contribute to uterine inversion, which can be complete or incomplete. **Complete inversion** results in the inverted fundus extending beyond the cervix and appearing at the vaginal introitus, whereas in an **incompletely inverted uterus**, the fundus does not extend beyond the external cervical os. Uterine inversion can cause hemorrhage and shock out of proportion to observed bleeding and must be managed as an obstetrical emergency. An anesthesiologist must be called to the delivery room as soon as a diagnosis of uterine inversion is made. The ready availability of GA is paramount. Large-bore iv access with rapid infusion systems and appropriate fluid resuscitation must be initiated emergently. Blood and blood products should be available for administration as indicated.

Frequently, **uterine replacement** can be accomplished with iv tocolytics, such as terbutaline, MgSO₄, and, more recently, NTG; however, GA with a volatile agent may be necessary. Delay in treatment of uterine inversion is associated with increased blood loss. Initiate iv resuscitation and iv tocolytics (or anesthesia). Three primary methods for uterine replacement are the Johnson, Huntington, and Haultain procedures. Usually, the **Johnson method** is attempted first. Leave the placenta attached. Cup the inverted fundus in the palm of the hand, and elevate the inverted prolapsed fundus. Apply persistent upward pressure to the ring of inversion with the tip of the operator's fingers. Once the fundus is replaced, keep a fist inside the uterus and apply external fundal massage. Continue uterotonic treatments. Attempt manual removal of the placenta at this stage. Watch for reinversion. Laparotomy must be performed if uterine replacement with the Johnson method is unsuccessful. The **Huntington procedure** involves grasping the round ligaments and applying upward traction on them while an assistant exerts upward pressure on the uterus via a hand in the vagina. If the inverted uterus is trapped below the cervical ring, the **Haultain procedure** is used. This procedure involves making a longitudinal fundal incision posteriorly to allow easier reinversion of the fundus. Hydrostatic and laparoscopic reduction of uterine inversion has also been described.

Usual preop diagnosis

Uterine inversion

ERAS: Preop enhanced recovery options are limited by the emergent nature of this surgery.

■ SUMMARY OF PROCEDURES		
	Manual Reinversion	Huntington/Haultain
Position	Dorsal lithotomy	Supine
Incision	None	Pfannenstiel's or midline longitudinal
Unique considerations	Prompt O ₂ and iv fluid resuscitation; use of blood and blood products as necessary.	⇐
Antibiotics	Cefazolin 2 g iv q8h; total 3 doses	⇐
Surgical time	30 min	1–2 h

EBL	250–3000 mL	⇐
Postop care	±ICU. ARDS may necessitate mechanical ventilation. Monitor for acute renal failure 2° hypoxia and hypovolemia.	⇐
Mortality	Rare	⇐
Morbidity	Febrile morbidity	⇐
	Clinical shock	⇐
	Infectious morbidity from blood transfusion	⇐
Pain score	2–5	3–7

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive age
Incidence	1/2000–6000 deliveries
Etiology	Unknown
Associated conditions	Fundal implantation of the placenta; primiparity; intrapartum oxytocin; placenta accreta; therapy of preeclampsia with MgSO ₄ ; macrosomic fetus

ANESTHETIC CONSIDERATIONS

PREOPERATIVE

Uterine inversion is an obstetric emergency as it usually is accompanied by hemorrhage and shock. Management requires prompt recognition and action. These patients often present in shock out of proportion to blood loss because of the parasympathetic effect of traction on the ligaments supporting the uterus. Immediate resuscitation almost always is necessary. Blood loss varies from 250 to 3000 mL and is easily underestimated. Patient condition will improve as soon as the uterus is replaced, and therefore, surgical treatment should not be delayed. Consider activation of MTP if available.

Respiratory	FRC returns to normal almost immediately after delivery. Tests: As indicated from H&P

Cardiovascular	Massive hemorrhage and pain usually present with complete inversion. Before induction, insert large-bore iv, and rapidly begin fluid resuscitation to treat ↓ BP. Blood transfusion may be necessary, although is seldom available until after surgery.
Gastrointestinal	Postpartum patients continue to be at risk for acid aspiration, although it is not known exactly when normal GI function returns. Precautions for prevention of acid aspiration should be followed as discussed in Cesarean Section.
Neurologic	Local anesthetic requirements for neuraxial anesthesia remain decreased compared to nonpregnant patients for 36–48 h after delivery.
Laboratory	Hct; Plt; consider coagulation studies and other tests as indicated from H&P. T&C for 2–4 U PRBC; keep 2 U ahead. MTP with O-negative PRBC if necessary.
Premedication	Na citrate 30 mL po within 30 min of GA induction. If time permits, other agents to decrease risk of aspiration pneumonitis, as discussed in Cesarean Section
ERAS	Enhanced recovery options are limited by the emergent nature of this surgery.

INTRAOPERATIVE

Anesthetic technique

Attempted uterine replacement and induction of anesthesia should not await intravascular volume replacement. Bleeding usually stops when the uterus is replaced. Uterotonic therapy should be discontinued, and iv NTG (50–200 mcg) will often provide uterine relaxation to facilitate replacement and can be administered in the labor suite. Other tocolytics, such as terbutaline or Mg^{++} , also may assist in restoration of the uterine fundus. **Do not give oxytocin before uterine replacement.** If regional anesthesia (e.g., epidural or spinal) was used for delivery, replacement of the uterus may be accomplished with little further anesthetic intervention, with or without NTG. If regional anesthesia was not used for delivery, iv analgesia and sedation with small doses of fentanyl (25–50 mcg iv), ketamine (10–20 mg iv), or midazolam (0.5–1 mg iv)

may allow reduction. Sometimes, however, GA is required, often with increasing concentrations of volatile agents to facilitate uterine replacement. GA also provides the opportunity for a laparotomy to correct the inversion if other measures fail.

Induction	Rapid-sequence induction with ketamine (1.0 mg/kg) or etomidate (0.3 mg/kg) preferred. Higher dose of ketamine may adversely increase uterine tone.	
ERAS	Verify preincision antibiotics. Assess volume status (see Appendix K). Attempt to maintain normovolemia and normothermia.	
Maintenance	Sevoflurane and desflurane are effective uterine relaxants. ↓ BP should be treated with fluids + vasopressors and transfusion ± blood and blood products as necessary.	
Emergence	Extubation should be delayed until patient is fully awake and protective airway reflexes have returned. Consider iv acetaminophen. PONV prophylaxis should include ondansetron.	
Blood and fluid requirements	Significant blood loss IV: 16–18 ga × 2	Possible continued blood loss after reduction of the uterus due to uterine atony. See Goal-Directed Fluid Management, Appendix K , p. K-4.
Monitoring	Standard monitors	Arterial line may be useful, if time allows.
Positioning	✓ and pad pressure points ✓ eyes	
Complications	Uterine atony Massive blood loss	Rx: ↓volatile anesthetic concentrations. Oxytocin infusion, uterine massage, ± methylergonovine 0.2 mg im (may ↑ BP), PGF _{2α} (Hemabate) 250 mcg/im/intrauterine (may →

cardiovascular
collapse/bronchospasm);
Cytotec (misoprostol) 600–
1200 mg rectal/sublingual

▼ POSTOPERATIVE

Complications	Bleeding	
Multimodal pain management	Oral or parenteral opioids: oral opioids (e.g., oxycodone 5–10 mg) or iv opioids (e.g., morphine 2–4 mg iv, up to 20 mg; hydromorphone 0.2–0.4 mg iv) instituted in the recovery room. Avoid NSAIDs if significant bleeding. Use acetaminophen for opioid-sparing analgesia.	Emphasize multimodal approach combining nonopioid and bolus/PCA opioids (see Appendix K, p. K-6).
ERAS	PONV protocol (see Appendix B, p. B-4 and p. B-6 .)	Encourage early enteral nutrition and mobilization.
Tests	Hct, coagulation profile	

Suggested Readings

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CHAPTER 8.4

Laparoscopic Procedures with and Without Robotic Assistance for Gynecologic Surgery

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LAPAROSCOPIC SURGERY FOR ENDOMETRIOSIS

SURGICAL CONSIDERATIONS

Description

Endometriosis is defined as the presence of endometrial glands and stroma in ectopic locations. Intervention usually is indicated for intractable pain, infertility, or impaired function of the gastrointestinal (GI) or genitourinary (GU) tracts or involvement of the thoracic cavity. GU endometriosis may range from superficial involvement of

peritoneum overlying the ureters (known as extrinsic disease) and bladder to deep infiltrating endometriosis (DIE) penetrating through ureteral and bladder mucosa. Scarring and fibrosis can cause ureteral obstruction and hydronephrosis with consequent renal insufficiency and eventual failure. Patients with GI endometriosis may have thickening of the rectovaginal septum, suggesting obliteration of the posterior cul-de-sac or rectosigmoid involvement. Adhesions may make rectovaginal examination difficult or painful. Pelvic structures may be immobile (known as the “frozen pelvis”), suggesting adhesions are fixing bowel or bladder to the uterus. In severe cases of GI endometriosis, sigmoidoscopy should be performed to rule out malignancy and to determine whether endometriosis has penetrated through the bowel mucosa, in which case bowel resection may be required.

Laparoscopy (Fig. 8.4-1) is the most appropriate surgical technique for the diagnosis and treatment of endometriosis. Data from animal and clinical studies suggest that laparoscopic surgery is more effective for adhesiolysis, reduces blood loss, and causes fewer de novo adhesions compared to laparotomy. Special consideration must be given to the patient’s history of abdominal or pelvic surgery, pelvic inflammatory disease (PID), and endometriosis. These factors will affect the choice of surgical approach as there may be extensive adhesive disease. Alternative port entry sites, such as Palmer point (midclavicular line, 4–5 cm below left costal margin), may be chosen to avoid adhesions at the umbilicus from prior surgeries. Conservative surgical management with excision of endometriosis is indicated for women who desire future fertility and whose disease is responsible for their symptoms of pain or infertility. Surgery improves fertility and pregnancy outcomes (preterm birth, placenta previa, small for gestational age, cesarean section, and miscarriage rates) and offers at least a temporary pain relief. **Bilateral oophorectomy** might be necessary to decrease estrogen levels that sustains and stimulates the ectopic endometrium to prevent future recurrence of disease. **Hysterectomy** with **bilateral salpingo-oophorectomy (BSO)** may be indicated for patients with severe symptoms who have not responded to medical or conservative surgical treatment and who do not desire fertility (see **Laparoscopic hysterectomy**). It is important to weigh these benefits against the risks of premature iatrogenic surgical menopause in women.

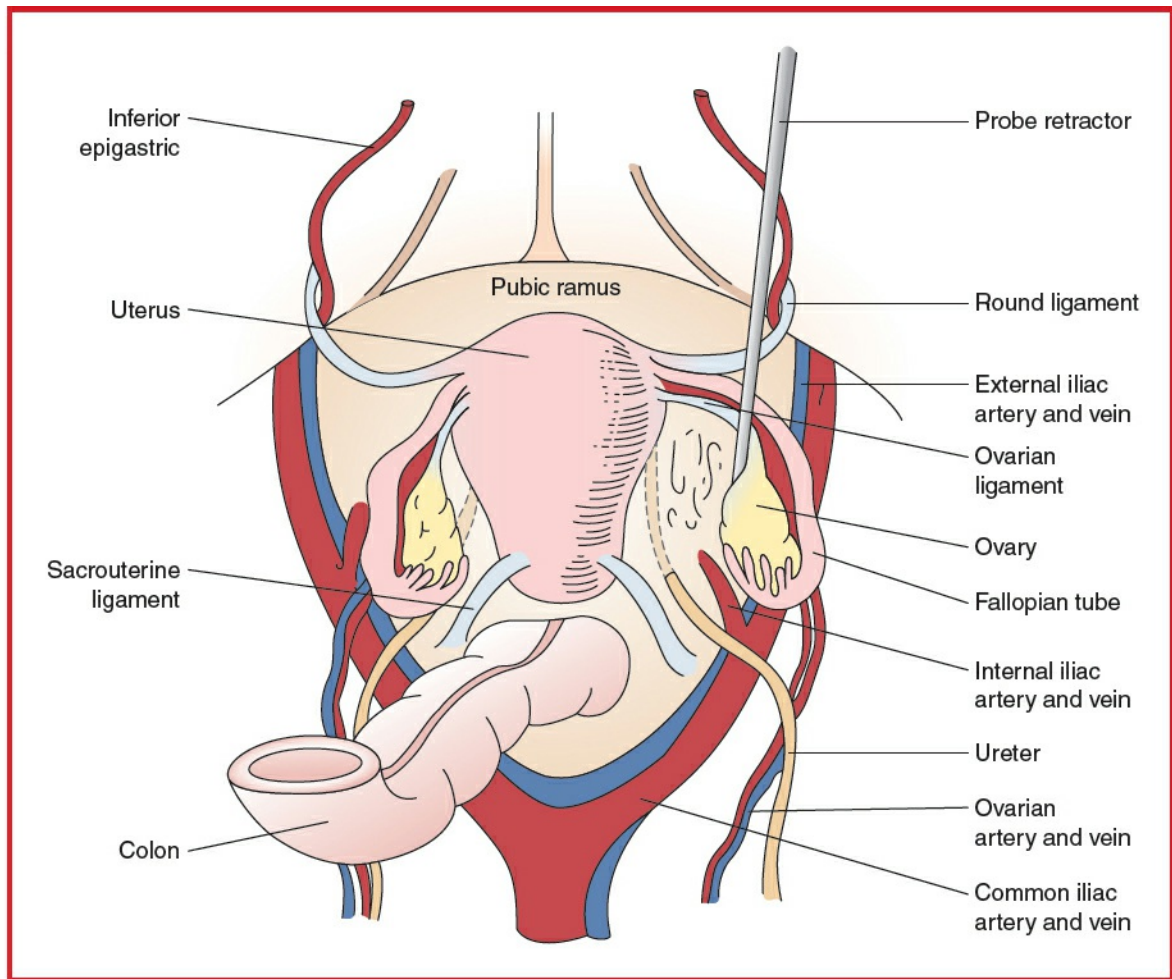


Figure 8.4-1 Laparoscopic view of the female pelvis.

Initial steps of laparoscopy

Entry into the abdomen, particularly placement of the initial port, has the potential of being the most hazardous maneuver in laparoscopic surgery, with over 50% of injuries to bowel and major vessels occurring during initial port placement. Although rare, both visceral and vascular injuries can lead to significant morbidity and potential mortality. The risk of any entry-related complications, including extraperitoneal insufflation or vascular and visceral injury, has been reported to increase with the number of attempts, ranging from 0.8% to 16.3% on entries requiring a single attempt, 16–37% with two attempts, and 44–64% with three or more attempts.

A Foley catheter should be placed before the beginning of the procedure to allow continuous drainage of the bladder, thereby reducing the likelihood of trocar injury to the bladder. The patient should be completely relaxed and supine on the operating table. An OG/NG tube should be placed to decompress the stomach to avoid trocar injury to the stomach. In a patient with no history of pelvic surgery, the direct trocar method with or without Optiview® may be used with an infraumbilical incision because this is the

anatomical area closest to the fascia and peritoneum and involves the least risk of injury to retroperitoneal structures. This area has the least amount of subcutaneous fat, both layers of fascia are fused, and there is no rectus muscle. Once the incision is made, towel clips on either side of the umbilicus are placed, the abdominal wall is lifted and the trocar is placed through the skin incision. Using an infraumbilical incision and inserting the trocar at 90° facilitate access to the abdominal cavity and decrease the risk of injury to the major pelvic vessels. This technique of direct trocar insertion may not be suitable for patients who have had prior laparotomy or laparoscopy due to risk of adhesions at the umbilicus. Another method of entry uses the Veress needle to obtain pneumoperitoneum before trocar placement. The majority (96.3%) of gynecologic laparoscopies are initiated using the Veress needle to achieve pneumoperitoneum. An infraumbilical incision is made, the abdominal wall is lifted, and the Veress needle is inserted through the skin incision at a 90° angle while connected to a low-flow CO₂ insufflation. This method allows for the abdomen to be insufflated before trocar placement, thereby facilitating ease of primary trocar placement. A mapping technique can be used after establishing pneumoperitoneum with the Veress needle to check for periumbilical adhesions. With this technique, a spinal needle attached to a water-filled 10 cc syringe is inserted adjacent to the primary port placement site. Aspiration of air indicates the site is clear of adhesions. Aspiration of bowel contents would indicate the presence of bowel adhesions to the anterior abdominal wall. The patient is then placed in steep Trendelenburg position. Two lateral 5-mm accessory ports are placed 2 cm cephalad and 2 cm medial to the anterior superior iliac spine under direct visualization, taking care to avoid the inferior epigastric vessels. A third 5-mm port is placed 2–4 cm above the pubic symphysis, again under direct visualization. Filmy adhesions of the bowel or omentum to the anterior abdominal wall or uterus can be lysed using CO₂ laser, bipolar cautery, monopolar scissors, or sharp dissection.

Pelvic endometriosis

Treatment of peritoneal endometriosis ranges from laser ablation of superficial peritoneal implants to excision and dissection of deeply embedded, fibrotic areas, known as DIE. Scarring from endometriosis that has penetrated the peritoneum to involve deeper structures destroys normal surgical planes and distorts anatomical relationships, and patients are at risk for ureteral or vascular injury at the time of surgery. Identification of ureters and blood vessels is critical before treatment of the pelvic sidewall disease. Although different modalities have been used, hydrodissection in combination with high-power superpulse or ultrapulse CO₂ lasers is the best option for endometriosis treatment. Carbon dioxide laser is much more precise, with less thermal spread and risk of unintentional injury compared to electrocautery (1.5 mm vs.

7.5 mm). Because the CO₂ laser does not penetrate water, hydrodissection of the retroperitoneal space increases the distance between the peritoneum and underlying vital structures such as blood vessels, ureter, or intestine. A 0.5-cm peritoneal incision is made in an unaffected area adjacent to the endometriotic lesion. The tip of the suction irrigator is inserted into the peritoneal opening, and LR or NS solution is injected into the retroperitoneal space. The entire endometriotic lesion is excised with a 1–2 cm margin.

Ovarian endometriosis

Type 1 endometriomas <2 cm are difficult to remove and can be resected using laser or bipolar electrocautery. Larger Type I endometriomas may require excision using a combination of blunt and sharp dissection with laparoscopic graspers and scissors. For Type II endometriomas, the procedure begins with lysis of periovarian adhesions using CO₂ laser or monopolar scissors. Suspicious areas are biopsied and sent to pathology. A plane is developed between the cyst wall and ovary by grasping the wall and separating it from ovarian stroma. Difficult areas where endometriosis has embedded through the cyst wall, and disrupted planes, require hydrodissection with vasopressin solution and bipolar cautery to control bleeding in the ovarian stroma. In some cases, it is necessary to remove a portion of the ovary attached to the cyst wall until a plane can be found. Redundant or bleeding ovarian tissue is reapproximated with suturing to close the defect.

Bladder endometriosis

If the lesions are superficial, hydrodissection and vaporization are adequate for treatment. Using hydrodissection, the areolar tissue between the bladder serosa and muscularis beneath the implants is dissected. The lesion is grasped and excised using the laser, ensuring complete excision with a 1–2 cm margin. Traction allows the small blood vessels supplying the surrounding tissue to be coagulated as the lesion is resected. Frequent irrigation is necessary to remove char, ascertain the depth of vaporization, and ensure that the lesion does not involve the muscularis or mucosa. Endometriosis extending to the muscularis but without mucosal involvement can be treated laparoscopically, and any residual or deeper lesions may be treated successfully with hormonal therapy. When endometriosis involves full bladder wall thickness, segmental bladder resection may be required. Cystoscopy is performed simultaneously, and bilateral ureteral stents may be inserted for ureteral identification, especially if the lesion is near the trigone or <2 cm from the ureteral orifice. The lesion is excised by first holding the bladder dome near the midline with laparoscopic graspers. An incision is made 5 mm beyond the lesion with the CO₂ laser. The specimen is completely

excised and removed from the abdominal cavity. After again identifying the ureters and examining the bladder mucosa, a watertight one- or two-layer closure using barbed suture is performed. Cystoscopy is performed to identify any leaks.

Patients may be discharged the same day and instructed to take prophylactic antibiotics while an indwelling catheter is present. The catheter is removed 7–14 d later, after a voiding cystourethrogram or computed tomography urogram with iv contrast is performed prior to Foley catheter removal.

Appendectomy

The appendix (a common site for endometriosis, found in 8% of women with gastrointestinal endometriosis) is identified, mobilized, and examined laparoscopically. Adhesions may be lysed using either the CO₂ laser or laparoscopic scissors. The mesoappendix is then coagulated using laparoscopic bipolar cautery and scissors or a vessel-sealing device. At this time, a laparoscopic GI stapler is introduced through a 10- to 12-mm port and placed across the mesoappendix and appendix. It is imperative to closely examine the placement of the stapler and its proximity to the rectum. After adequate placement is observed, the entire appendix and mesoappendix are stapled and cut in a single motion. The specimen is removed, and the staple line is inspected for hemostasis. Copious amounts of irrigation fluid is used to minimize bacterial contamination. Additional broad-spectrum prophylactic antibiotics to cover gastrointestinal flora (i.e., metronidazole) are given.

Thoracic endometriosis

Diaphragmatic endometriosis is more common than realized and has been shown to co-occur in 50–80% of cases with pelvic endometriosis. Thoracic endometriosis can be detected on the diaphragm, chest wall, pleura, and lung parenchyma. It is more commonly found on the right side and on the thoracic diaphragm. Patients may present with catamenial (menses-related) chest pain, shortness of breath, hemoptysis, pneumothorax or hemothorax, and pulmonary nodules, known as thoracic endometriosis syndrome. The diaphragm should be routinely inspected for endometriosis in all patients. Placing the patient in a steep reverse Trendelenburg position, the use of an atraumatic liver retractor or grasping forceps to retract the liver and cutting the falciform ligament near the liver may also assist with visualization of the diaphragm. Medical management is the first-line approach, but patients frequently require surgical excision if medical management fails. Treatment is not advised in asymptomatic patients with incidentally discovered thoracic endometriosis, due to the risk of injury to the diaphragm, phrenic nerve, lungs, or heart. A DLT or bronchial blocker may be used for single-lung ventilation of the unaffected side. In addition to the usual accessory trocars

in the right and lower quadrants, additional trocars are placed in the upper quadrant (right or left depending on location of endometriotic lesions), similar to the arrangement for a laparoscopic cholecystectomy. Laparoscopically identified endometrial lesions on the diaphragm can be visualized and treated with hydrodissection and CO₂ laser fulguration or excision, but this requires an experienced gynecologist and thoracic surgeon. If large lesions are visualized, then pulmonary wedge resection of deep parenchymal nodules or diaphragm resection with concurrent VATS is the preferred method of treatment. Lesions found during VATS are treated either by bipolar diathermy, CO₂ laser, or sharp dissection. Treatment with CO₂ laser is preferred as it is much more precise and has a smaller depth of penetration with less thermal spread and risk of unintentional injury. Electrocautery of lesions near the heart may induce arrhythmias, while tissue necrosis due to electrocautery may cause diaphragmatic fenestrations. Postop, a chest tube to decompress the pneumothorax or hemothorax may be used.

Robotic assistance

The available evidence demonstrates the feasibility and safety of robotic-assisted laparoscopic surgery in benign and malignant gynecologic diseases, with equivalent outcomes compared with standard laparoscopy. The robot does require additional costs, as well as longer surgical and anesthesia time, but may be particularly useful in cases of extensive endometriosis involving the ureters, bladder, or bowel. The robotic arm is another tool that can enable a surgeon to perform complex laparoscopic procedures. With the advent of robotic-assisted laparoscopic surgery, the previously described procedures can be performed with three-dimensional visualization, improved magnification, and greater operative flexibility and comfort. The difference is location and sometimes size of the trocars used for the robotic arms as well as possible increased operative time. Additionally, patients are placed in steep Trendelenburg for the entirety of procedure.

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, and the possibility of urinary drain postop. See [Appendix K: Enhanced Recovery After Surgery, p. K-1](#).

■ SUMMARY OF PROCEDURES

Ovarian
Endometriosis

Pelvic/GI/GU Endometriosis

Position	Dorsal lithotomy, legs in Allen stirrups ± steep Trendelenburg	⇐
Incision	Infraumbilical; lateral; midline suprapubic	⇐
Special instrumentation	CO ₂ laser or another cutting instrument; bipolar Kleppinger forceps; LigaSure; harmonic scalpel; monopolar scissors or hook; suction irrigator; uterine manipulator	± Ureteral stenting, cystoscope, proctoscope, sigmoidoscope
Antibiotics	Cefoxitin or cefazolin 2 g iv before incision	⇐
Surgical time	45 min–3 h, depending on extent of disease	1–4 h
Closing considerations	Close fascia of all 10- to 12-mm ports; subcuticular closure for 5-mm trocar sites	⇐
EBL	Minimal	⇐
Postoperative care	Mild disease: PACU →home; otherwise discharge on POD1-2	⇐
Mortality	0.08–0.2/1000	⇐
Morbidity	Overall complication rate: 2.5% Conversion to laparotomy: 0.42% Venous gas embolism Peroneal nerve injury Vascular injury: 0.2–0.4/1000	⇐ ⇐ ⇐ ⇐ ⇐ ⇐

	Bowel injury: 0.4/1000 Urinary tract injury: rare	
Pain score	4–8	4–8

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive-age women	20s–40s
Incidence	10–15% of all reproductive age women 25% of all gynecologic laparotomies	1–11% of women with known endometriosis
Etiology	Numerous theories (see above)	Extensive disseminated endometriosis
Associated conditions	Infertility; pelvic pain; bladder or bowel symptoms; dysmenorrhea; dyspareunia	Pelvic pain; GI or GU symptoms

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Laparoscopic Hysterectomy.

Suggested Readings

1. Nezhat C, Falik R, McKinney S, et al: Pathophysiology and management of urinary tract endometriosis. *Nat Rev Urol* 2017; 14:359–72.
2. Nezhat C, Li A, Falik R, et al: Bowel endometriosis: diagnosis and management. *Am J Obstet Gynecol* 2018; 218(6):549–62.
3. Nezhat C, Modest AM, King LP: The role of the robot in treating urinary tract endometriosis. *Curr Opin Obstet Gynecol* 2013; 25(4):308–11.
4. Nezhat C, Nezhat F, eds: *Nezhat's Video-Assisted and Robotic-Assisted Laparoscopy and Hysterectomy*. Cambridge University Press, New York: 2013.
5. Nezhat C, Stevens A, Balassiano E, et al: Robotic-assisted laparoscopy vs conventional laparoscopy for the treatment of advanced stage endometriosis. *J Minim Invasive Gynecol* 2015; 22(1):40–4.

LAPAROSCOPIC SURGERY FOR ECTOPIC PREGNANCY OR ADNEXAL MASS

SURGICAL CONSIDERATIONS

Description

Ectopic pregnancy is defined as a pregnancy occurring outside the uterus and affects 1–2% of pregnancies. The majority of ectopic pregnancies occur in the ampullary region of fallopian tubes (95–97%); the remainder occurs in the cornua (2–4%), ovary (0.1%), cervix (0.1%), or abdomen (0.03%). Ectopic pregnancy remains a leading cause of maternal morbidity and mortality and is responsible for 9% of pregnancy-related deaths in the United States. Predisposing factors include history of tubal ligation or other tubal surgery, pelvic inflammatory disease (PID), smoking, prior ectopic pregnancy, IUD in situ, and history of in vitro fertilization (IVF) or other treatments for infertility.

Patients typically present with lower quadrant pain, vaginal bleeding, and an $\uparrow$ β -hCG or positive pregnancy test. The imaging of choice is a transvaginal ultrasound. A complex adnexal mass with a sonolucent center may be seen on ultrasound. The presence of an intra-uterine gestational sac (double decidual sign) when β -hCG >1500–2000 (known as the discriminatory zone) effectively rules out an ectopic pregnancy as the incidence of heterotopic pregnancy (pregnancy simultaneous in the uterine cavity and fallopian tube) is extremely rare. Treatment options for an ectopic gestation include expectant management, operative laparoscopy, or a trial of medical management with intramuscular methotrexate. A 2007 Cochrane review found no difference in success rates, tubal patency, and fertility rates between **laparoscopic salpingostomy** and medical treatment with methotrexate. In cases where the size of the ectopic pregnancy is too large for conservative medical management (>3.5 cm, β -hCG >5000, or embryonic cardiac motion) and the patient is hemodynamically stable, operative management is reasonable.

Expectant management is possible for patients with low (<200) and falling β -hCG levels, with no evidence of adnexal mass on ultrasound, and who are asymptomatic. β -hCG levels should be checked every 48 h until they decrease and then weekly until they reach zero. The patient must be counseled on the risk of tubal rupture and the need for immediate surgical intervention in that case. Methotrexate or surgical management should be considered if that patient experiences increasing pain or rising β -hCG levels.

Access to the abdomen is obtained in the usual fashion (e.g., through a Veress needle or direct trocar insertion, as described in previous section). **Ruptured tubal pregnancies**

can be treated endoscopically if the bleeding has ceased or can be controlled, and the surgeon is proficient in laparoscopic techniques. Actively bleeding vessels are identified and cauterized, and forced irrigation is used to dislodge clots and trophoblastic tissue. A large-bore suction irrigator can be used to clear large amounts of clot and hemoperitoneum. Depending on their size, the products of conception (POCs) are removed through either a 5- or 10-mm trocar sleeve or placed into an endoscopic bag for removal; for larger POCs a minilaparotomy can be performed. Copious irrigation should follow to ensure hemostasis and identify and remove any remaining trophoblastic tissue. Trophoblast is invasive, and residual tissue may implant into bowel, bladder, peritoneum, or other abdominal structures and cause significant future morbidity.

For an unruptured ectopic pregnancy, the tube is identified and stabilized using laparoscopic forceps. To minimize bleeding, 5–7 mL of a solution of 50 IU **vasopressin** in 100 mL NS is injected into the mesosalpinx just below the ectopic pregnancy and over the antimesenteric surface of the tubal segment containing the gestation. Intravascular injection of vasopressin solution can precipitate acute arterial HTN, bradycardia, or even death; therefore, care must be taken to avoid intravascular injection. A linear incision is made with the monopolar needle over the thinnest portion of the tube. Trophoblastic tissue usually protrudes through the incision. The suction irrigator tip is inserted into the incision, and forceful irrigation dislodges POCs from its implantation site.

In a ruptured tubal or isthmic pregnancy, **resection of the tubal segment (salpingectomy)** containing the gestation is preferable to **salpingostomy**. A salpingostomy should be performed in patients who desire future fertility unless the affected tube is severely damaged and the contralateral tube is normal. If a salpingostomy is performed, serial β -hCG levels should be checked until they reach zero to ensure complete evacuation of products of conception. **Total salpingectomy** can be performed by progressive coagulation and cutting the mesosalpinx, which is separated from the uterus using bipolar coagulation and scissors, LigaSure, or laser. The isolated tube segment containing the ectopic pregnancy is removed intact in an endoscopic bag or in sectioned parts through the 10-mm trocar sleeve. At the completion of the procedure, the abdomen is irrigated and inspected, and incisions are closed in the usual fashion.

Hydrosalpinx

A hydrosalpinx is a fluid-filled fallopian tube, which occurs after an inflammatory process (e.g., PID, tubo-ovarian abscess [TOA], or endometriosis) damages the serosa

and mucosa of the tube. This results in occlusion at the fimbriated end and accumulation of tubular fluid and can occur unilaterally or bilaterally. Currently, for women with hydrosalpinges who are to undergo IVF, the recommendation is laparoscopic salpingectomy before IVF treatment. Bilateral salpingectomy has been shown to improve IVF success rates. It is theorized that the fluid in the fallopian tube causes inflammation and is toxic to embryos. The mechanical force from back flow of fluid may also prevent implantation of the embryo. The procedure is performed as described earlier for resection of a tubal segment.

Tuboovarian abscess

TOA is a severe complication of pelvic inflammatory disease (PID). PID results from an ascending infection from the vagina or cervix, leading to salpingitis. If this is not recognized early and adequately treated with antibiotics, this infection can then spread to the ovaries, resulting in unilateral or bilateral TOAs. The treatment of choice for TOA is medical management with antibiotics or percutaneous drainage if medical management fails or the TOA is very large. If percutaneous drainage is not possible, laparoscopic drainage may be performed. Laparoscopic access to the abdomen is performed with an umbilical port and two lateral ports. Using a suction irrigator and laparoscopic forceps, the entire abdominal cavity is examined for purulent material. Once the abscess is localized, it is drained with the suction irrigator. Care is taken to lyse adhesions as needed using blunt dissection and hydrodissection. Adhesions encountered in this situation are usually filmy and can be disrupted without the need for sharp dissection. Meticulous dissection of the abscess and surrounding adhesions is vital for the success of this procedure. Once the ovary is freed, a small incision is made to allow for copious irrigation. The laparoscopic forceps can be used to separate any adhesions at the fimbriated end of the fallopian tubes. The procedure is completed with copious irrigation of the abdominal cavity. In cases where the TOA is chronic, laparoscopy will show that the walls of the abscess are denser, and surrounding adhesions may cause structures to be matted together. In this case, sharp dissection with laparoscopic scissors may be needed, but care should be taken as planes of dissection can be difficult to define.

Usual preoperative diagnosis

Ectopic pregnancy; adnexal mass

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, recovery plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Dorsal lithotomy, legs in Allen stirrups, steep Trendelenburg
Incision	Infraumbilical, lateral suprapubic, midline suprapubic
Special instrumentation	CO ₂ laser, laparoscopic instrumentation, uterine manipulator
Unique considerations	Intraperitoneal hemorrhage and hemodynamic instability are potential risks. Patient should be cross-matched for 2 units of PRBCs.
Antibiotics	Cefoxitin 1–2 g iv before incision
Surgical time	45 min to 2 h
Closing considerations	Fascia of 10- to 12-mm trocar sites should be closed in layers. Smaller 5-mm ports closed in a subcuticular fashion.
EBL	100 mL, severe hemorrhage if ruptured
Postoperative care	The patient may be admitted overnight for observation. Quantitative β -hCG should be followed until undetectable to rule out the possibility of retained trophoblastic tissue.
Mortality	5/1000. For any laparoscopy, 0.08–0.2/1000
Morbidity	Overall complication rate: 2.5% hemorrhage and need for transfusion Conversion to laparotomy: 4.2/1000 Air embolism: rare Unintended puncture of viscera: rare Puncture of a major vessel: rare Insufflation of incorrect site: rare
Pain score	3–5

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive-age females
Incidence	1/100 pregnancies

Etiology

Distorted tubal or uterine anatomy

Associated conditions

History of pelvic inflammatory disease, endometriosis, tubal damage, IUD, and IVF

 ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Laparoscopic Hysterectomy.

Suggested Readings

1. Nezhat C, Cho J, King LP, et al: Laparoscopic management of adnexal masses. *Obstet Gynecol Clin N Am* 2011; 38(4):663–76.
2. Nezhat C, Nezhat F, eds: *Nezhat's Video-Assisted and Robotic-Assisted Laparoscopy and Hysterectomy*. Cambridge University Press, New York: 2013.

LAPAROSCOPIC MYOMECTOMY

 SURGICAL CONSIDERATIONS**Description**

Uterine myomata, or fibroids, are the most common uterine neoplasm, affecting ~20–25% of women of reproductive age. Their growth is influenced by many factors, such as estrogen and progesterone or synergistically with growth hormone and human placental lactogen in pregnancy. The severity of symptoms depends on the number, size, and location of the tumors. Symptoms may include constipation, pelvic or abdominal pressure, dysmenorrhea, dyspareunia, infertility, pregnancy complications, urinary frequency, and most commonly, menorrhagia. Patients often present with profound anemia from menorrhagia or menometrorrhagia, the usual indications for surgery. Other indications include rapidly changing size, ureteral compression, hydro-ureter, or hydronephrosis. Size, number, and location are the primary factors that will determine surgical approach to myomata.

For women who have completed childbearing, the vast majorities are offered a **hysterectomy**. It is associated with a high degree of patient satisfaction but is not free from complications. Anatomic distortions from a myomatous uterus pose an increased risk of injury to surrounding structures. For women wanting to maintain their fertility, a **myomectomy** is offered. The simplest approach is a combination of **laparoscopy** and **minilaparotomy** for removal of myomas. This approach limits operative time. The three

goals of **laparoscopic myomectomy** are (a) minimizing blood loss, which can be severe, (b) minimizing postop adhesion formation, and (c) maintaining uterine wall integrity. Preop treatment with GnRH analogues to shrink fibroids has been shown to decrease the size of the myoma and reduce the need for transfusion but can cause fibroid degeneration and disrupt surgical planes. For a large myomatous uterus, in which a pure laparoscopic procedure may be tedious and difficult, a **laparoscopically assisted myomectomy (LAM)** is an alternative. Although myomectomy may be performed to preserve fertility, postop adhesion formation often jeopardizes this goal. Adhesion formation can be minimized by using a single, vertical, anterior midline uterine incision. Patients who have had myomectomies that enter the endometrial cavity will require cesarean sections with future pregnancies.

The abdomen is entered in the usual fashion for laparoscopy (e.g., through Veress needle or direct trocar insertion, followed by insufflation and insertion of accessory trocars). To reduce blood loss in pedunculated myomas, diluted **vasopressin** (3–5 mL) is injected into the base of the stalk where it joins the uterine wall. Intravenous vasopressin can cause $\uparrow$ BP, myocardial ischemia, dysrhythmias, or cardiac arrest and should be avoided. The pedicle is coagulated and cut with bipolar electrocautery. For subserosal or intramural myomas, diluted vasopressin is injected between the myometrium and the myoma pseudocapsule, and an incision is made on the serosa overlying the myoma using the CO₂ laser or monopolar needle or hook. As the incision is made, the myometrium is retracted away with graspers to expose the tumor. Vessels are coagulated before cutting. After the myoma is removed, the myoma bed is copiously irrigated, and bleeding points are coagulated again. The myometrial defect is closed in multiple layers to ensure all the dead space is eliminated. After closure, the serosa is irrigated. Uteroperitoneal fistulae may follow laparoscopic myomectomy because complete laparoscopic approximation of all layers of the myometrium is impossible, and the use of electrocoagulation for hemostasis inside the uterine defect also may increase the risk of fistula formation. Failure to close the dead space can also cause postop hematoma, infection, and pregnancy complications.

Removal of the specimen is frequently the most challenging aspect of the operation. The myoma can be removed either by morcellation of the specimen manually with a scalpel or by extending the suprapubic incision via a minilaparotomy. Alternatively, a **posterior culdotomy** may be performed and the myoma removed via the vagina. After the myoma is removed, the incision can be closed using laparoscopic suturing. **Minilaparotomy** or **culdotomy** facilitate removal but increase postop wound complication risks, such as infection or hernia formation. After myoma removal, the abdomen and pelvis are irrigated, the patient is taken out of the Trendelenburg position, and any fluid that might

have tracked into the upper abdomen is suctioned. The ports are closed in the usual fashion.

Usual preoperative diagnosis

Uterine myoma, fibroids

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, discussion of carb-loading DVT prophylaxis, postop pain management options, recovery plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Dorsal lithotomy; steep Trendelenburg to move bowel out of the operative field
Incision	Infraumbilical; 5 mm later and 5–12 mm midline suprapubic; minilaparotomy via extension of suprapubic incision or posterior culdotomy for removal of large myomata
Special instrumentation	CO ₂ laser; laparoscopic instrumentation
Unique considerations	For large myomata: pretreatment with GnRH analogues (3–6 mo), intraoperative injection of dilute vasopressin (1 IU in 100 mL LR) into myometrium to decrease bleeding
Antibiotics	Cefoxitin 1–2 g iv before incision
Surgical time	1–4 h
Closing considerations	Close fascia of all 10- to 12-mm ports; 5-mm ports are closed in a subcuticular fashion. Fascial closure essential in minilaparotomy. Posterior culdotomy requires laparoscopic suturing or vaginal closure.
EBL	100–600 mL
Postoperative care	Usually PACU → home. Occasionally 24-h hospital stay; early ambulation; clear liquids POD #0; gradually advance diet after discharge home

Mortality	0.08–0.2/1000
Morbidity	Rates 1.3–5.9/100 Peroneal nerve damage from positioning Severe bleeding with possible need for transfusion Uteroperitoneal fistulae Infertility GI/GU injury Air embolism Puncture of a major vessel Insufflation in the wrong place Need for emergent laparotomy Uterine rupture in pregnancy: 2% Adhesion formation
Pain score	4–6

■ PATIENT POPULATION CHARACTERISTICS

Age range	Reproductive-age females; myomas shrink in postmenopausal women and are less symptomatic.
Incidence	20–25% of all women
Etiology	Benign transformation and proliferation of a single smooth muscle cell
Associated conditions	Menorrhagia; anemia; ureteral obstruction; pelvic pain or pressure

ANESTHETIC CONSIDERATIONS

See Anesthetic Considerations following Laparoscopic Hysterectomy.

Suggested Readings

1. AAGL Advancing Minimally Invasive Gynecology Worldwide: AAGL practice report: morcellation during uterine tissue extraction. J Minim Invasive Gynecol 2014; 21:517–30.
2. Kho KA, Anderson TL, Nezhat CH: Intracorporeal electromechanical tissue morcellation: a critical review and recommendations for clinical practice. Obstet Gynecol 2014; 124:787–93.
3. Kho KA, Brown DN: Surgical treatment of uterine fibroids within a containment system and without power morcellation. Clin Obstet Gynecol 2015; 59:85–92.

4. Kho KA, Nezhat CH: Evaluating the risks of electric uterine morcellation. *JAMA* 2014; 311(9):905–6.
5. Nezhat C, Nezhat F, eds: *Nezhat's Video-Assisted and Robotic-Assisted Laparoscopy and Hysterectomy*. Cambridge University Press, New York: 2013.

LAPAROSCOPIC HYSTERECTOMY

SURGICAL CONSIDERATIONS

Description

Hysterectomy is the second most common gynecologic operation, after **cesarean section**. The indications for **hysterectomy ± salpingo-oophorectomy** include leiomyomata (38%), malignancy (15%), ovarian tumors (10%), abnormal bleeding (13%), adenomyosis (9%), pelvic pain or adhesions (5%), endometriosis (3%), and uterine prolapse (1%). Other less common indications include parametrial disease, pelvic infection, and complications of pregnancy and delivery. Selection of surgical approach for hysterectomy requires consideration of the patient's age, medical history, history of prior pelvic surgery, the presence or possibility of adhesions or endometriosis, uterine size, adnexal pathology, and the presence or amount of uterine prolapse.

Laparoscopic hysterectomy offers many advantages, including shorter recovery time, rapid return of bowel function, less pain, less bleeding and adhesion formation, and a lower wound complication rate. The disadvantages are higher cost and the surgical expertise required. The most commonly performed procedure is the **laparoscopically assisted vaginal hysterectomy** (LAVH), in which hysterectomy is begun by laparoscopy, but four or more steps are performed vaginally. Variants include **total laparoscopic hysterectomy** (TLH), in which all steps are performed laparoscopically; **supracervical hysterectomy**, where the cervix is not removed; and **vaginally assisted laparoscopic hysterectomy** (VALH), in which four steps are performed laparoscopically and the procedure is completed vaginally. Combinations are usually performed, depending on surgical findings. Mechanical bowel preparation is not advised. Consultation with a urologist, colorectal surgeon, and oncologist are sought as necessary.

Access to the abdomen is obtained in the usual fashion (e.g., through a Veress needle or direct trocar insertion or through special assist device, followed by insufflation and insertion of accessory trocars). **Diagnostic laparoscopy** is performed, adhesions lysed,

and endometriotic lesions treated. The course of the ureters is noted through the peritoneum from the pelvic brim medial to the external iliac arteries and followed inferiorly until they are no longer visible at the level of the cardinal ligaments. When ureters cannot be identified clearly because of severe scarring or endometriosis, they are dissected retroperitoneally, and the dissection proceeds as for a radical hysterectomy. At the cardinal ligaments, the peritoneum is opened above or below the ureter, and hydrodissection is performed to lift the peritoneum off the ureter without damaging it. Routine hysterectomy using hydrodissection to identify tissue planes and limit blood loss can be performed following identification of the ureters.

If the ovaries are to be spared, the utero-ovarian ligament, proximal tube, and mesosalpinx are cauterized and cut progressively, and the posterior leaf of the broad ligament is opened with blunt dissection or hydrodissection. The bladder is dissected free from the cervix and lower uterine segment. Next, the uterine vessels are identified, noted to be free of the ureters, desiccated, and cut. At the level of the cardinal ligaments, the ureters and descending branches of the uterine artery are close to one another and the cervix; therefore, cardinal ligament dissection and cautery must be precise to prevent bleeding and ureteral injury. A small uterus can be removed easily through the vagina. The use of **intracorporeal power morcellation** to remove a large uterus is no longer advised because morcellated particles may contain neoplastic tissue (known or occult) and disseminate throughout the abdominal cavity. Additionally, spread of morcellated benign tissue can cause disseminated peritoneal leiomyomatosis. A large uterus, instead, can be removed by extracorporeal **hand morcellation** in a bag and removed through the vagina or minilaparotomy incision. Pneumoperitoneum will be lost during this procedure, and care must be taken to keep instruments free of bowel or other abdominal structures.

If the procedure is to be completed entirely laparoscopically, pneumoperitoneum can be maintained by placing a glove containing two 4" × 4" sponges in the vagina. The vaginal wall is cut circumferentially, and the uterus is pulled to mid vagina, but not removed, to preserve the pneumoperitoneum. Alternatively, the uterus may be placed in a laparoscopic specimen bag. The suprapubic incision may be extended into a minilaparotomy incision for specimen removal. The vaginal cuff is closed transversely using laparoscopic sutures, and any coexisting cystocele or enterocele is repaired. After the uterus is removed and the vaginal cuff closed, the pelvic and abdominal cavities are reevaluated, irrigated, and cleared of blood and debris. The skin and fascia are closed in the usual fashion. **Cystoscopy** is performed at the end of the procedure to verify ureter function and bladder integrity.

Variant procedure

In patients with severe rectovaginal and vesical endometriosis, the retroperitoneal space is first entered using hydrodissection, and the external iliac vessels, hypogastric artery, and ureters are identified. In cases where extensive dissection and resultant blood loss are anticipated, coagulation or ligation of the hypogastric artery with laparoscopic clips may be performed. Endometriosis of the rectum, rectovaginal septum, and uterosacral ligaments is treated by vaporization, excision, or a combination of both. **Sigmoidoscopy or proctoscopy** with concurrent laparoscopic visualization of the pelvis may be necessary to rule out the presence of incidental enterotomy. **Cystoscopy with ureteral stenting** also may be helpful to identify anatomy and ureteral injury if it occurs.

After the ureter is identified along its course and entry into the bladder, the uterine vessels are retracted medially and separated from the ureter using a CO₂ laser. The uterus is retracted medially, and the ureter laterally as the cardinal and uterosacral ligaments are cauterized and cut, maintaining the ureter under direct visualization. After these vascular pedicles have been ligated and all endometriosis treated, the hysterectomy and specimen removal proceed as described earlier. In patients with a history of prior cesarean section, endometriosis, or anterior myomectomy, adhesions can distort anatomy and increase the risk of inadvertent cystotomy. In this case, it may be useful to perform a “reverse” inferior to superior vesicouterine fold dissection technique to mobilize the bladder. This technique utilizes blunt dissection within the broad ligament to visualize a new anatomical plane, lateral to the vesicocervical ligaments. By extending from this “new space” inferomedially across the anterior aspect of the cervix, a connection is developed inferior to the vesicouterine adhesions. Then, using an inferior to superior sweeping motion, the bladder can be easily dissected off of the vagina, cervix, and uterus from an unscarred plane, and the cervical cuff can be exposed for **colpotomy**.

After or at the time of closure of the vaginal cuff, some gynecologic surgeons perform a **culdoplasty** to close the posterior cul-de-sac. This is recommended to decrease the risk of enterocele formation and the potential development of vaginal vault prolapse. There are different types of culdoplasty, and they can be done vaginally or laparoscopically. The most common vaginal culdoplasty is the **McCall culdoplasty** where the uterosacral-cardinal complex is plicated and attached to the peritoneal surface of the posterior cul-de-sac to elevate the posterior vaginal cuff. After the uterus is removed, the patient is placed in a dorsal lithotomy position to allow for vaginal access. Before closure of the cuff, an absorbable suture is placed through the full-thickness of the posterior vaginal wall from outside to in and then passed through the left uterosacral ligament pedicle, the posterior peritoneum, the right uterosacral ligament pedicle, and back through the full

thickness of the posterior vaginal cuff from inside out. The two ends of the suture are then tied, bringing the uterosacral ligaments together in a procedure called a **Moschcowitz culdoplasty**. After the vaginal cuff is closed, a separate absorbable suture is used to close the posterior cul-de-sac. Suture is passed through one of the uterosacral ligaments, through the posterior peritoneum, through the other uterosacral ligament, and then through another portion of the posterior peritoneum and tied to form a purse string.

Another type of laparoscopic culdoplasty is the **Halban culdoplasty**: after the vaginal cuff is closed, interrupted sutures are placed vertically across the posterior cul-de-sac starting with the posterior peritoneum over the rectum and taking small portions of the peritoneum up to and including the vaginal cuff apex. Usually, three to four interrupted sutures are used. The most common complications of culdoplasty are injuries to the ureter. As a result of suturing of the uterosacral ligaments, which are very close to the ureters, the ureters are at risk for being obstructed by the culdoplasty stitch. Pyridium (phenazopyridine) should be given before the procedure to help better identify the ureter. **Cystoscopy** should also be performed after the culdoplasty to ensure bilateral ureteral patency. Another complication, although uncommon, is rectal injury. The rectum is directly under the posterior peritoneum and can occasionally get sutured into the culdoplasty. If this is suspected, then proctoscopy should be performed to evaluate the rectum.

Usual preoperative diagnosis

Leiomyomata; malignancy; ovarian tumors; abnormal bleeding; adenomyosis; pelvic pain or adhesions; endometriosis; uterine prolapse; parametrial disease; pelvic infection; complications of pregnancy and delivery

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, postop pain management options, recovery plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Dorsal lithotomy; legs in Allen stirrups; steep Trendelenburg
Incision	Infraumbilical; lateral suprapubic; midline suprapubic (5 or 10 mm)
Special	CO ₂ laser; laparoscopic instruments; uterine

instrumentation	manipulator
Unique considerations	Extensive ureterolysis and treatment of endometriosis may require addition of cystoscopy, ureteral stent placement, and/or sigmoidoscopy.
Antibiotics	Cefoxitin 2 g iv before incision
Surgical time	2–6 h. Operative time is increased in cases of extensive adhesiolysis or endometriosis.
Closing considerations	Close fascia of all 10- to 12-mm ports; 5-mm ports are closed in a subcuticular fashion. Fascial closure essential in minilaparotomy.
EBL	100–800 mL, depending on anatomy and difficulty of dissection
Postoperative care	PACU → home. Occasionally 24-h hospital stay
Mortality	0.08–0.2/1000
Morbidity	Overall complication rate: 2.5% Conversion to laparotomy: 4.2/1000 Air embolism: rare Peroneal nerve damage from positioning: rare Unintended puncture of viscera: rare Puncture of a major vessel: rare Insufflation of incorrect site: rare Urinary and ureteral trauma, including fistulae: 1.6%
Pain score	4–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	30s–90s
Incidence	20% of women <40 yr; 37% of women < age 65
Etiology	see Associated Conditions
Associated conditions	Myomata; abnormal uterine bleeding; adenomyosis; malignancy; pelvic pain; endometriosis

Suggested Readings

1. Nezhat C, et al: Reverse vesicouterine fold dissection for laparoscopic hysterectomy after prior cesarean deliveries. *Obstet Gynecol* 2016; 128(3):629–33.
2. Nezhat C, Li A, Falik R, et al: Bowel endometriosis: diagnosis and management. *Am J Obstet Gynecol* 2018; 218(6):549–62.
3. Nezhat C, Nezhat F, eds: *Nezhat's Video-Assisted and Robotic-Assisted Laparoscopy and Hysterectomy*. Cambridge University Press, New York: 2013.
4. Nezhat C, Falik R, McKinney S. et al: Pathophysiology and management of urinary tract endometriosis. *Nat Rev Urol* 2017; 14:359–72.

LAPAROSCOPIC SURGERY FOR VAGINAL VAULT SUSPENSION

SURGICAL CONSIDERATIONS

Description

Pelvic organ prolapse (POP) is a more common finding in older women, and there are both nonsurgical and surgical approaches to treatment. Nonsurgical options include pessaries and pelvic floor training and are often used for women who are not surgical candidates or who would instead not undergo surgery. A surgical approach to treating POP is more definitive and can be performed abdominally, vaginally, laparoscopically, or robotically. When being performed laparoscopically or robotically, the patient benefits from a shorter hospital stay, less postop pain, and faster recovery.

Sacrocolpopexy is considered the surgical standard for treatment of POP. It is suitable for patients with uterine prolapse or Stages II–IV vaginal vault prolapse and recurrence after previous operation, particularly for younger patients who are sexually active. This procedure aims to provide apical support for defects in the cardinal-uterosacral ligament complex and has the highest success rate. This procedure is not without unique complications. They include mesh and suture erosion, dyspareunia, and alterations to bowel or bladder function.

The abdomen is entered in the usual fashion for laparoscopy through a Veress needle or direct trocar insertion followed by insufflation and insertion of accessory trocars. If the patient has not had a hysterectomy, a supracervical hysterectomy is performed instead of a total hysterectomy, to reduce the risk of mesh erosion. The bowel is elevated from the pelvis for visualization. There are two main areas of dissection: pelvic and presacral. The bladder is mobilized and dissected away from the anterior surface of the cervix and vagina. In a similar fashion, the rectum is dissected and mobilized away from the

posterior surface of the vagina. It is important in this dissection to identify tissue planes and avascular spaces. Vasopressin injection may be used during the dissection to control bleeding and allow for better visualization. Attention is then turned to the presacral area. The peritoneum over the sacrum is tented up with laparoscopic graspers and incised. Frequently, hydrodissection in this area is beneficial for identification of the correct tissue planes. The dissection is carried down to and through the presacral adipose to the level of the anterior longitudinal ligament. Particular attention must be taken to avoid injury to the middle sacral vessels, the right ureter, and sigmoid colon. At this time, the peritoneum between the presacral dissection and the posterior vaginal dissection is opened. This will be closed later over mesh, making it retroperitoneal. A “Y”-shaped mesh is inserted through a trocar into the abdomen. The mesh is then secured to the posterior and anterior side of the vagina, taking full thickness bites and using laparoscopic suturing technique. The tail of the mesh is then secured to the anterior longitudinal ligament in a similar fashion, suspending the vaginal apex. The peritoneum is then closed over the mesh to help prevent bowel adhesions. The pelvis is then irrigated, and hemostasis confirmed. The ports are closed in the usual fashion. If a concurrent **hysterectomy** was performed, a **cystoscopy** is done at the end of the procedure to ensure bilateral ureteral patency.

Variant procedure

The vaginal posterior and anterior dissection, as well as the type of graft material used for the suspension are the two main areas that may vary during this procedure. Both of these factors are highly dependent on the surgeon.

ERAS: Preop counseling by the surgeon and anesthesiologist should include limiting NPO duration, alcohol and smoking cessation, review of nutritional status, discussion of carb-loading DVT prophylaxis, postop pain management options, recovery plan, and possibility of urinary drain postop. See [Appendix K, p. K-1](#).

■ SUMMARY OF PROCEDURE

Position	Dorsal lithotomy; legs in Allen stirrups; steep Trendelenburg
Incision	Infraumbilical; 5 mm lateral and 5–12 mm midline suprapubic
Special instrumentation	CO ₂ laser; laparoscopic instrumentation; uterine manipulator, bipolar Kleppinger forceps, suction irrigator

Unique considerations	Extensive ureterolysis and enterolysis may require addition of cystoscopy, ureteral stent placement, and/or sigmoidoscopy.
Antibiotics	Cefoxitin 1–2 g iv before incision
Surgical time	2–6 h, depending on anatomy and difficulty of dissection
Closing considerations	Close fascia of all 10- to 12-mm ports; 5-mm ports are closed in a subcuticular fashion. Fascial closure essential in minilaparotomy.
EBL	100–800 mL
Postoperative care	Usually PACU → home. Occasionally 24-h hospital stay; early ambulation
Mortality	Rare
Morbidity	Unintentional puncture of viscera: rare Puncture of a major vessel: rare Insufflation of incorrect site: rare Urinary and ureteral trauma, including fistulae: 1.6%
Pain score	4–9

■ PATIENT POPULATION CHARACTERISTICS

Age range	39s–90s
Incidence	20% of women >40 yo; 37% of women >65 yo
Etiology	See associated conditions
Associated conditions	Cystocele, rectocele, enterocele

Suggested Readings

1. Callewaert G, Bosteels J, Housmans S, et al. Laparoscopic versus robotic-assisted sacrocolpopexy for pelvic organ prolapse: a systematic review. *Gynecol Surg* 2016; 13:115–23.
2. Nezhat C, Nezhat F, eds: *Nezhat's Video-Assisted and Robotic-Assisted-Laparoscopy and Hysterectomy*. Cambridge University Press, New York: 2013.

ANESTHETIC CONSIDERATIONS

(Procedures covered: laparoscopic surgery for endometriosis, ectopic pregnancy, myomectomy, hysterectomy.)

PREOPERATIVE

Respiratory	There can be intraop respiratory compromise from ↑ intra-abdominal pressure 2° CO₂ insufflation; however, patients without significant respiratory disease tolerate the insufflation quite well. Look for diaphragmatic hernia or advanced pulmonary disease, which → ↑ risk of laparoscopic approach. Tests: As indicated from H&P
Cardiovascular	Insufflation of the abdomen (typically with pressures of 14–22 mm Hg) → ↑ SVR and ↓ venous return. ↑ PaCO₂ → ↑ dysrhythmias. These are usually well-tolerated in the otherwise healthy patient. Tests: As indicated from H&P
Gastrointestinal	Consider NPO status for emergent ectopic pregnancy patients. High BMI complicates steep Trendelenburg positioning during laparoscopic pelvic surgery. Prior pelvic and abdominal surgery and intra-abdominal disease including, endometriosis and pelvic inflammatory disease will complicate laparoscopic surgery and may force conversion to open procedure. The combination of ↑ intra-abdominal pressure + Trendelenburg position → ↑ aspiration risk. Note: mechanical bowel preparation no longer recommended. Normal NPO restrictions should apply. Test: Electrolytes
Hematologic	These patients often are having surgery for abnormal uterine bleeding → consider anemia, which may be severe. Tests: Hgb/Hct; consider T&S/T&C, especially if patient anemic preop

Laboratory	Other tests as indicated from H&P
Premedication	Anxiolysis: consider midazolam 1–2 mg iv Analgesia: consider acetaminophen 650–1000 mg po, gabapentin 300–600 mg po, and/or celecoxib 400 mg po (see Appendix C: Multimodal Perioperative Pain Management , p. C-1)
ERAS	Routine mechanical bowel prep not recommended. Check compliance with presurgical ERAS recommendations (see above) including NPO status. Consider preop acetaminophen 1000 mg, ± gabapentin 300 mg po. Consider aprepitant 40 mg po for patients with h/o severe PONV.

INTRAOPERATIVE

Anesthetic technique

GETA

Induction	Standard induction (see p. B-2). Rapid sequence induction usually is indicated for patients with an ectopic pregnancy. OGT/NGT to decompress stomach before laparoscopic trocar insertion.
ERAS	Goals include opiate-sparing techniques and multimodal analgesia (Appendix K , p. K-2 and p. K-5), measures to achieve normothermia, euvolemia w/ balanced salt solutions, and PONV prophylaxis. Verify preincision antibiotics. Assess volume status (see Appendix K). Maintain normovolemia and normothermia.
Maintenance	Standard maintenance (see p. B-3). Muscle relaxation helpful for initial trocar insertions and may be helpful throughout procedure. Consider ketamine for chronic pain patients (0.5 mg/kg bolus + 5 mcg/kg/min infusion).
Emergence	Standard emergence. PONV prophylaxis may include dexamethasone and ondansetron.

Blood and fluid requirements	Usually minimal blood loss though may be significant during some procedures. Discuss with surgeon ahead of time. IV: 18–16 ga × 1–2 BSS @ 2–6 mL/kg/h	Consider preop hypovolemia in setting of acute bleeding (ectopic pregnancy with rupture). Establish iv access accordingly. During surgery, large volumes of fluid are sometimes given intra-abdominally for irrigation and hydrodissection → fluid overload; therefore, carefully account for all fluids, and titrate iv fluids to maintain euvoemia. See Restricted Fluid Management, p. K-2.
Monitoring	Standard monitors (see p. B-1) Foley catheter	
Positioning	✓ and pad pressure points ✓ eyes	Sciatic and femoral nerve injuries may occur in the lithotomy position. Check to minimize hip abduction and external rotation. Carefully pad and secure knees and heels. Watch hands and finger carefully when arms tucked at patient's side.
Complications	Abdominal access injuries Pneumoperitoneum effects	Most complications occur during abdominal access. These include injuries to major and minor blood vessels and may lead to catastrophic and even fatal bleeding. Bowel and bladder injuries may occur and go undiagnosed until the postop period. Effects may include subcutaneous emphysema,

		mediastinal emphysema, pneumothorax, shoulder pain, CO₂ retention, air embolism following venous injury, and dysrhythmias. Physiologic changes induced by pneumoperitoneum including ↑HR, MAP, PVR, and SVR, and ↓VC, venous return, preload, and CO may be poorly tolerated in patients with advanced pulmonary and/or cardiac disease.
	Bradydysrhythmias	May be profound. Halt surgery, consider deflating pneumoperitoneum, and treat with atropine (0.5 mg) or glycopyrrolate (0.2–0.4 mg) if persistent or severe.
	Abdominal wall hematoma	
	Nerve injury	See positioning.
	Hypothermia	
	Neuropathies	These can be long cases, with the patient in lithotomy position. Make sure the pressure points are padded well, and, if the arms are out, relieve stress on the brachial plexus.
	Fluid overload	✓ fluid volume entering and exiting the abdomen. Fluid absorption → fluid overload → CHF, edema

POSTOPERATIVE

Complications	PONV VTE Hypothermia	↑↑ Risk for PONV. See p. B-6. See p. B-7.

Multimodal pain management	Consider ketorolac 15–30 mg iv at closing ± LA into surgical wounds	Patients may complain of shoulder pain due to diaphragmatic irritation. Consider acetaminophen 1000 mg po/iv q8h, ±gabapentin 100–200 mg po tid, ibuprofen 200–800 mg po/iv q6h, or hydromorphone 0.2–0.5 mg iv for breakthrough pain.
ERAS	Goals include opiate-sparing and multimodal postop analgesia	Encourage early enteral nutrition, Foley catheter removal, and mobilization. Consider DVT prophylaxis. Chewing gum may promote return of bowel function.

Suggested Readings

1. Børglum J, Gögenür I, Bendtsen TF: Abdominal wall blocks in adults. *Curr Opin Anaesthesiol* 2016; 29(5):638–43.
2. Chapman JS, Roddy E, Ueda S, et al: Enhanced recovery pathways for improving outcomes after minimally invasive gynecologic oncology surgery. *Obstet Gynecol* 2016; 128(1):138–44.
3. Gerges FJ, Kanazi GE, Jabbour-Khoury SI: Anesthesia for laparoscopy: a review. *J Clin Anesth* 2006; 18(1):67–78.
4. Nelson G, Altman AD, Nick A, et al: Guidelines for postoperative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part II. *Gynecol Oncol* 2016; 140(2):323–32.
5. Nelson G, Altman AD, Nick A, et al: Guidelines for pre- and intra-operative care in gynecologic/oncology surgery: enhanced recovery after surgery (ERAS®) Society recommendations—Part I. *Gynecol Oncol* 2016; 140(2):313–22.